Fitting Mathematical Models to Biological Data using Non-Linear Least-Squares (nlls)

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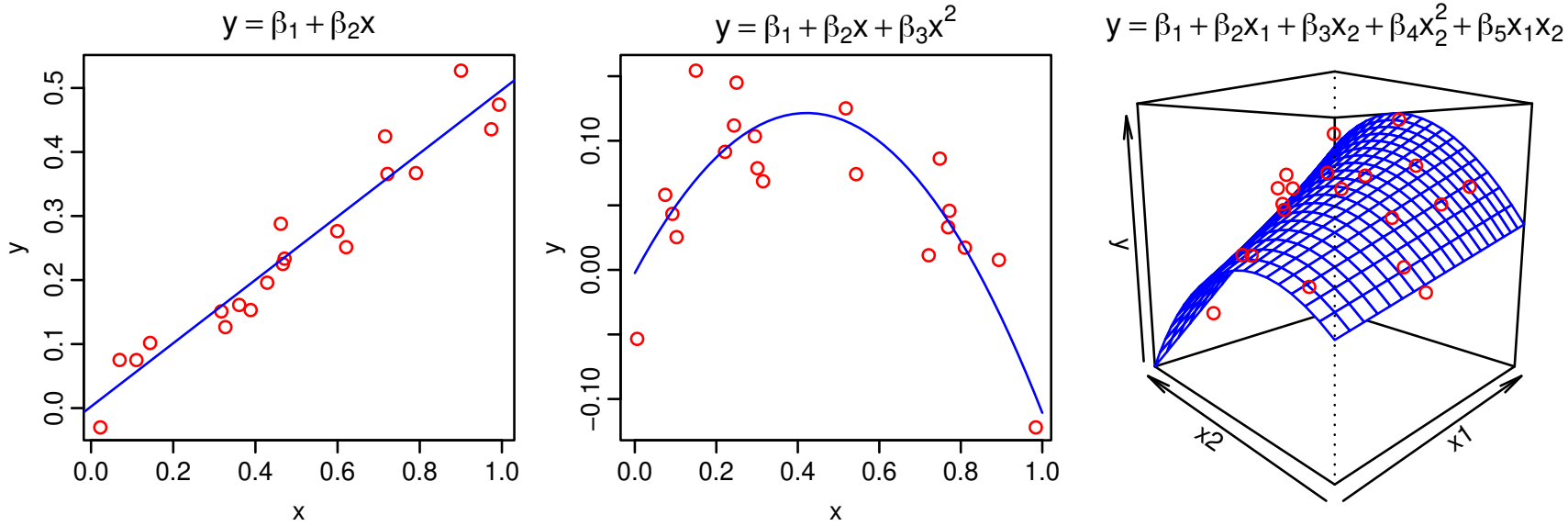
OUTLINE

- Why nlls?
- The nlls fitting method
- Practicals (in `r`) overview

Why NLLS?

LINEAR MODELS

- These are *all* good *Linear Models* (really?!):



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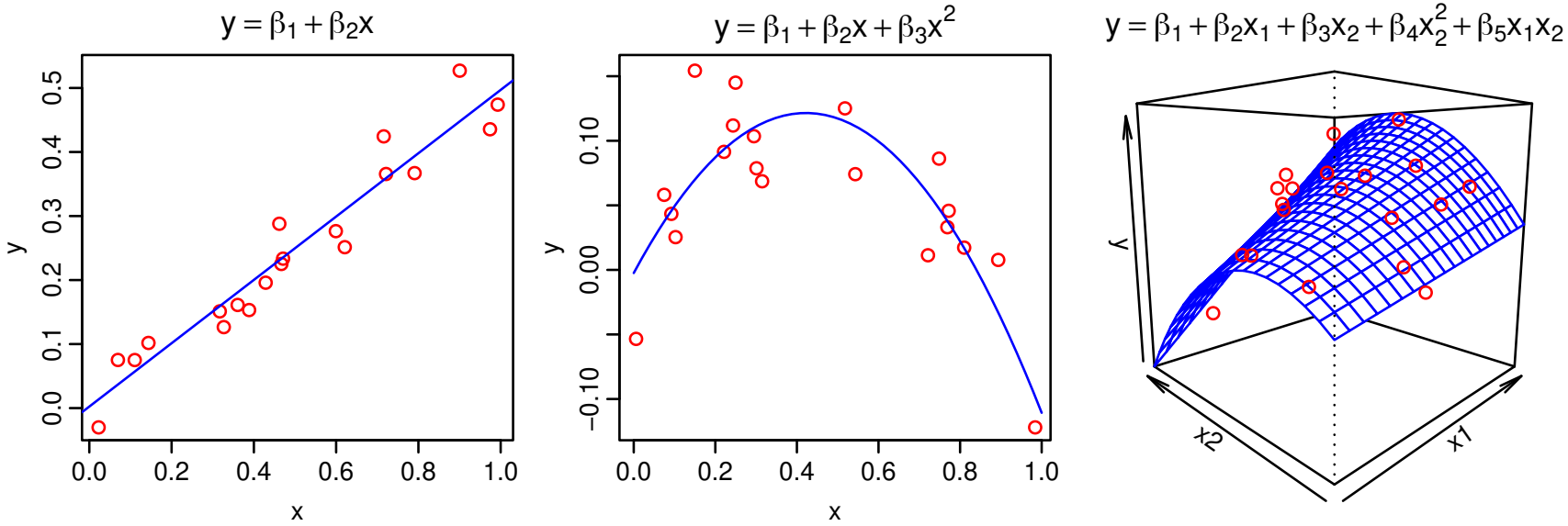

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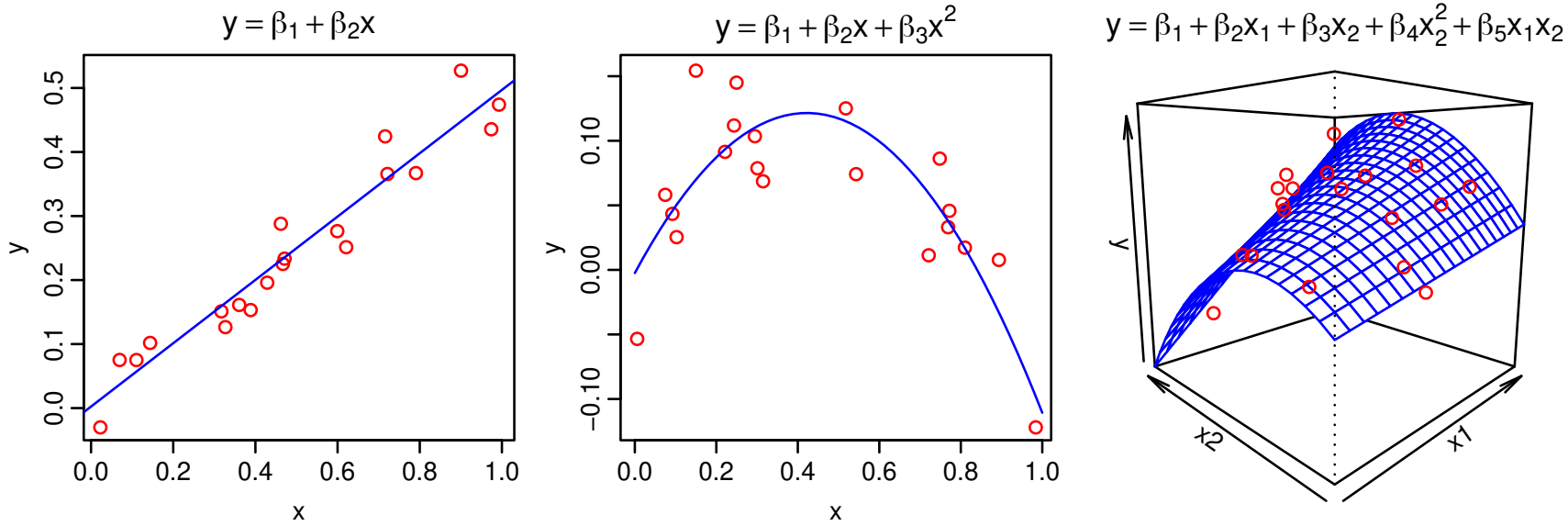

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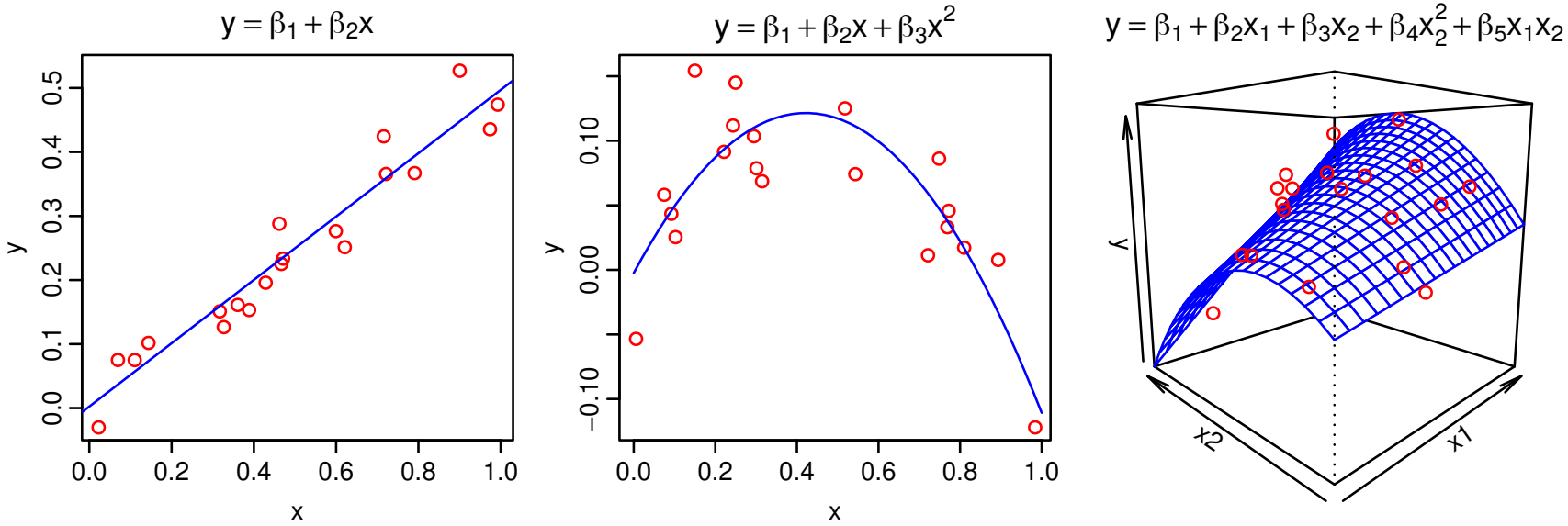

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- Easily fitted using *Ordinary Least Squares* (OLS)
- Linear models can *include curved responses* (e.g. Polynomial regression)

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  In all of these, at least one term is non-linear (e.g., $x_i^{\beta_2}$, $e^{\beta_2 x_i}$, etc.)

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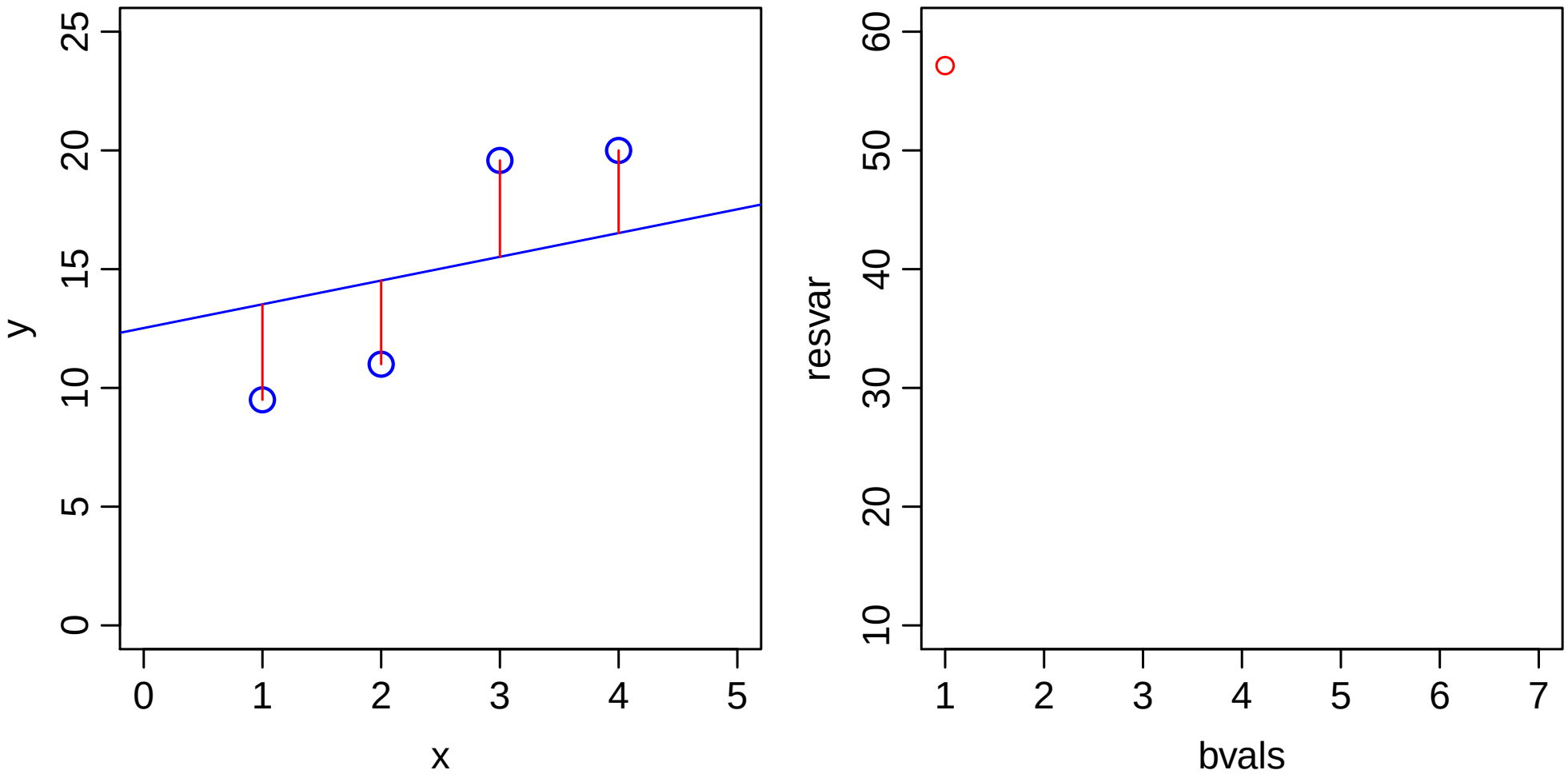
- Let's picture this using a simple (OLS) example; fitting the model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$...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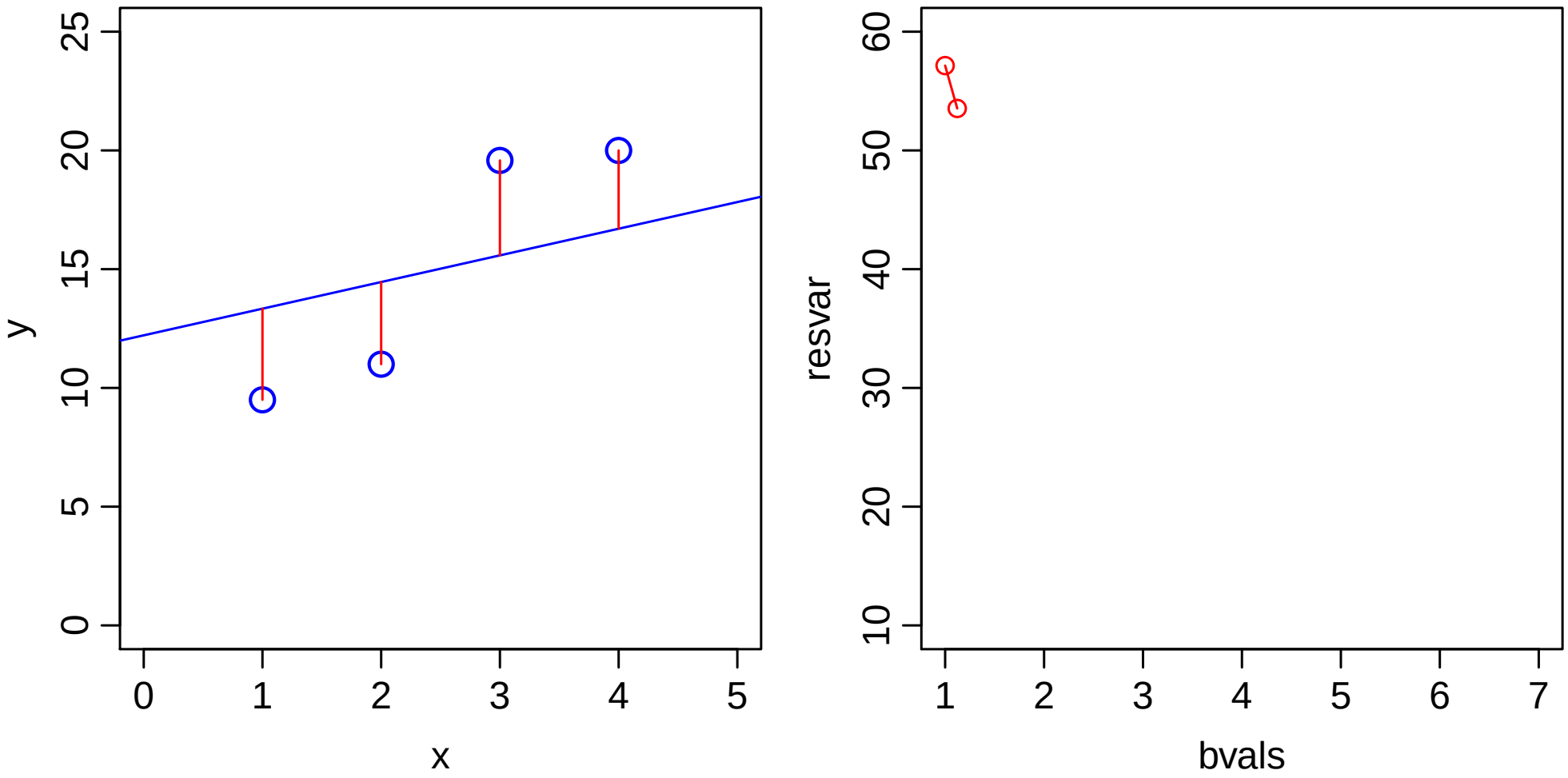
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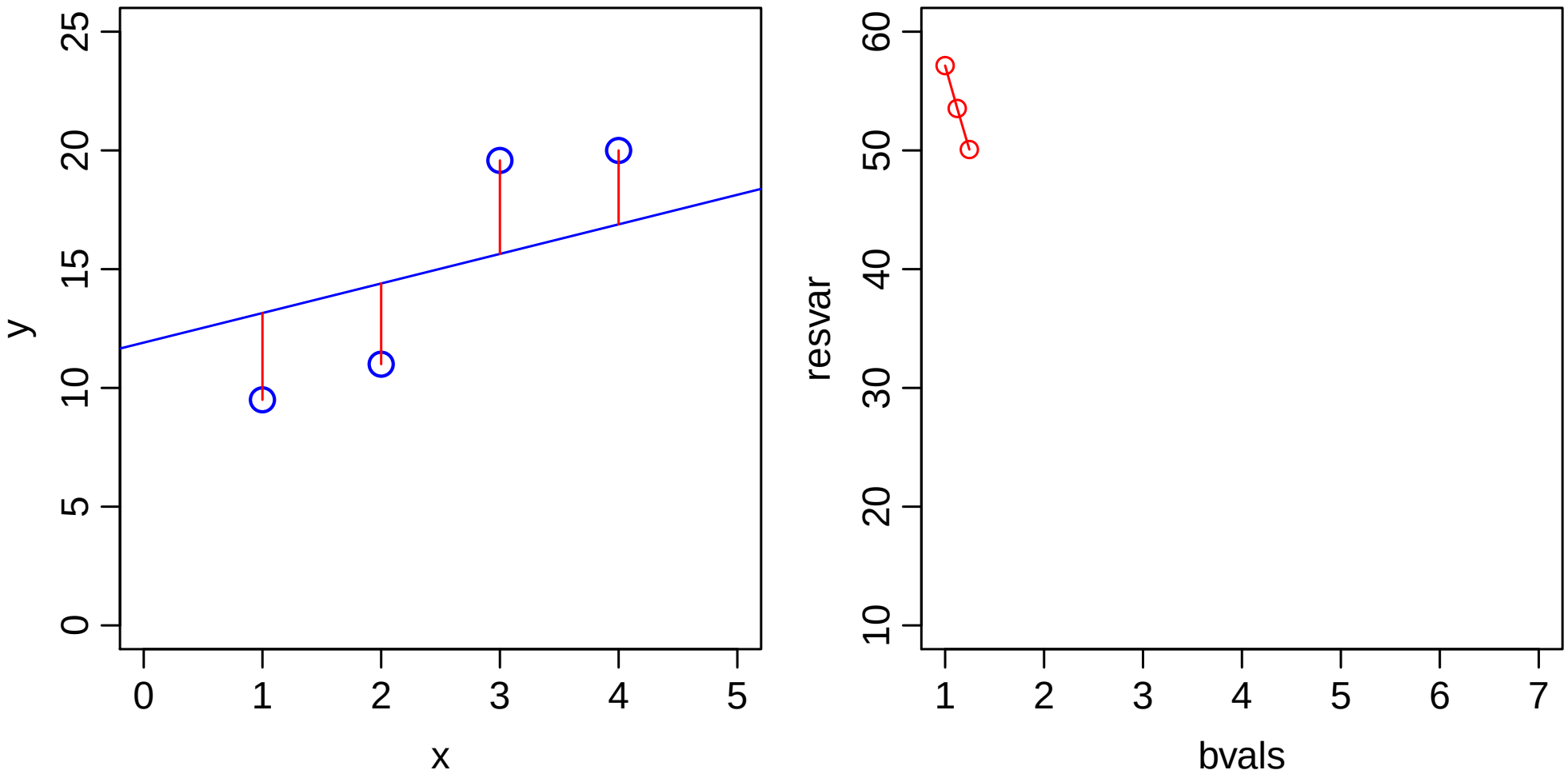
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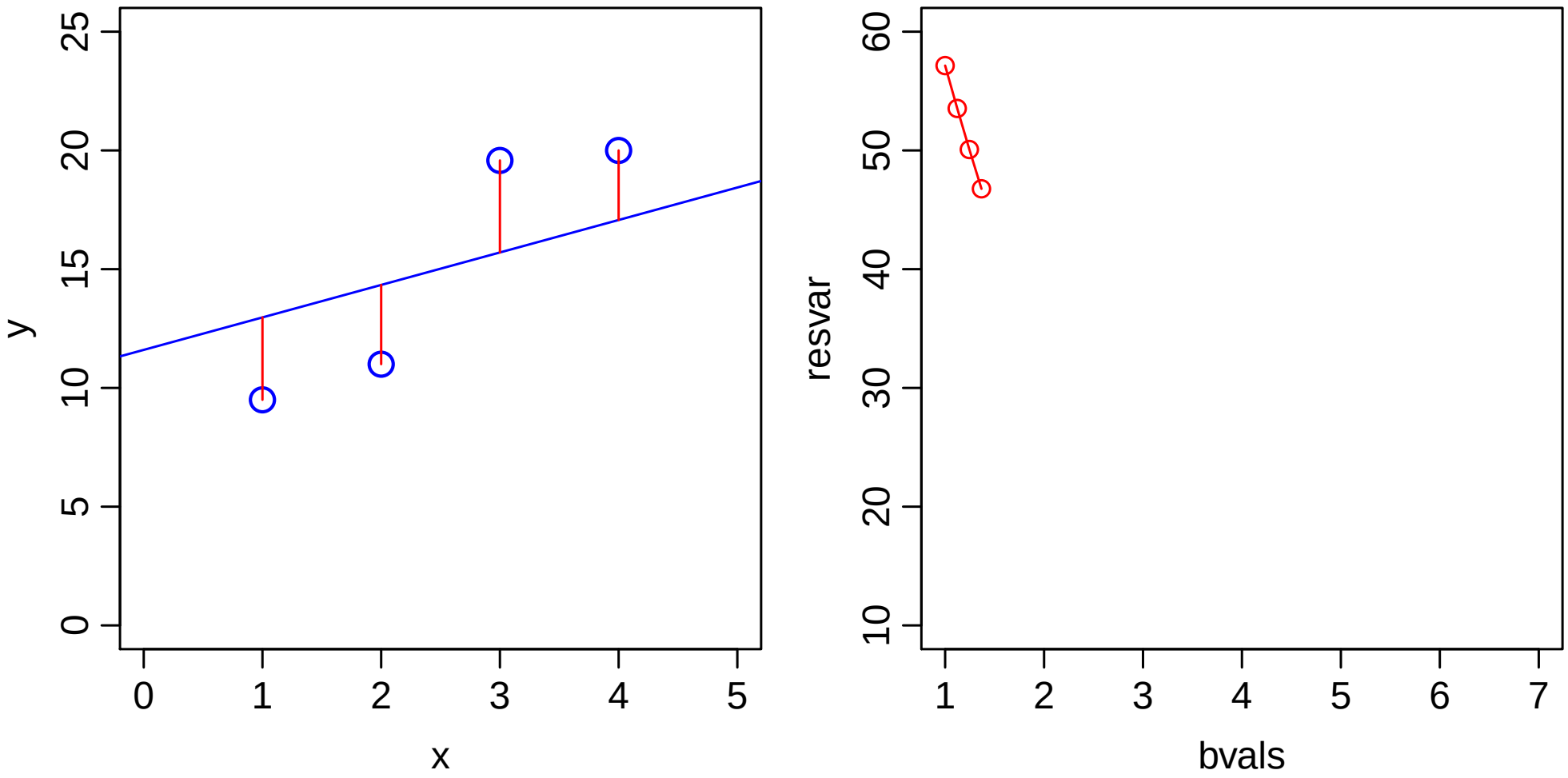
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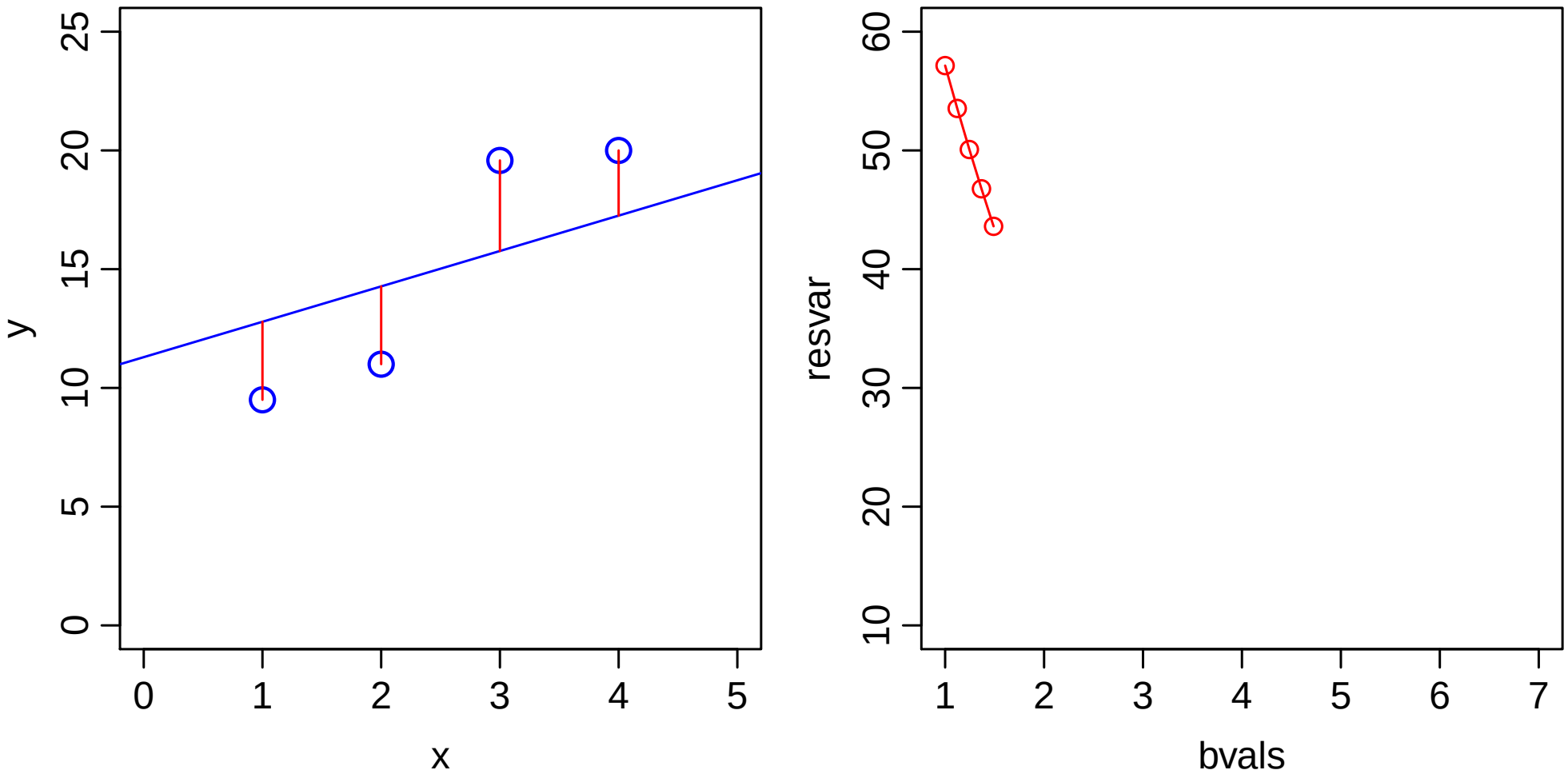
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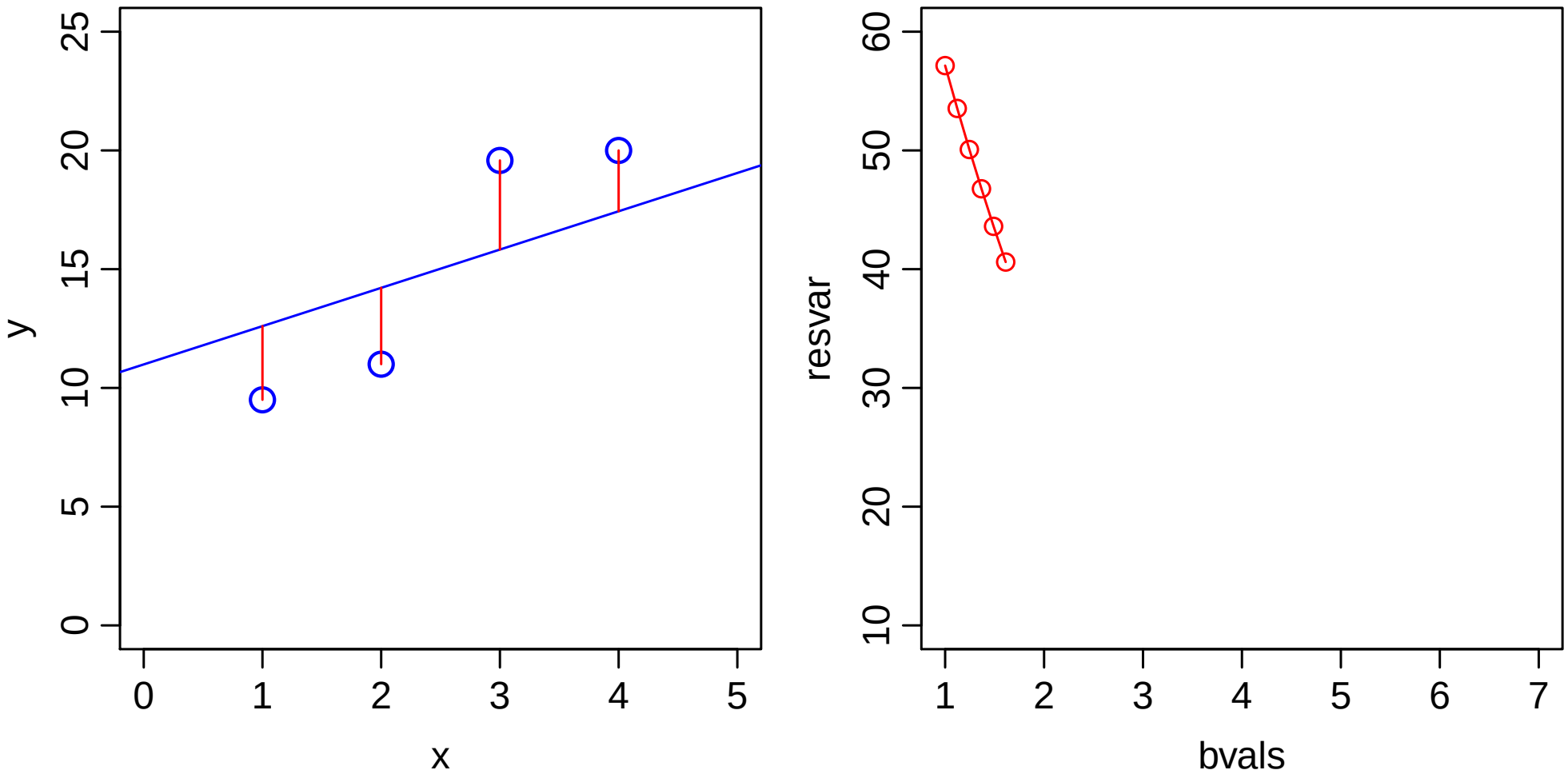
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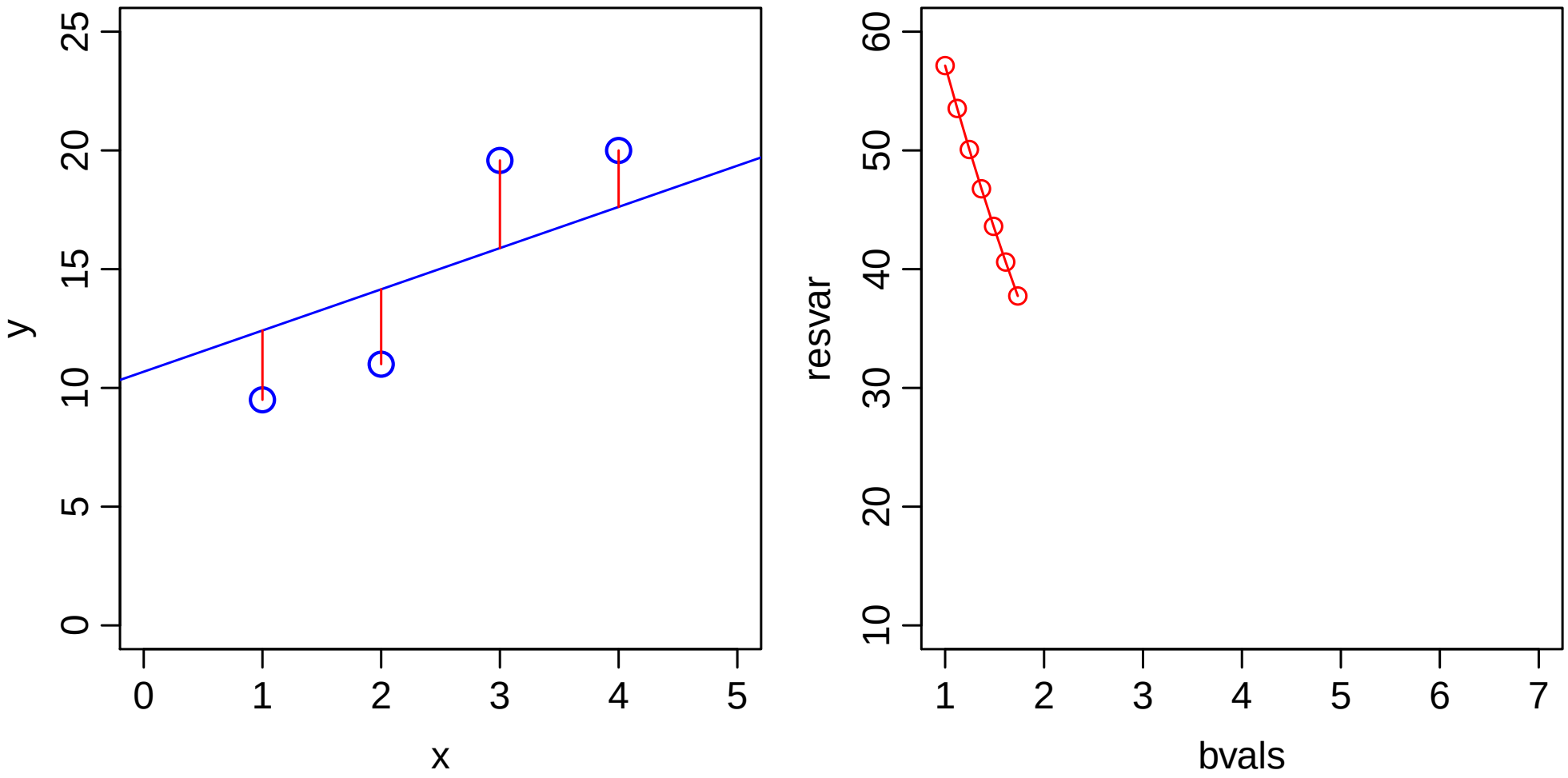
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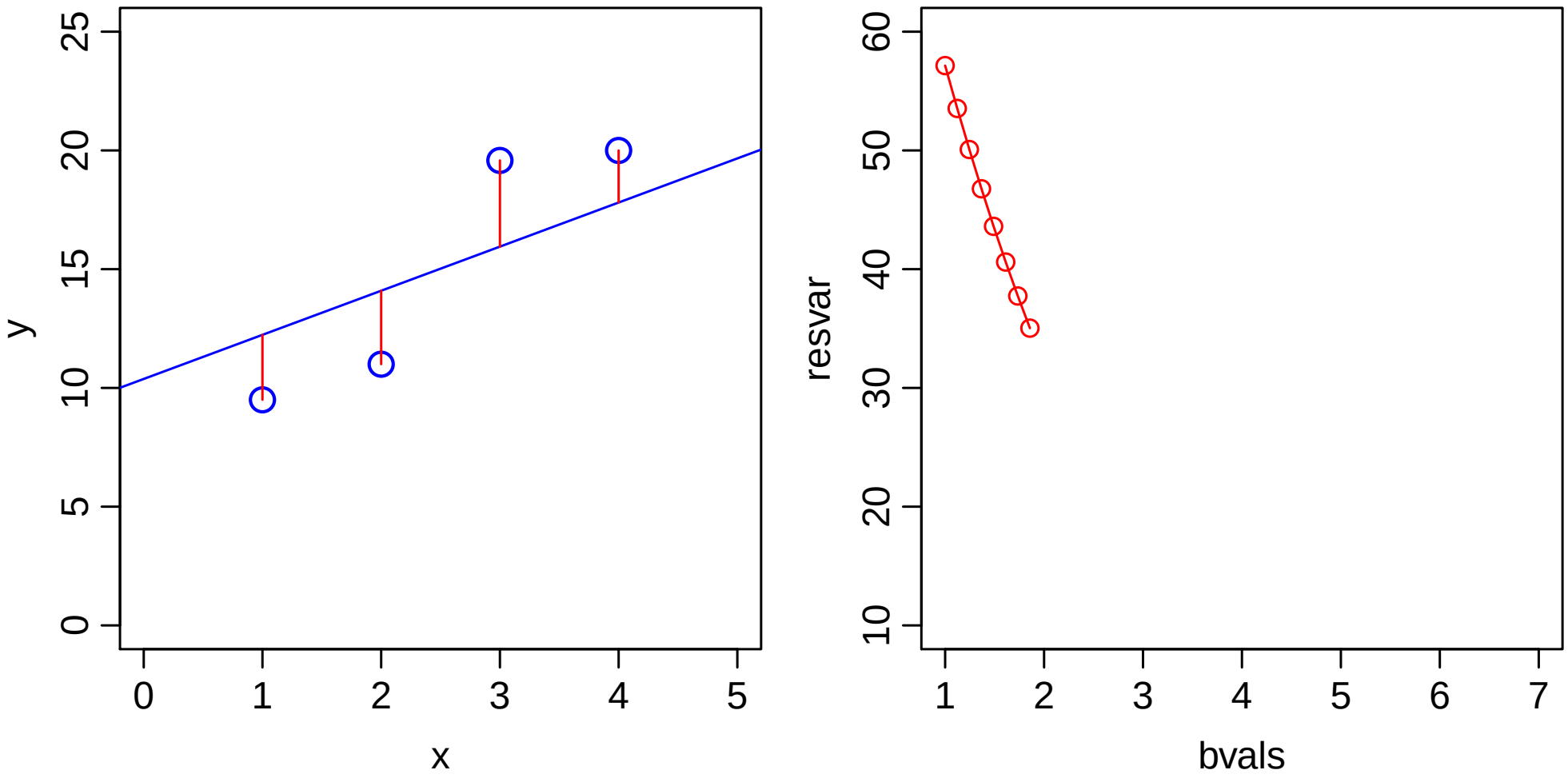
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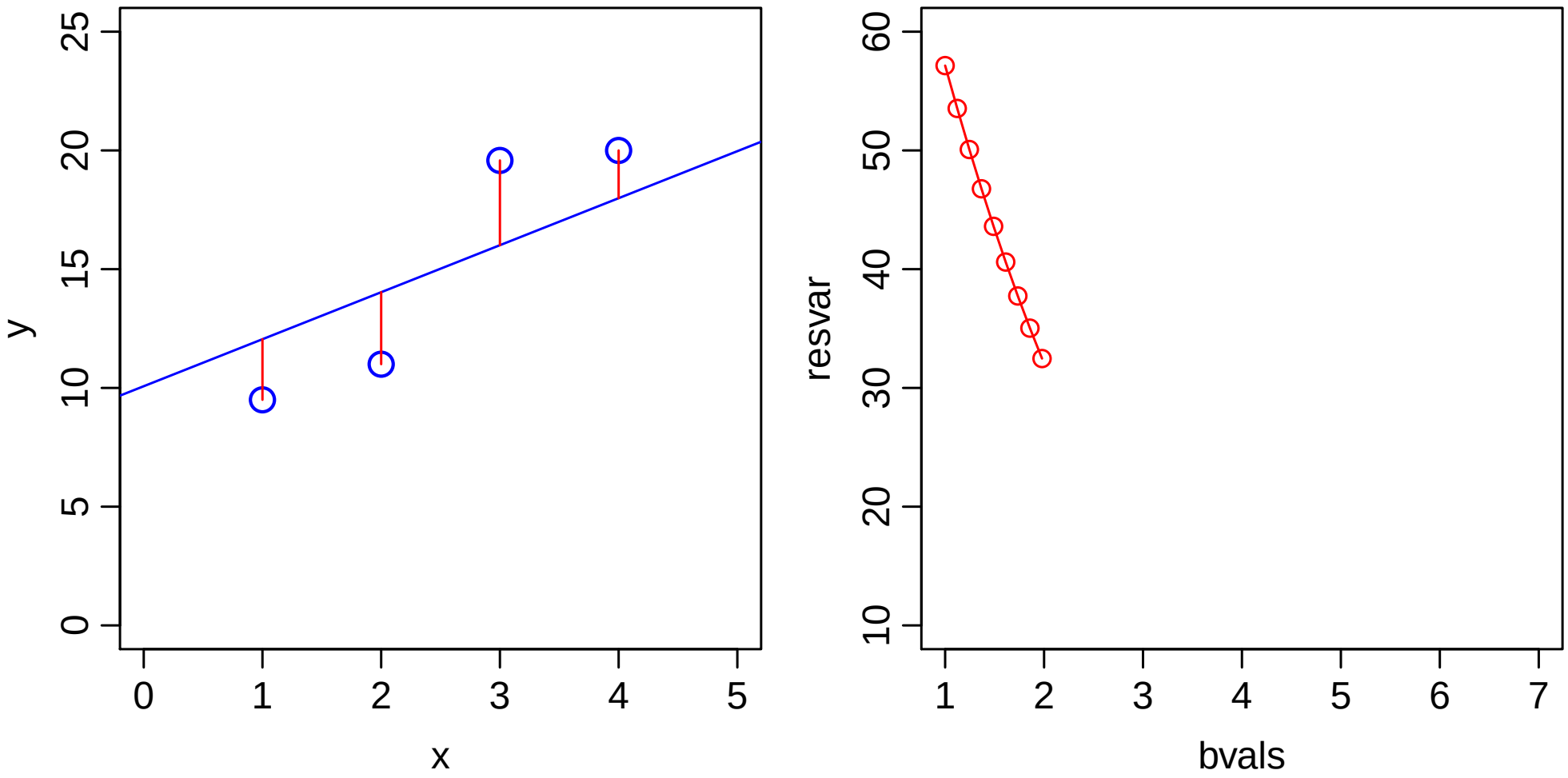
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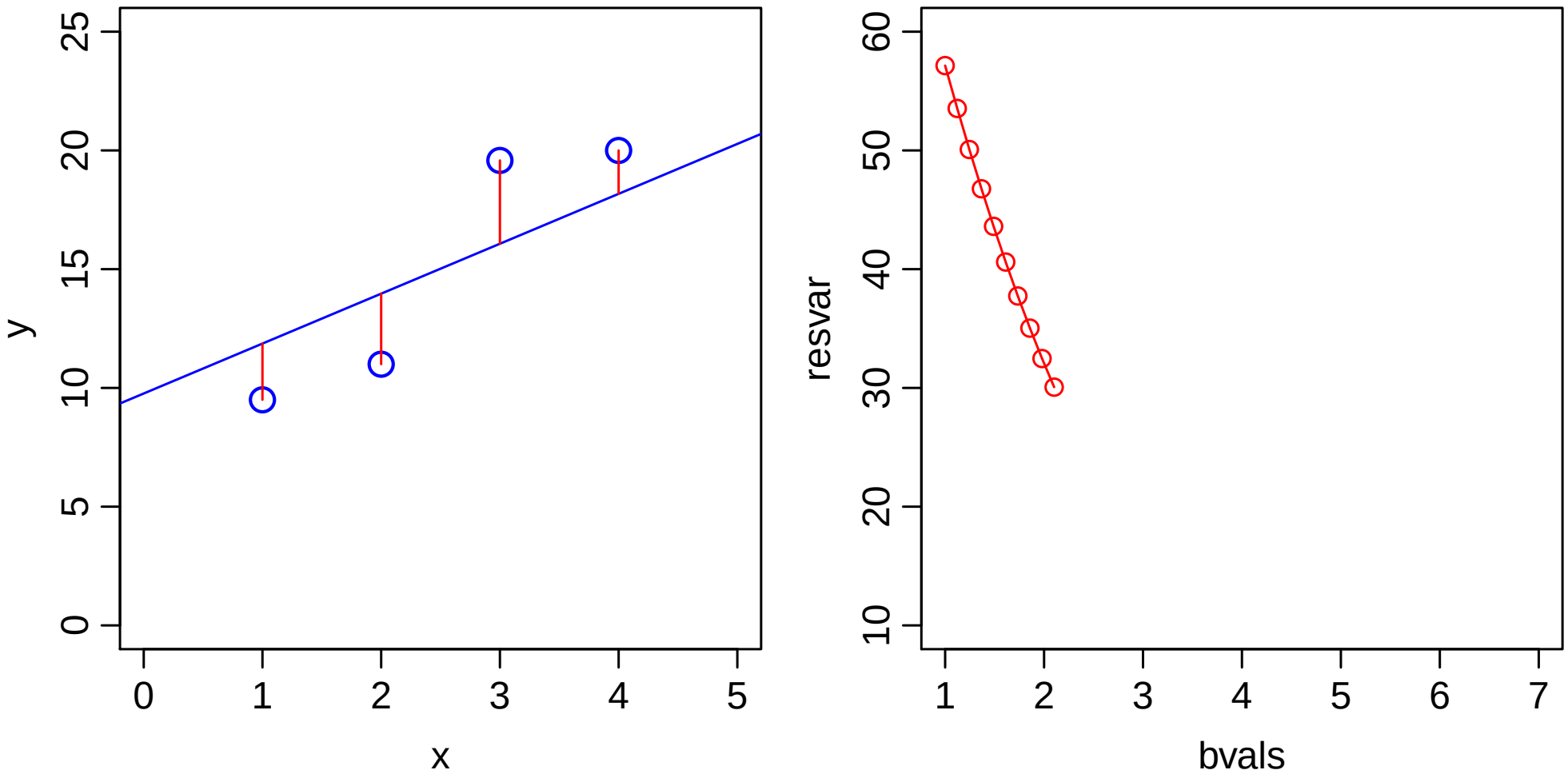
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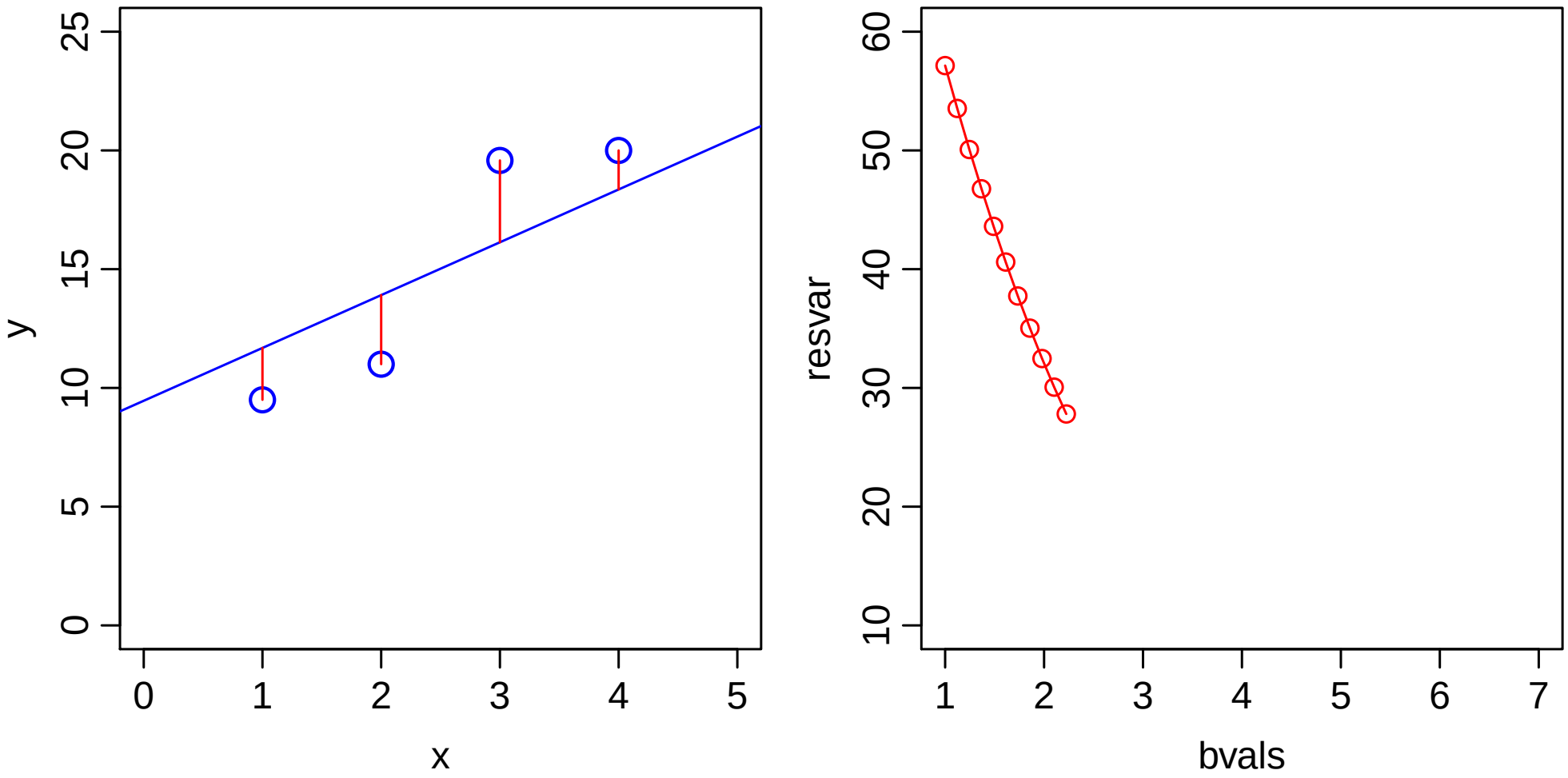
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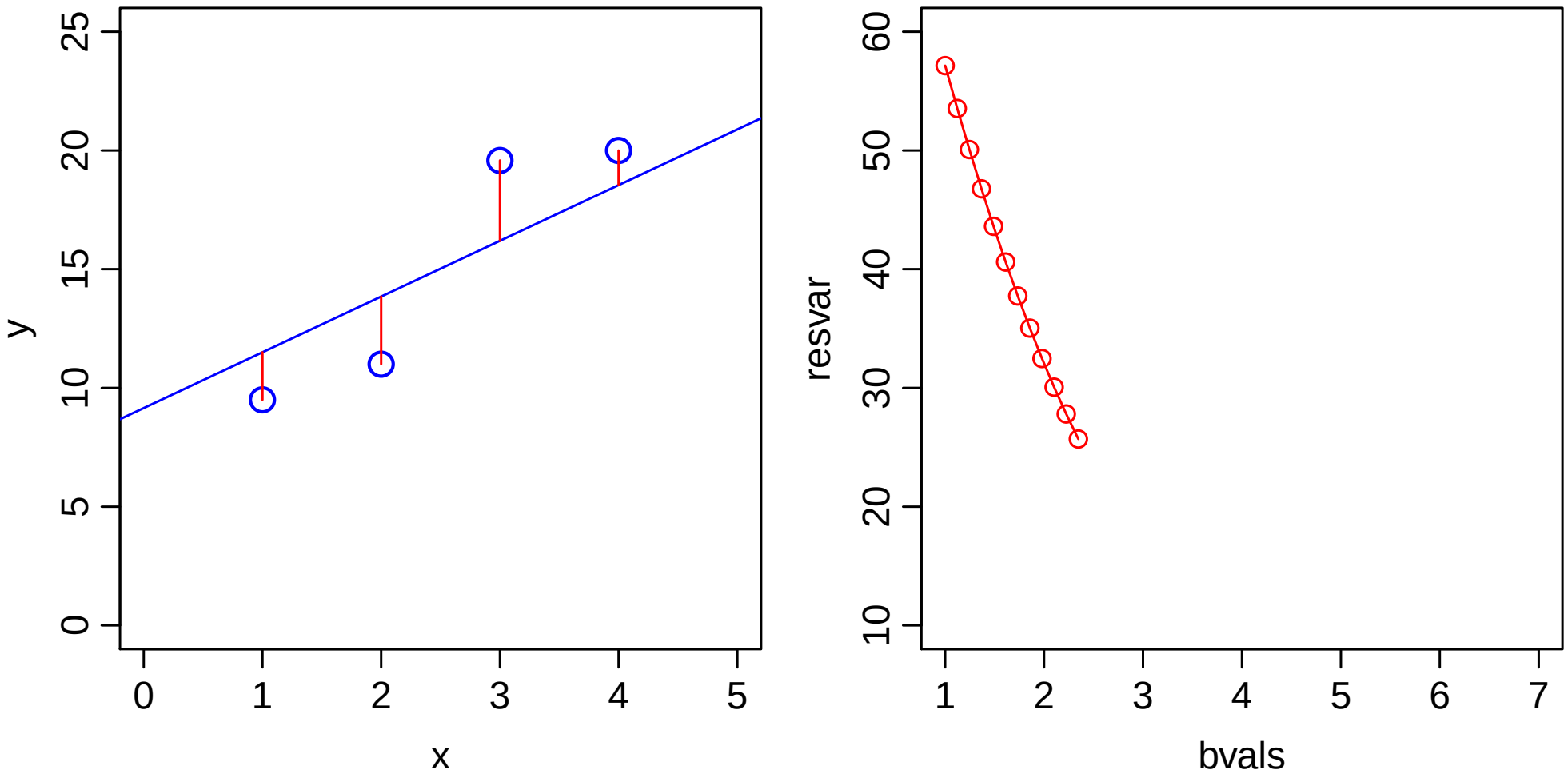
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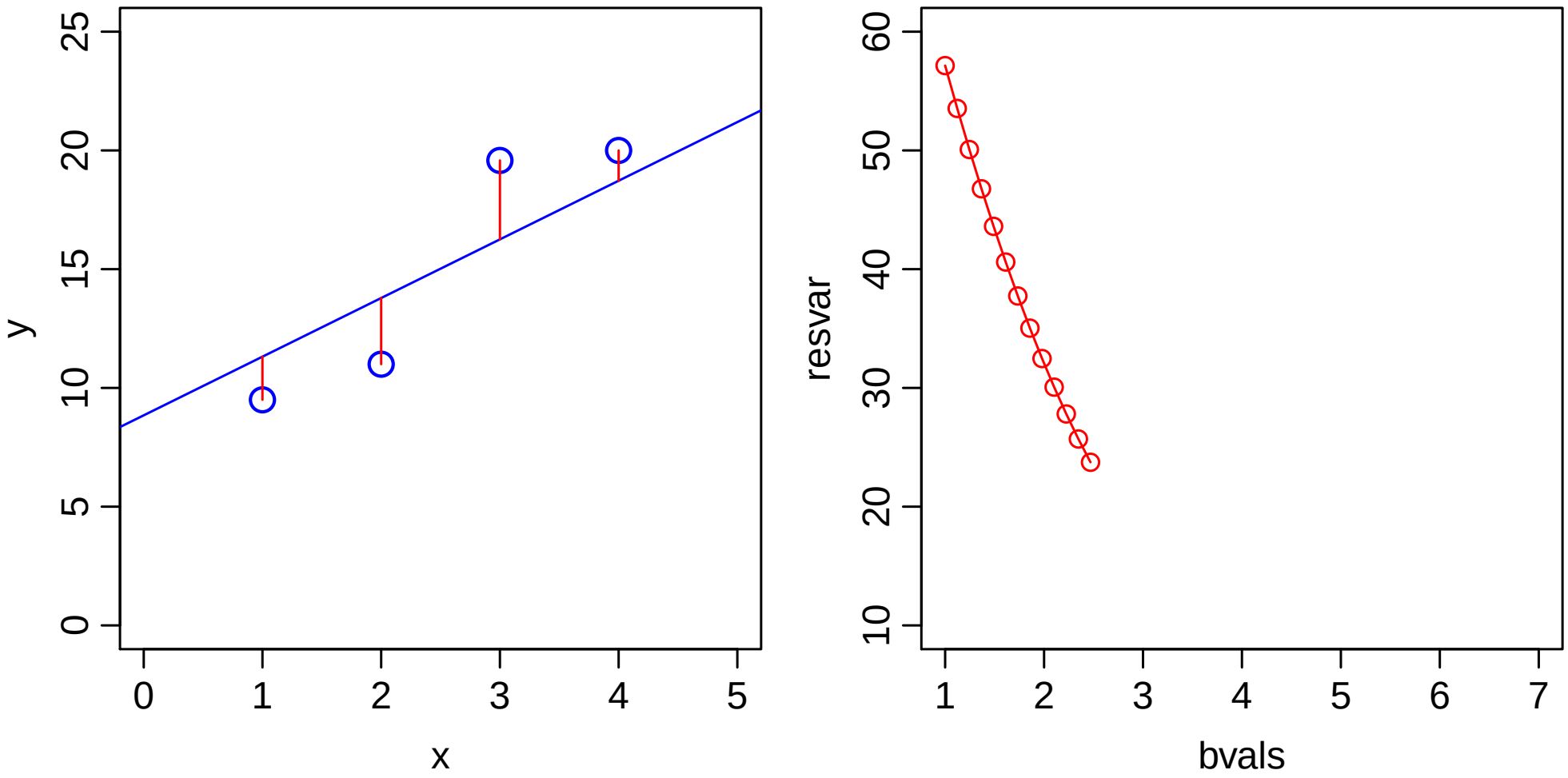
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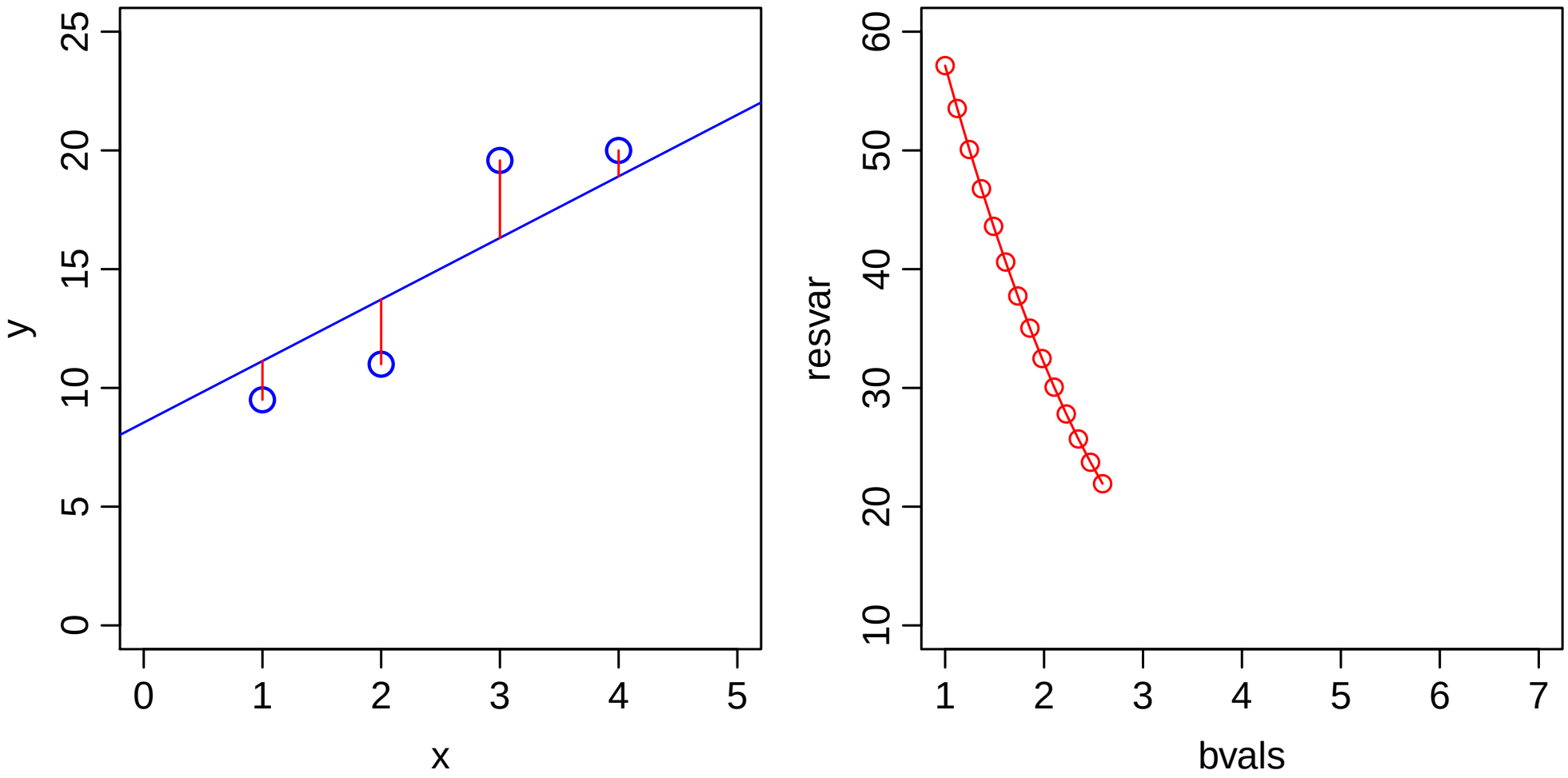
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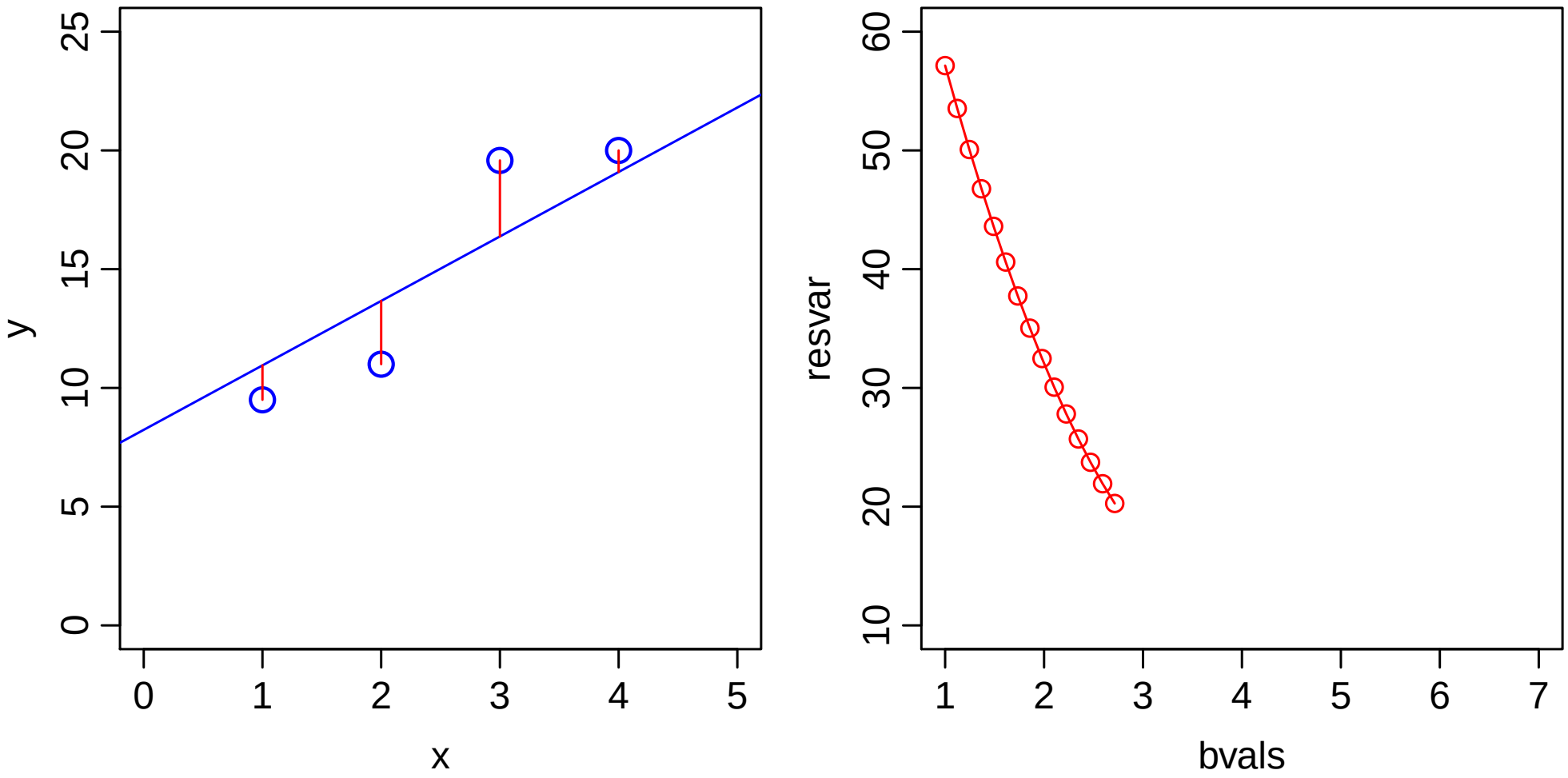
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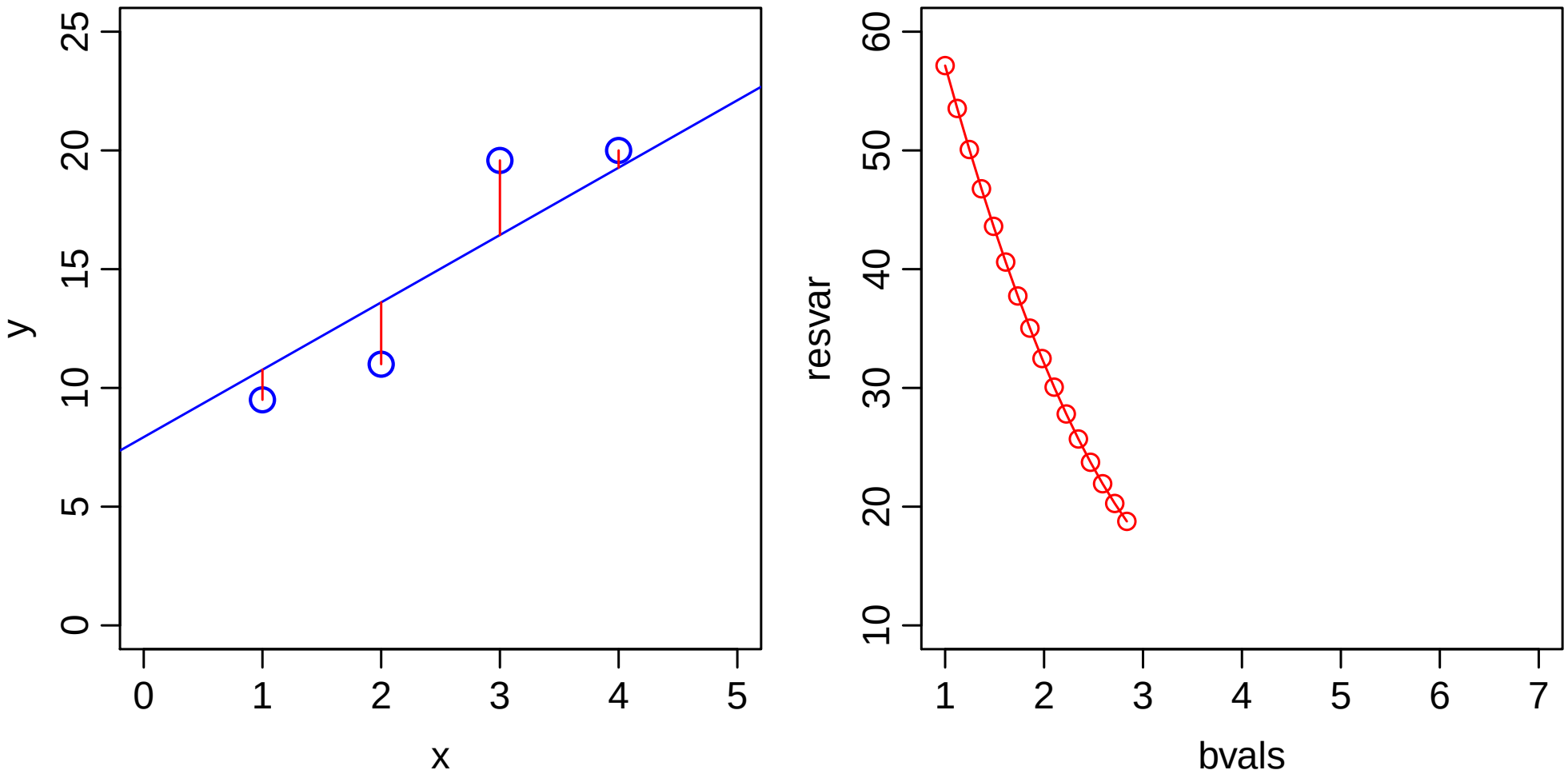
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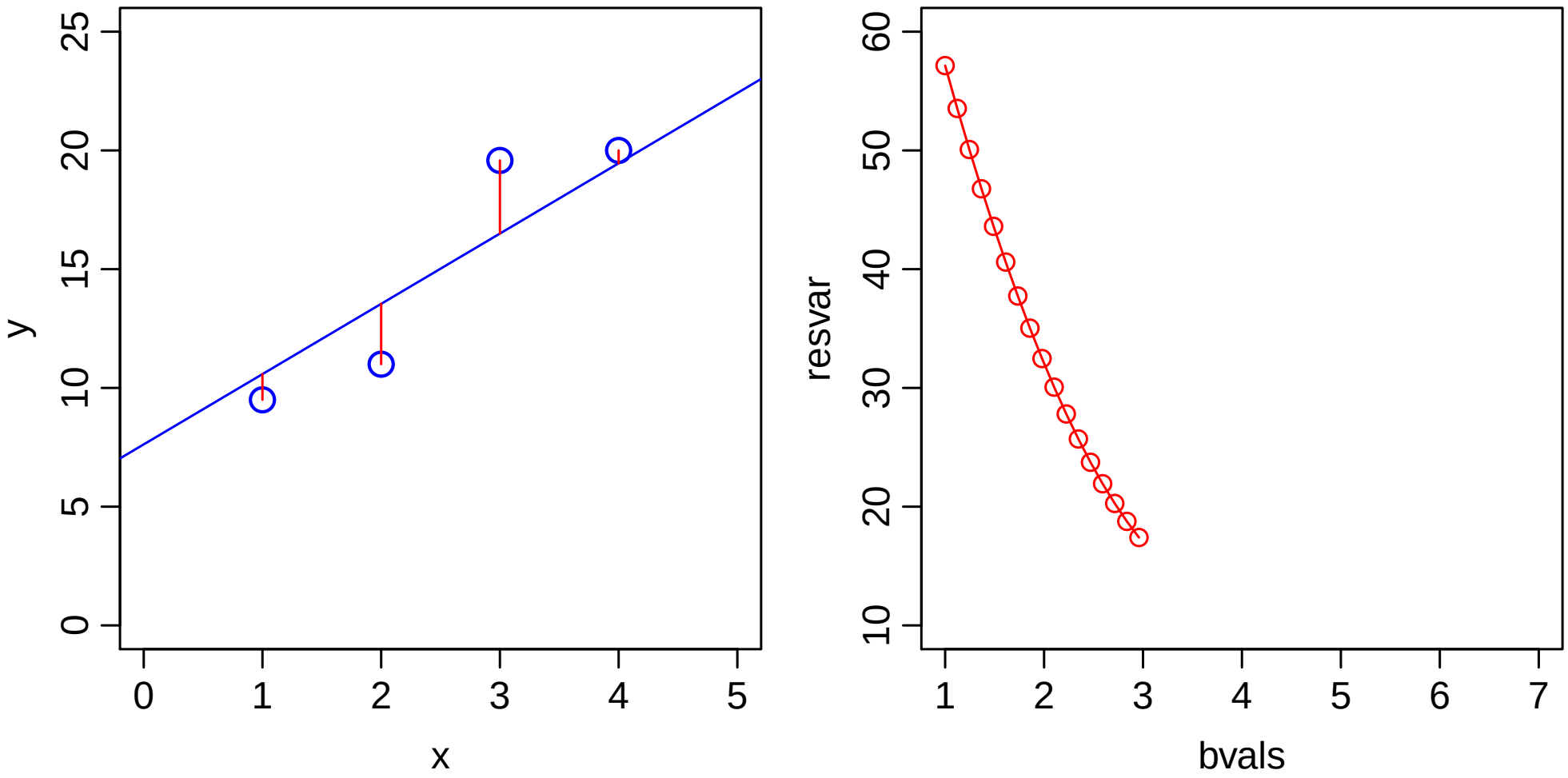
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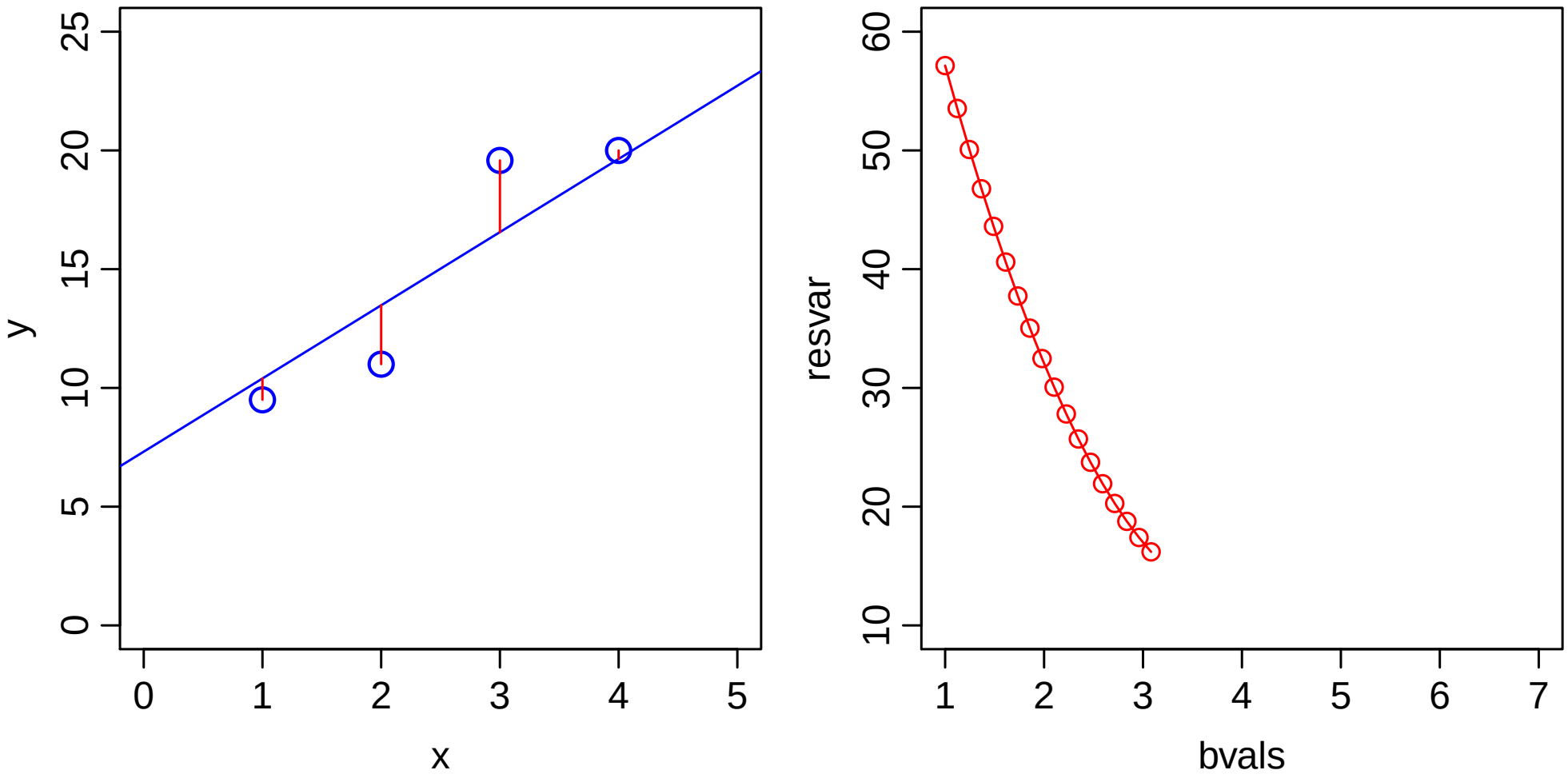
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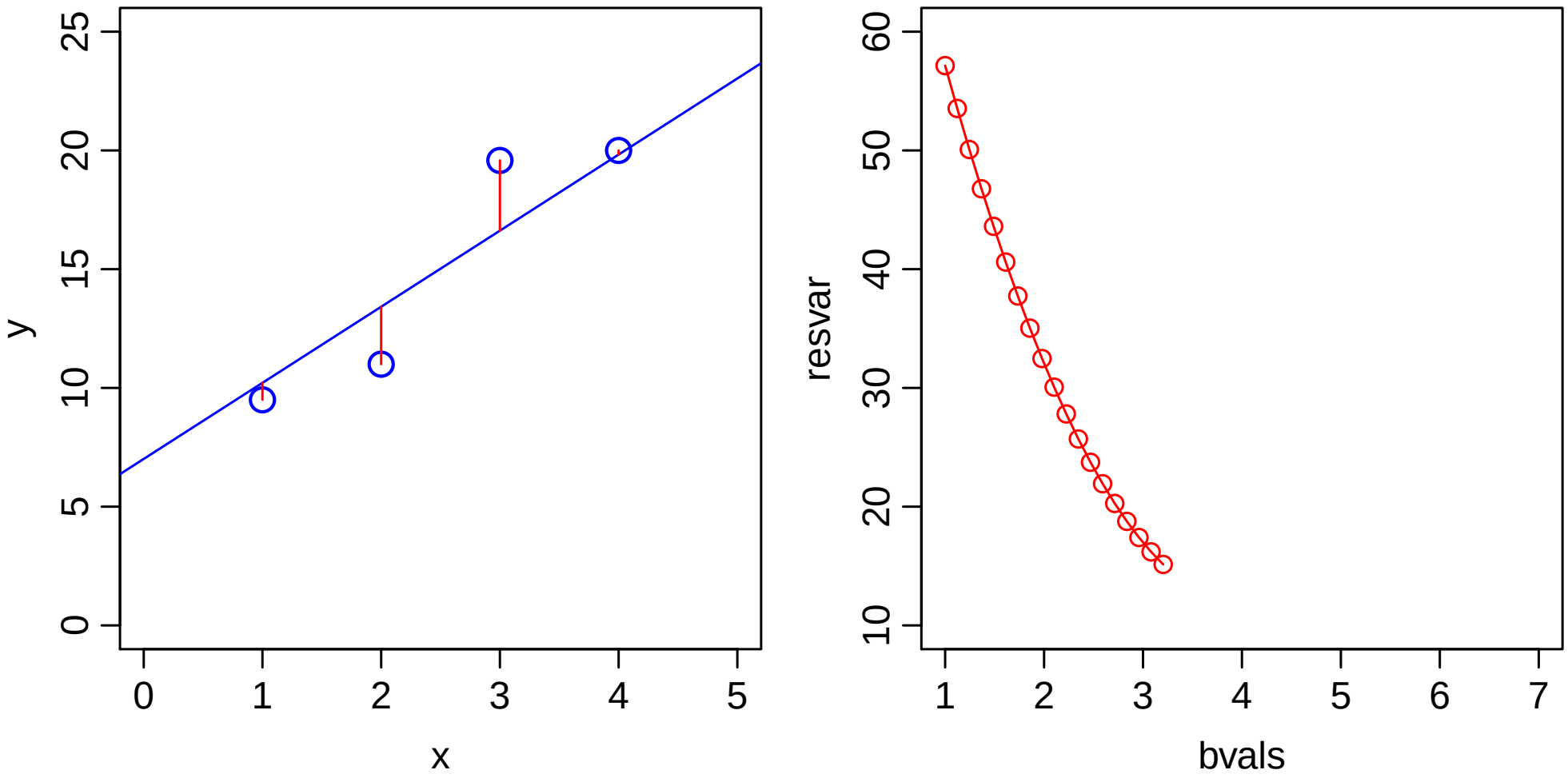
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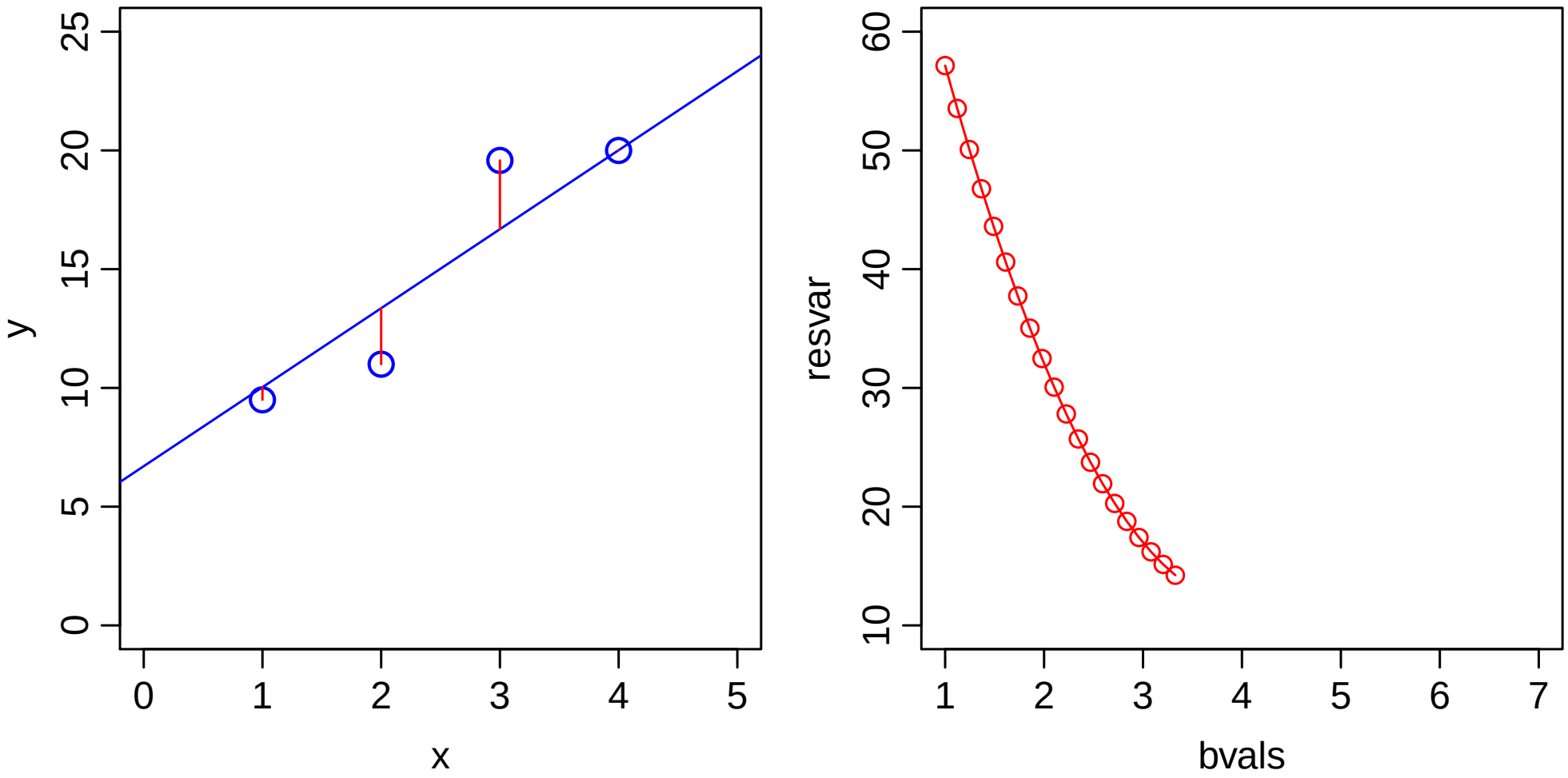
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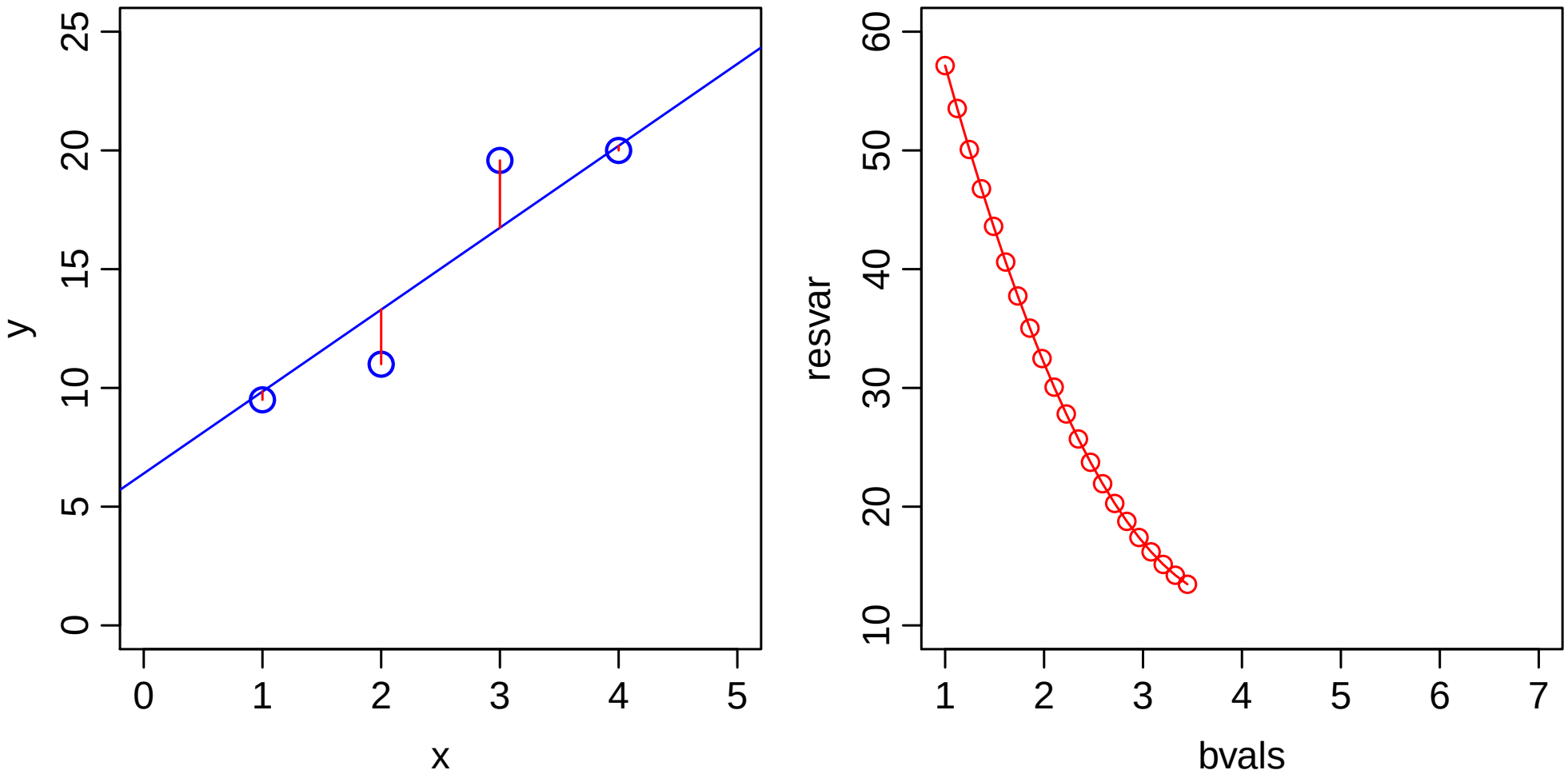
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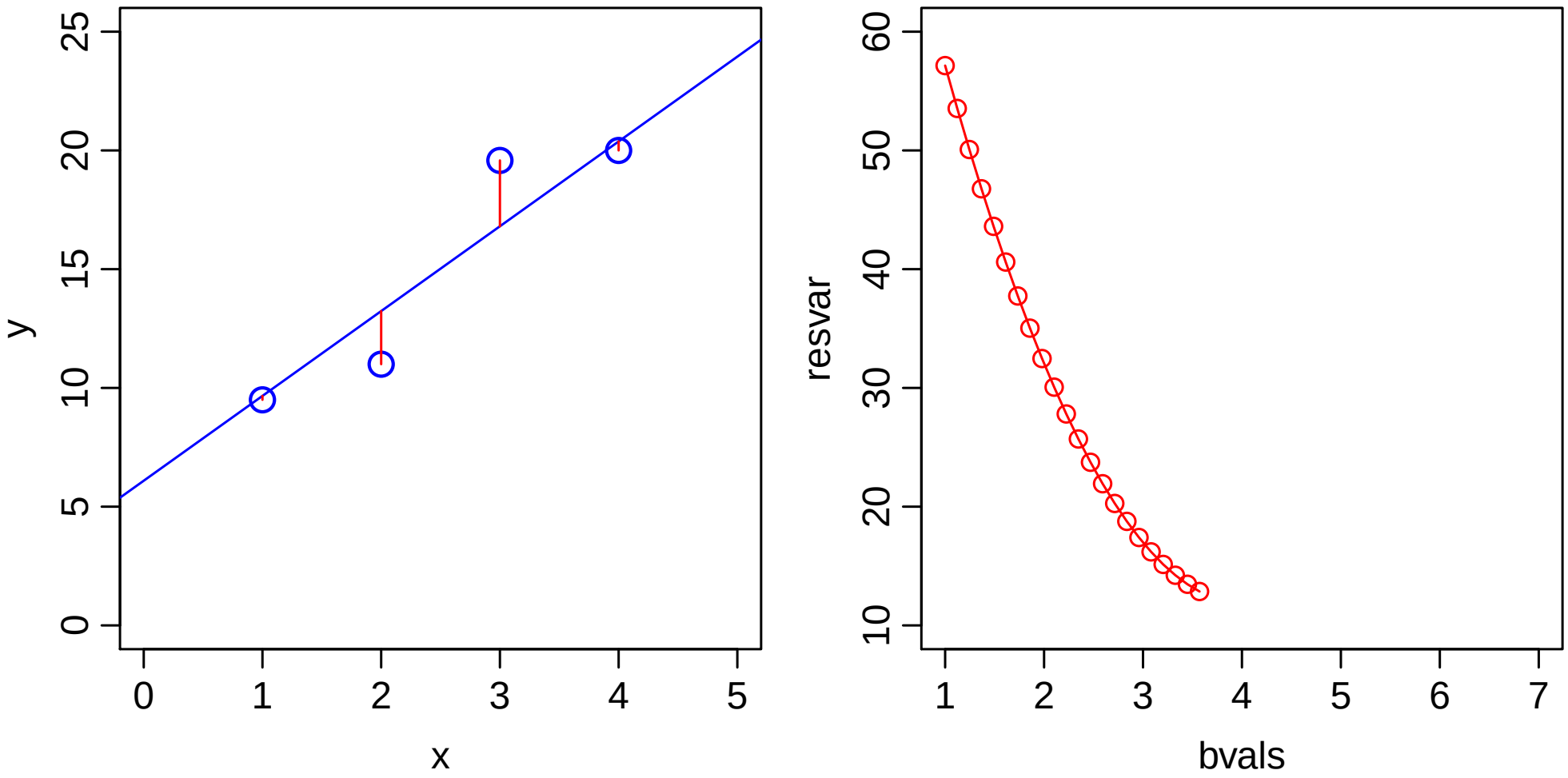
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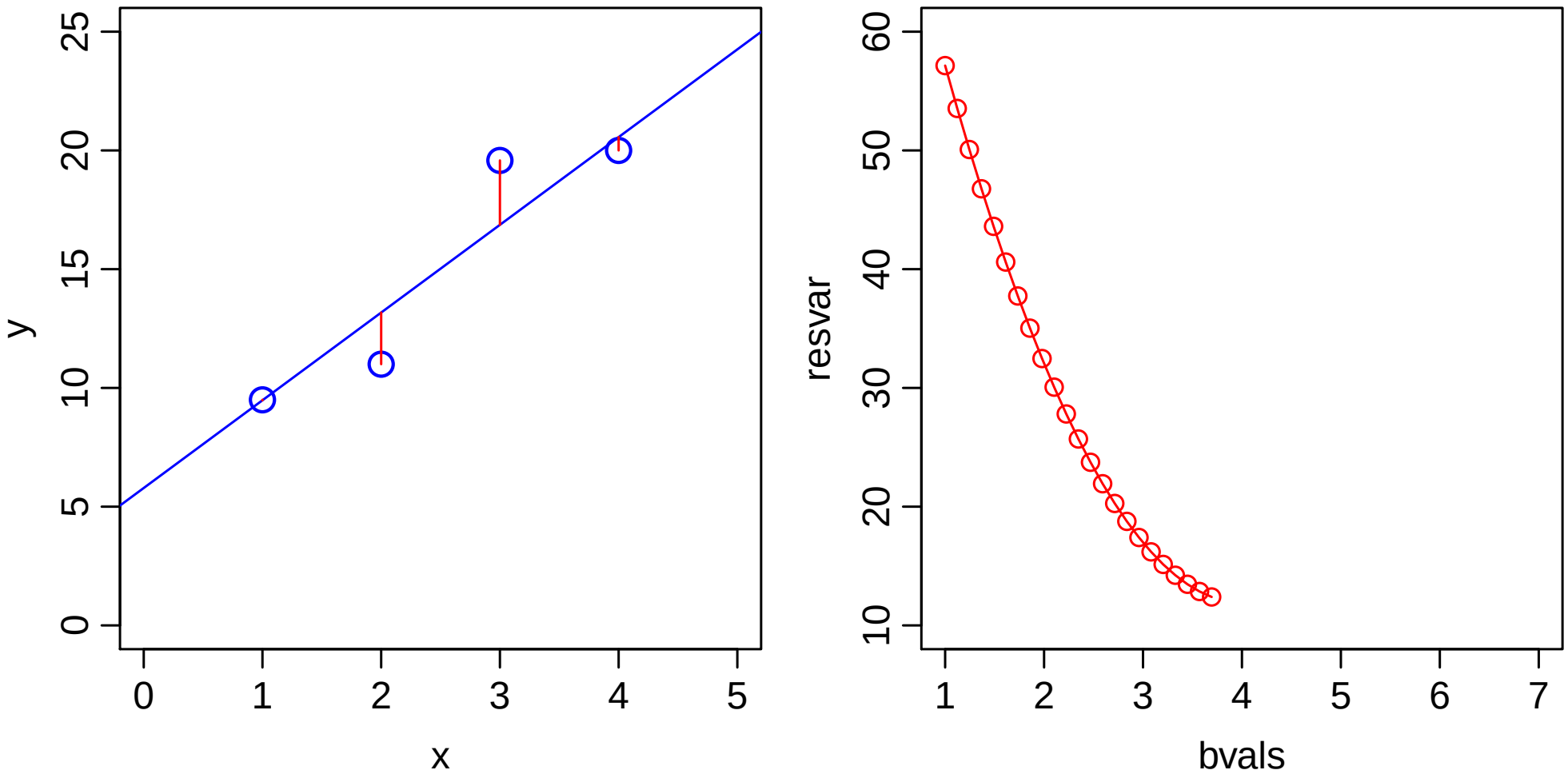
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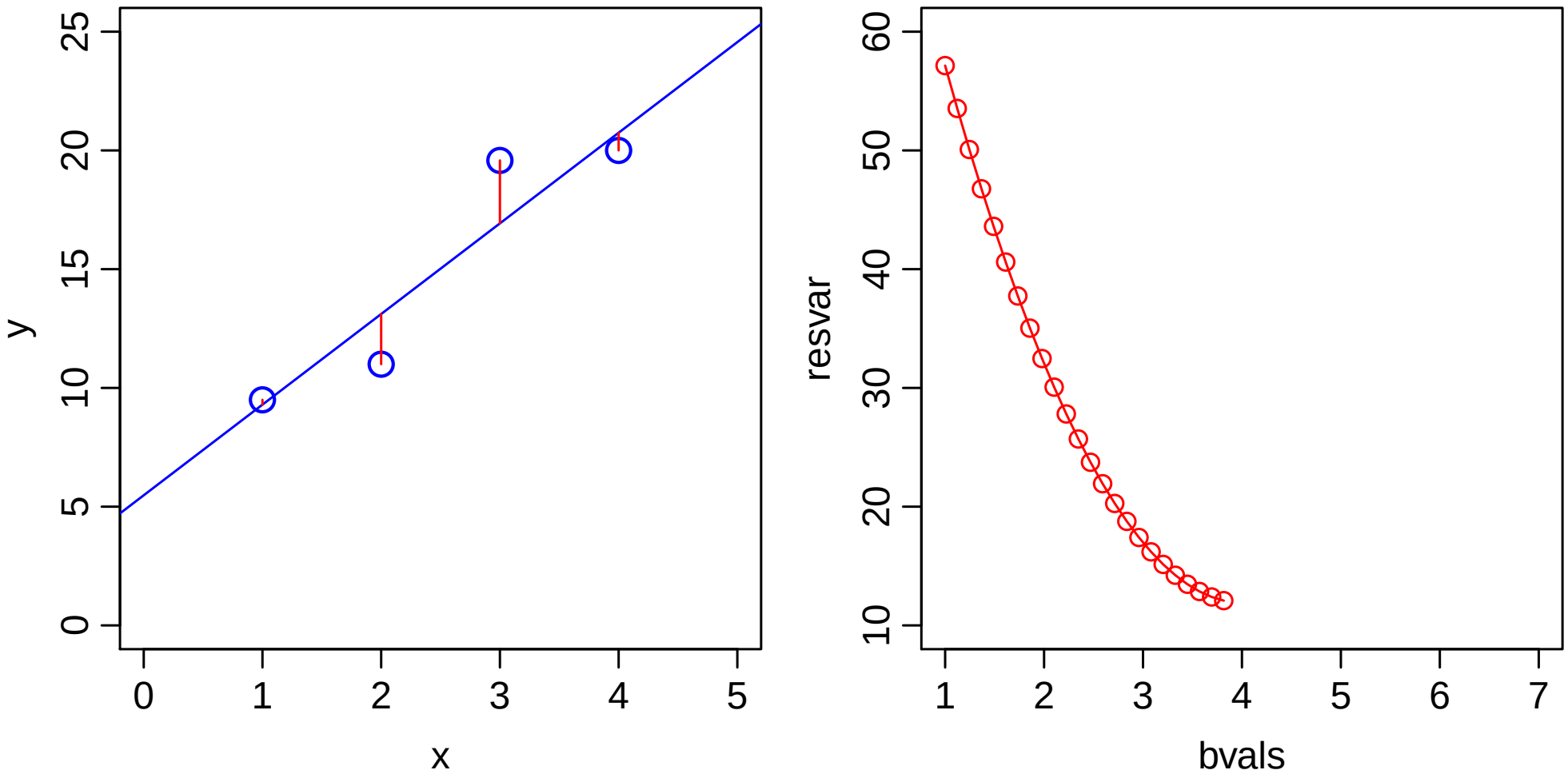
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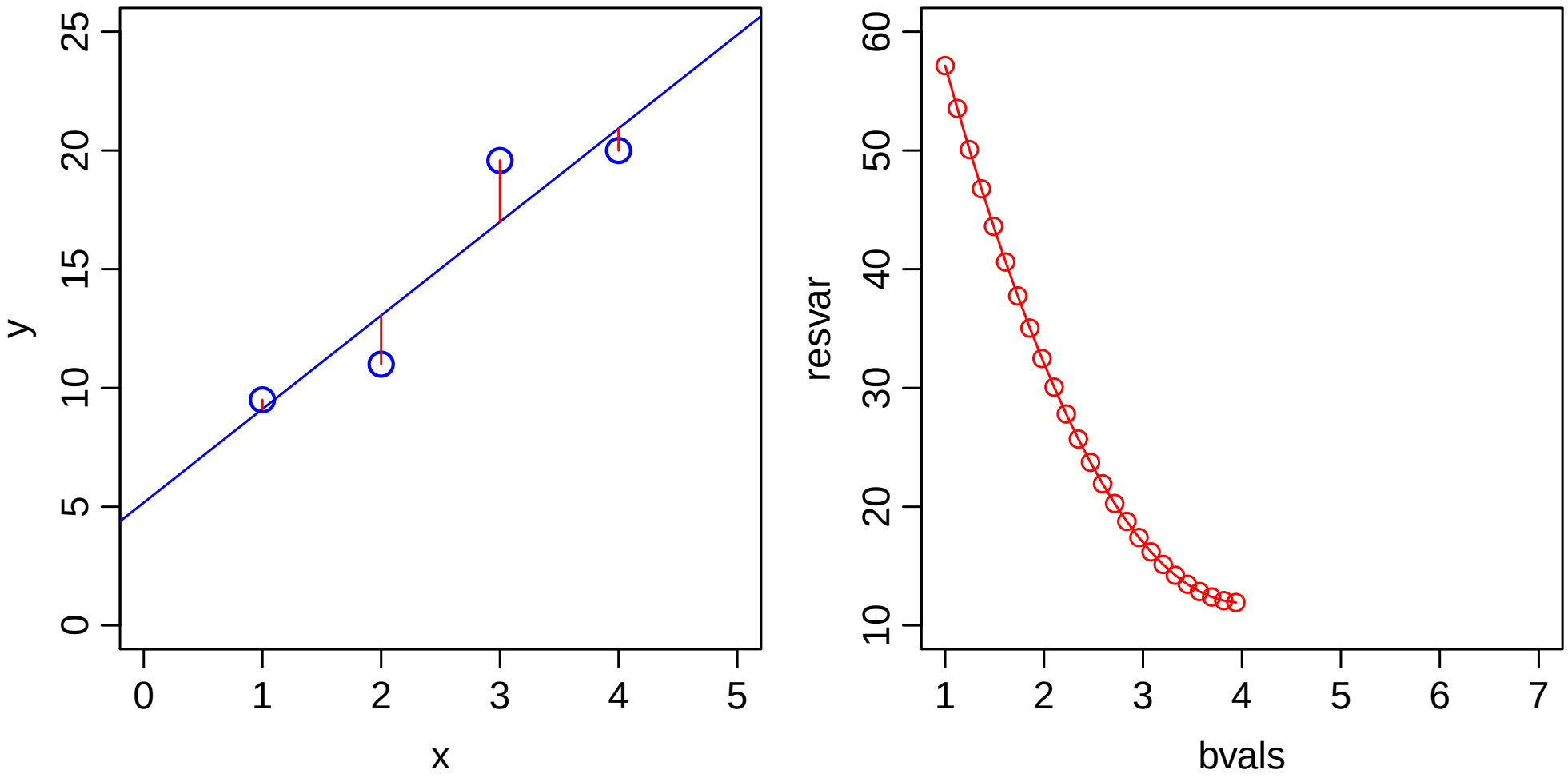
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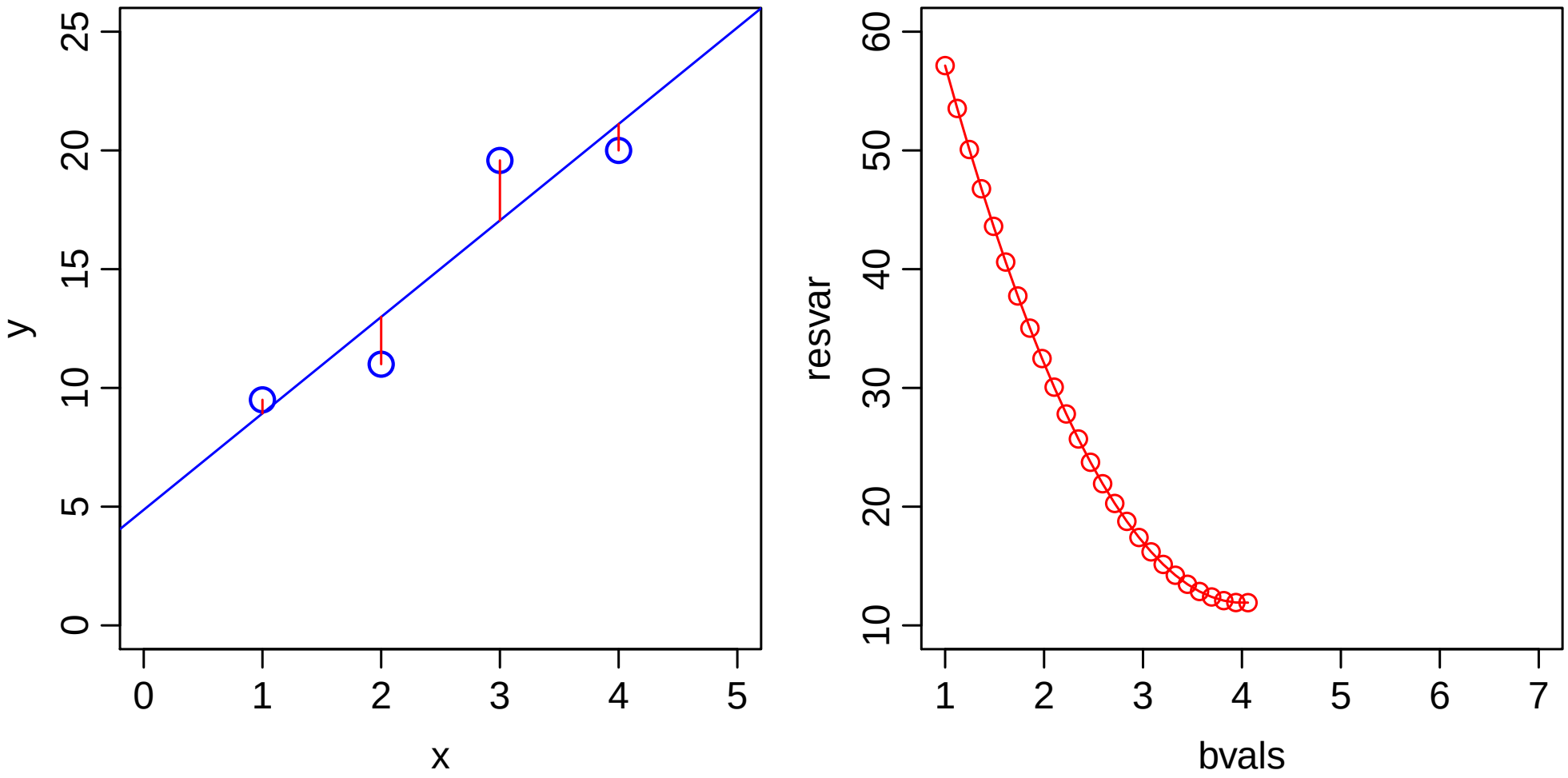
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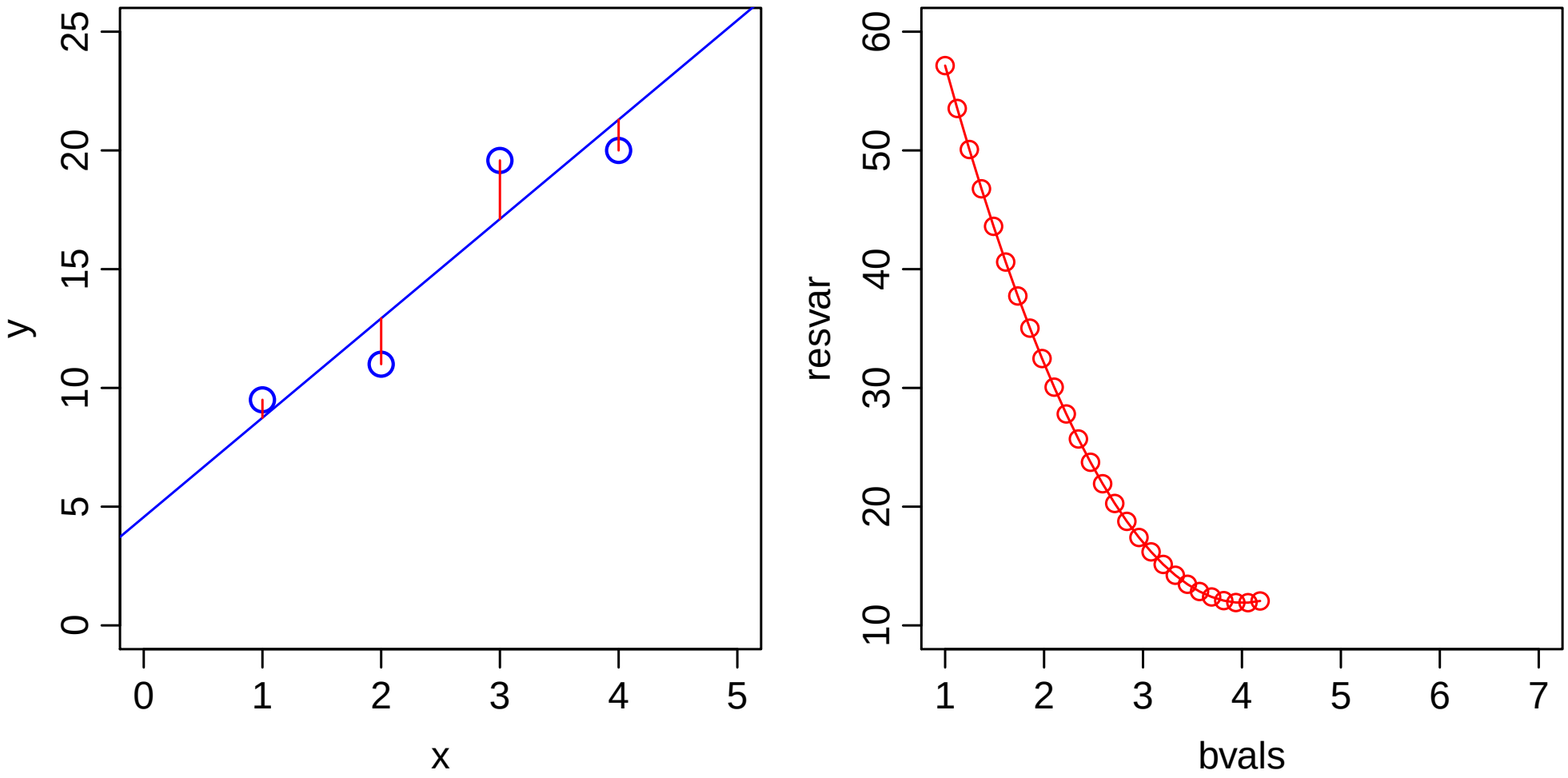
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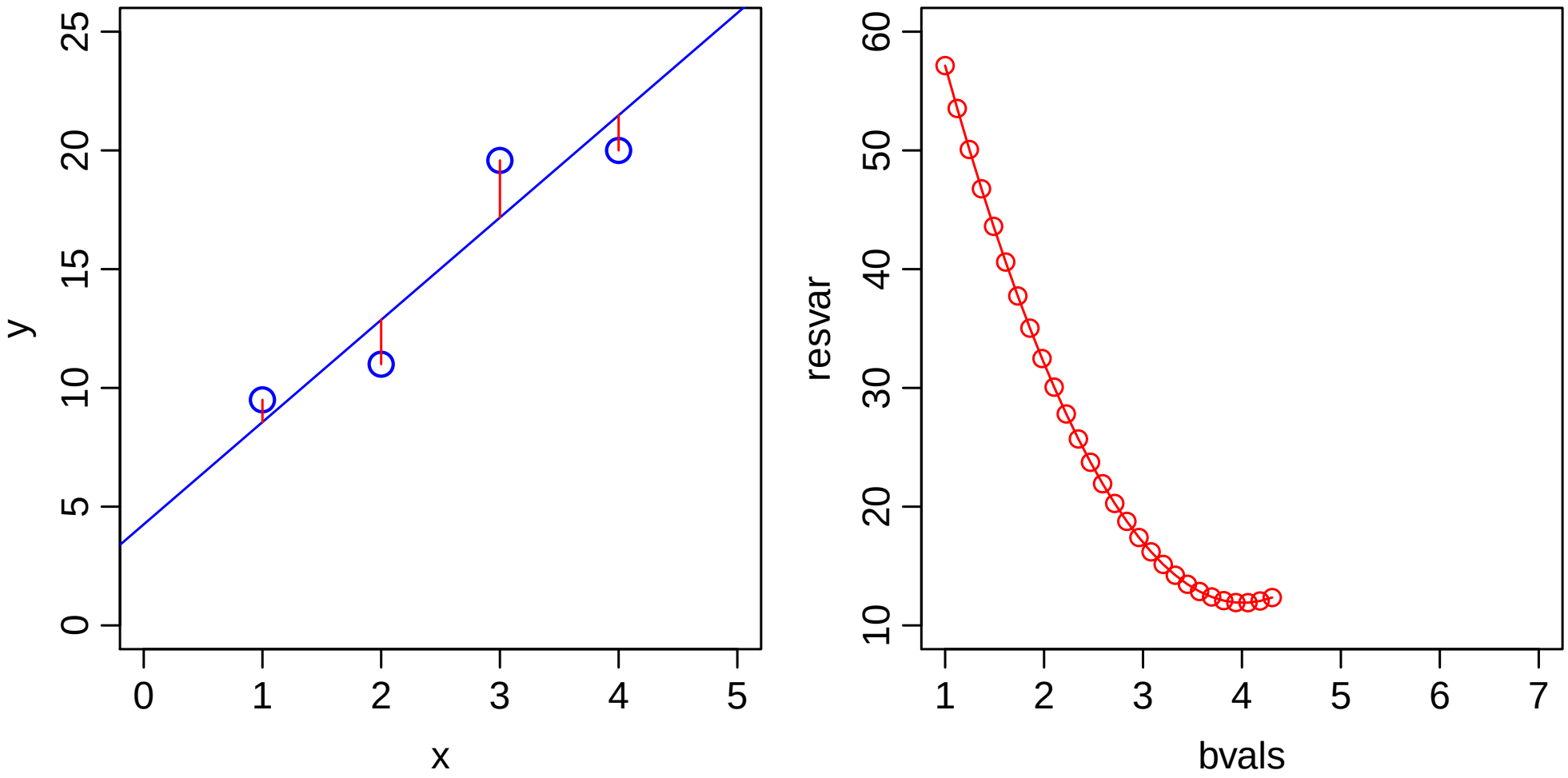
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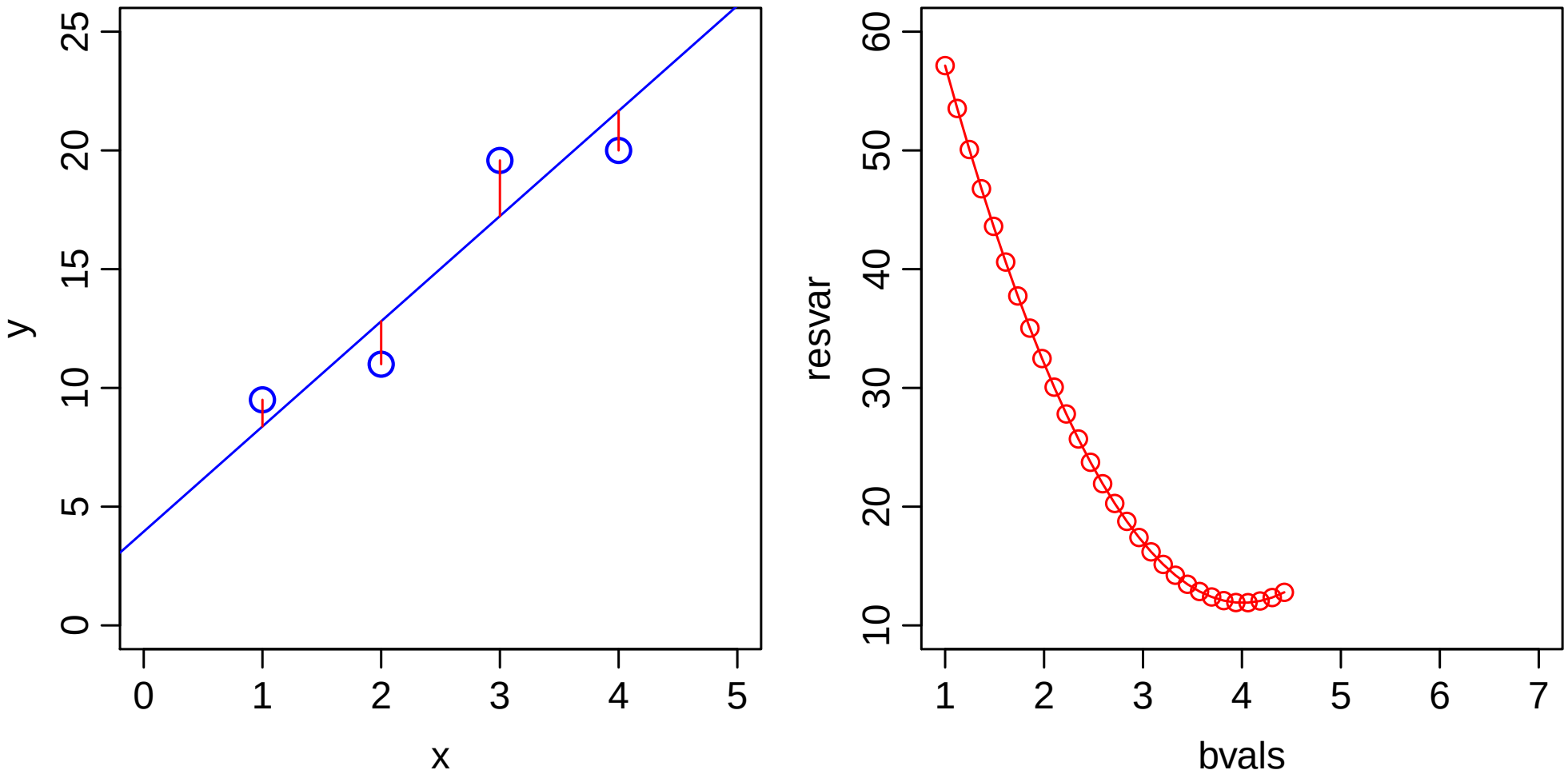
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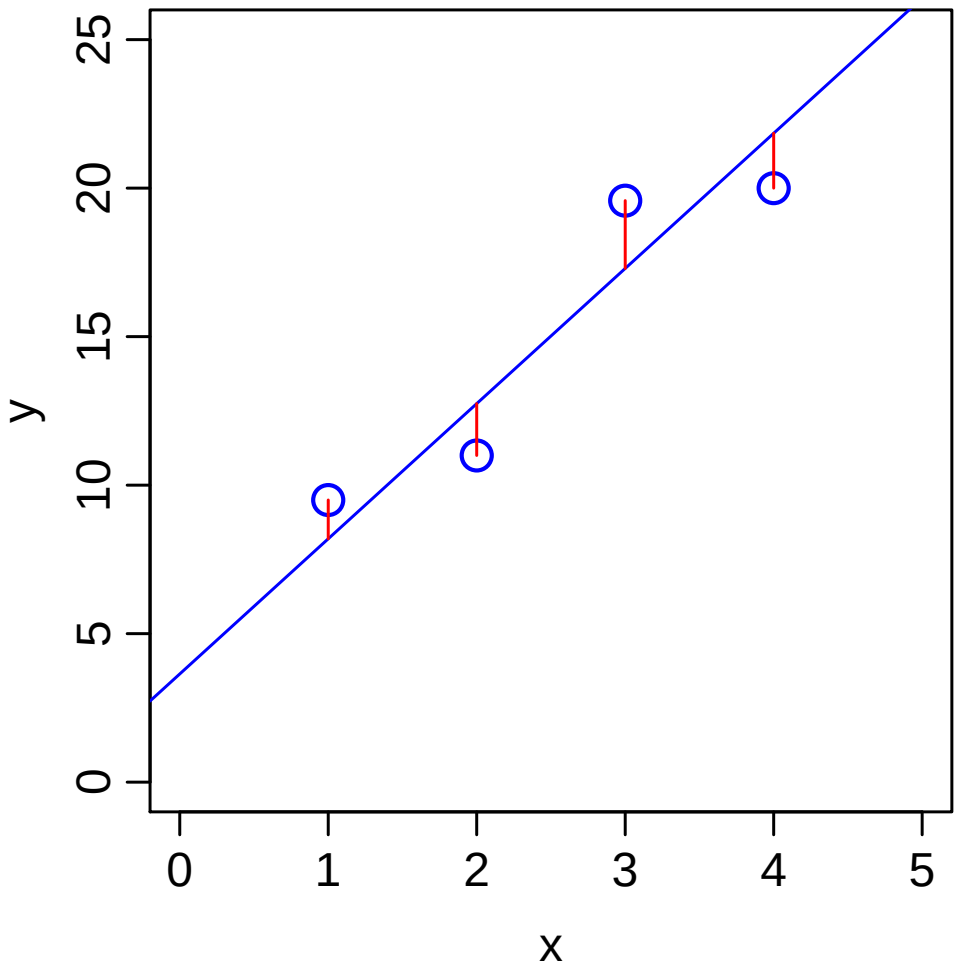
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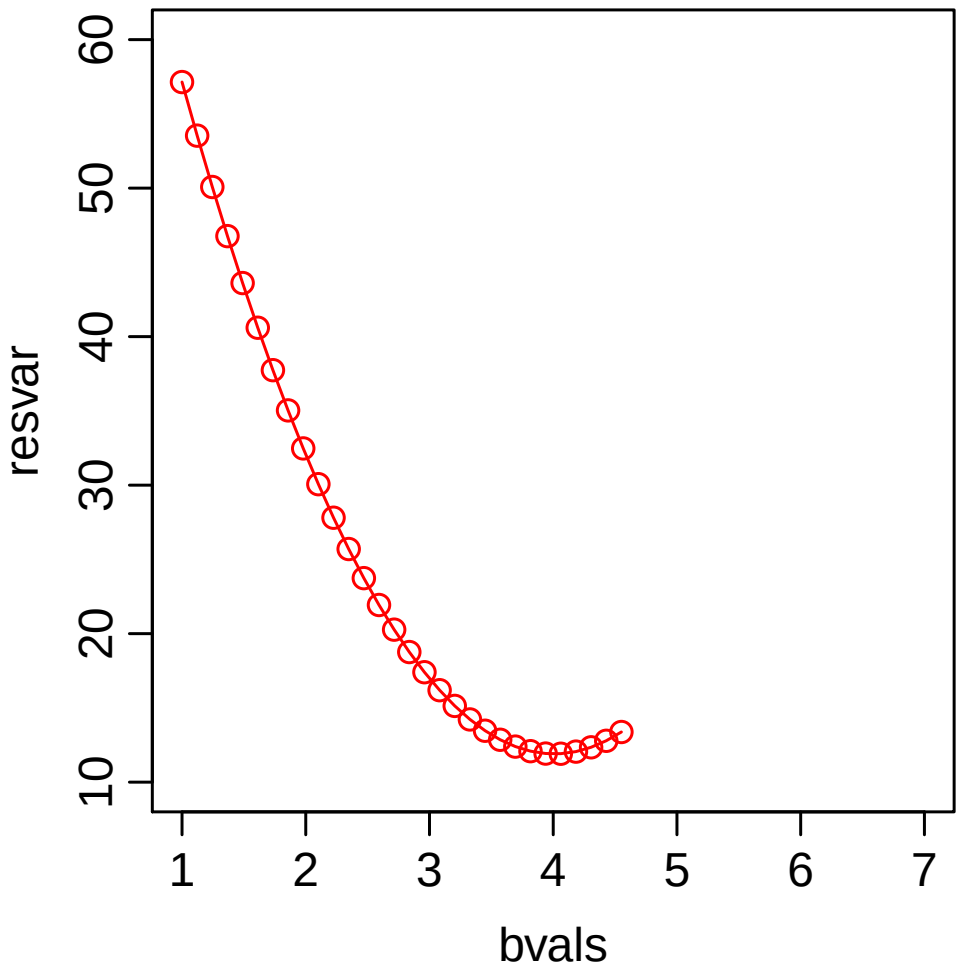
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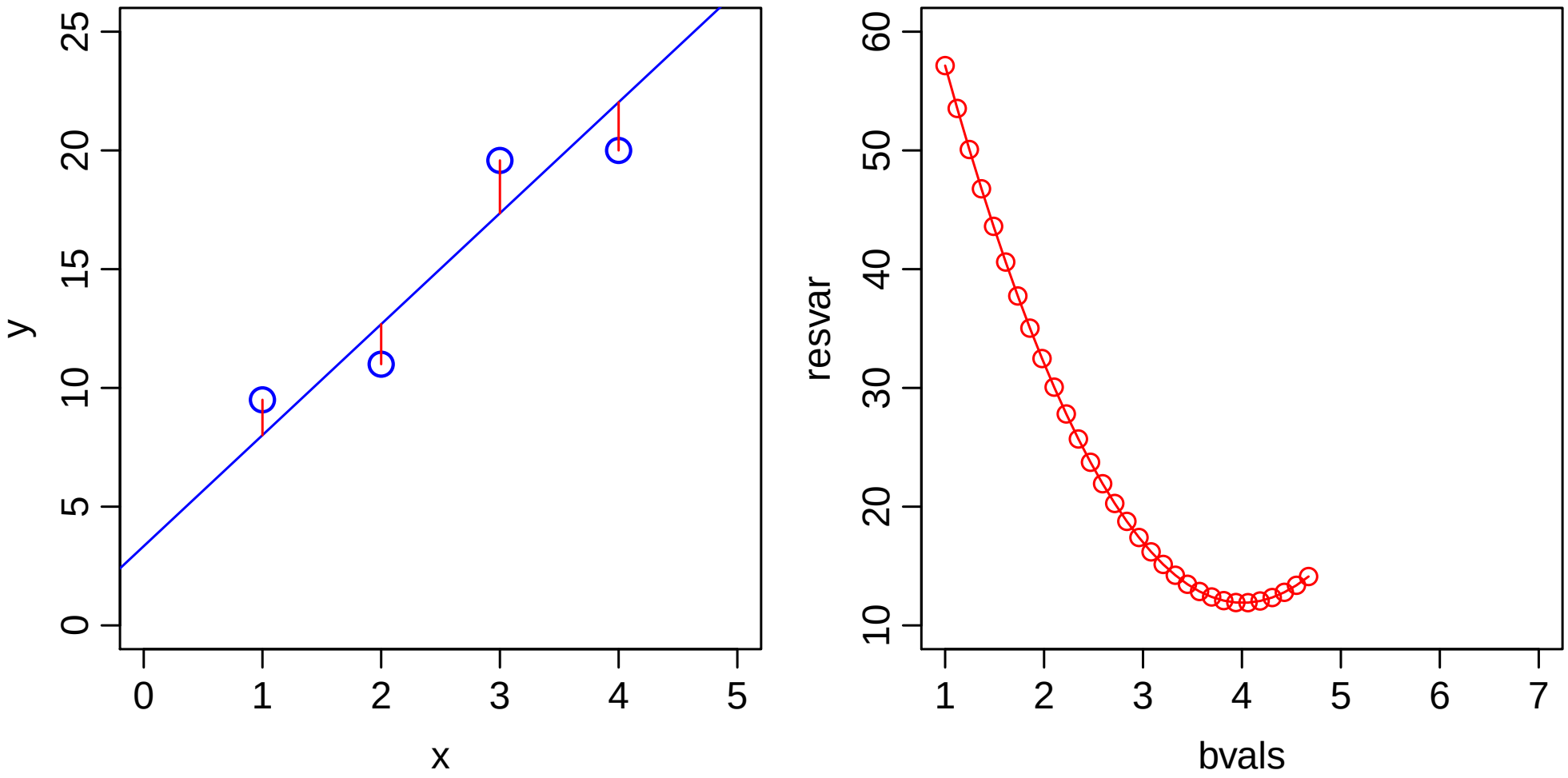
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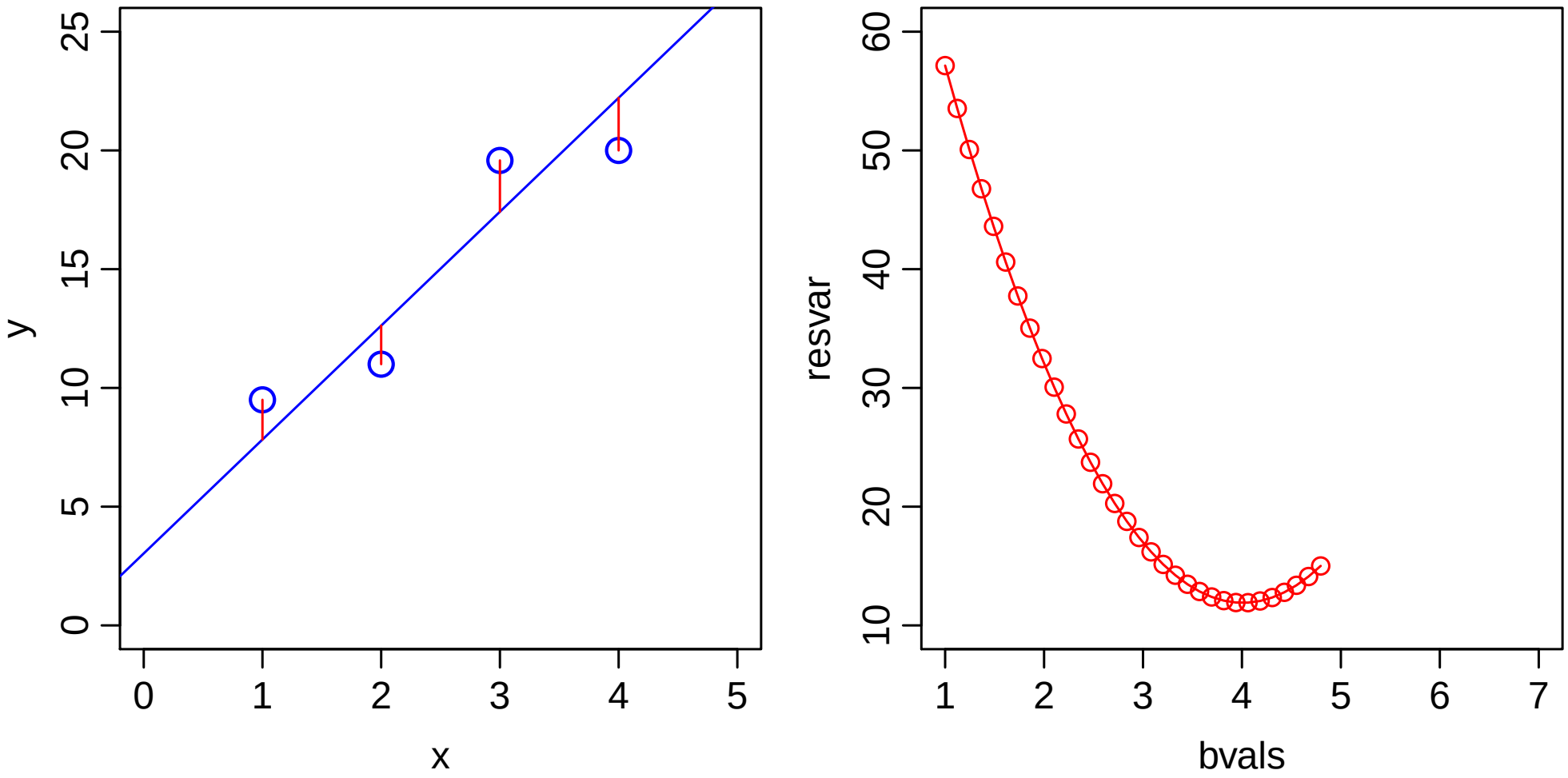
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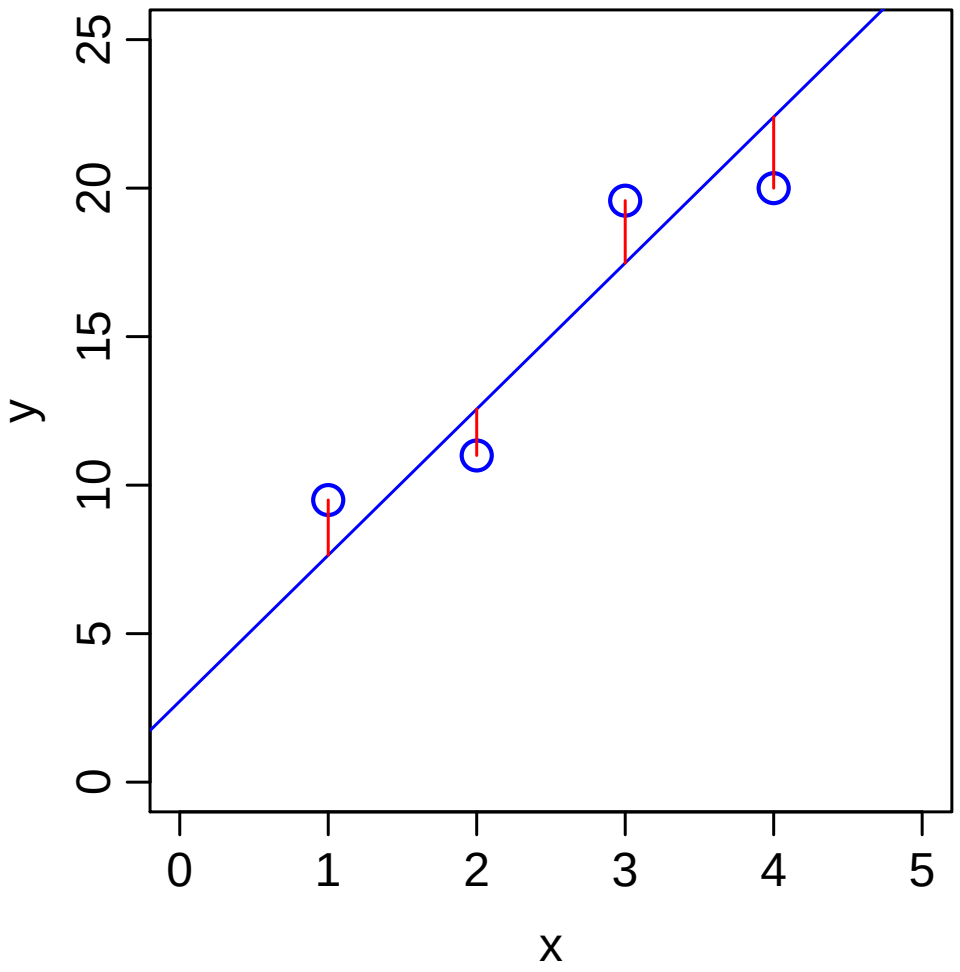
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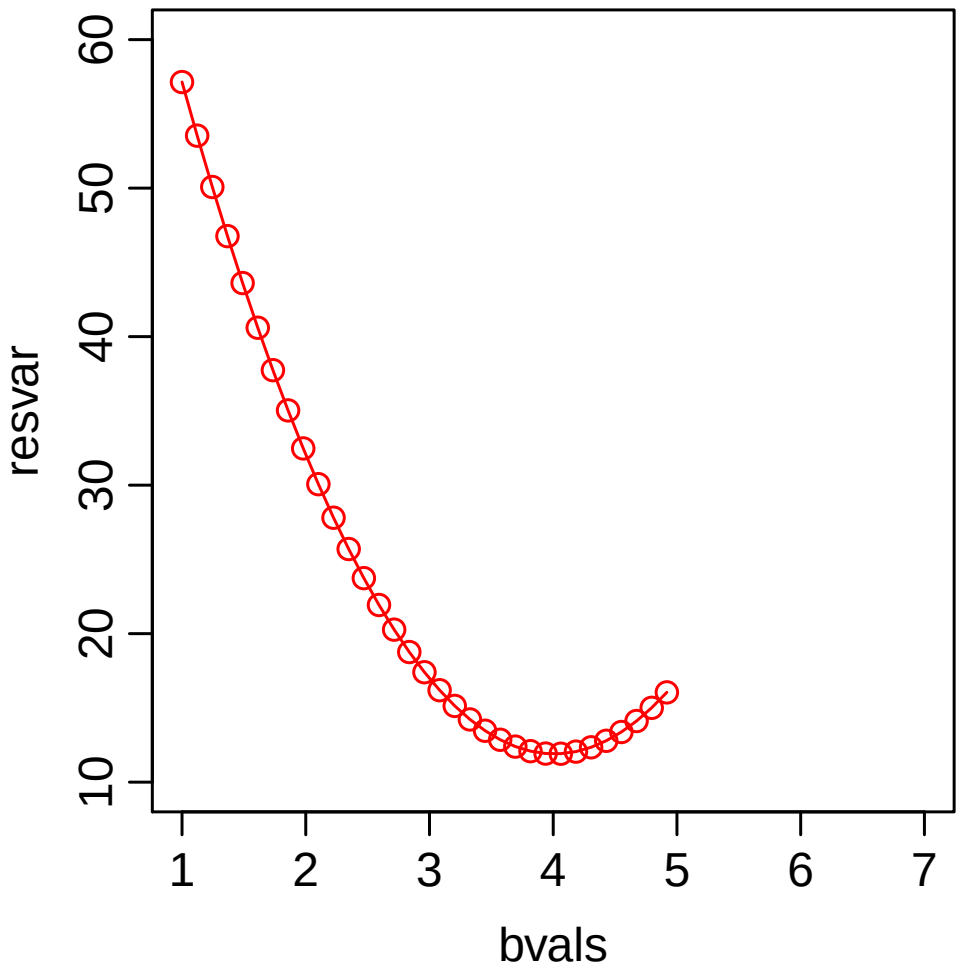
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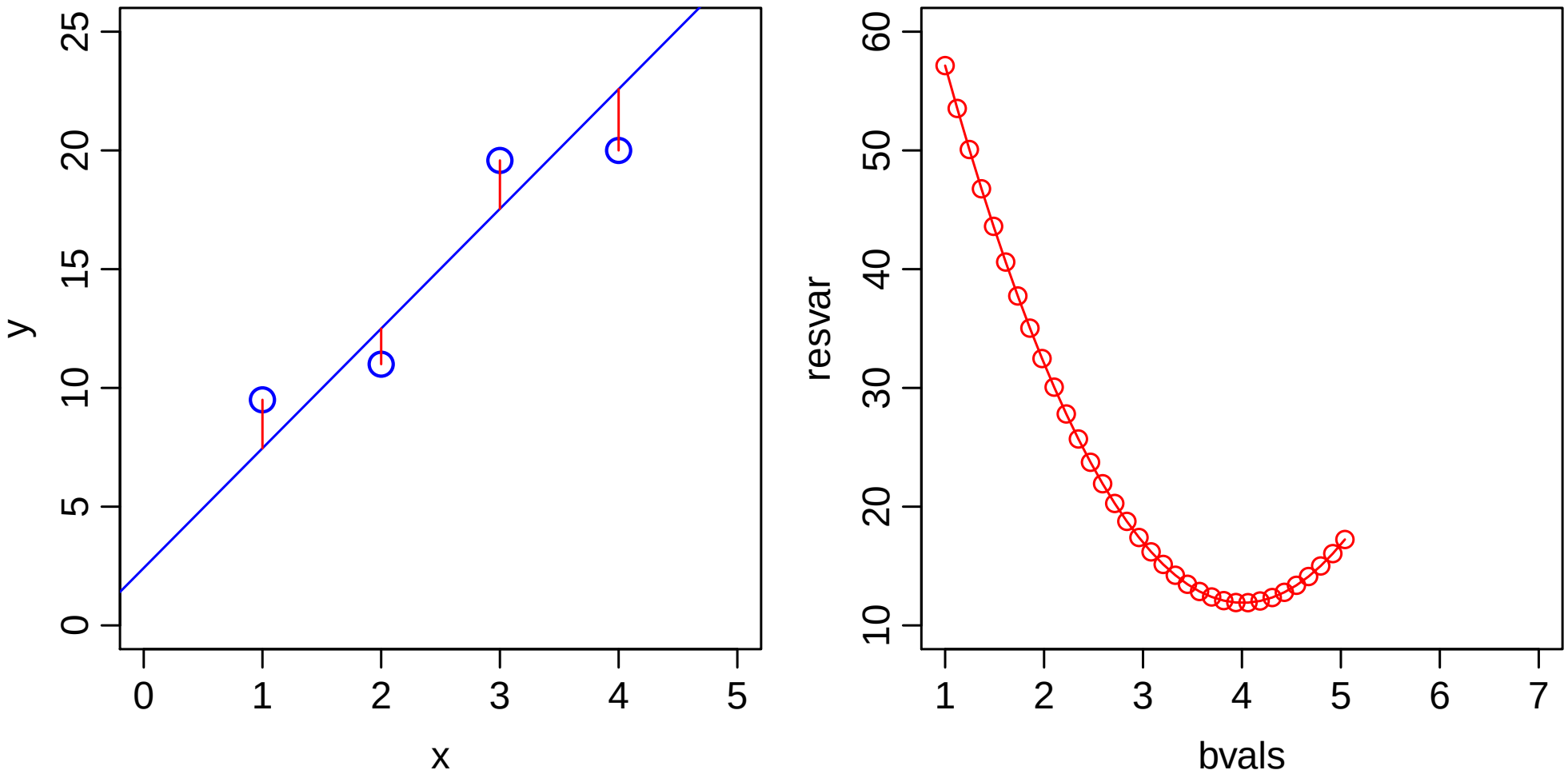
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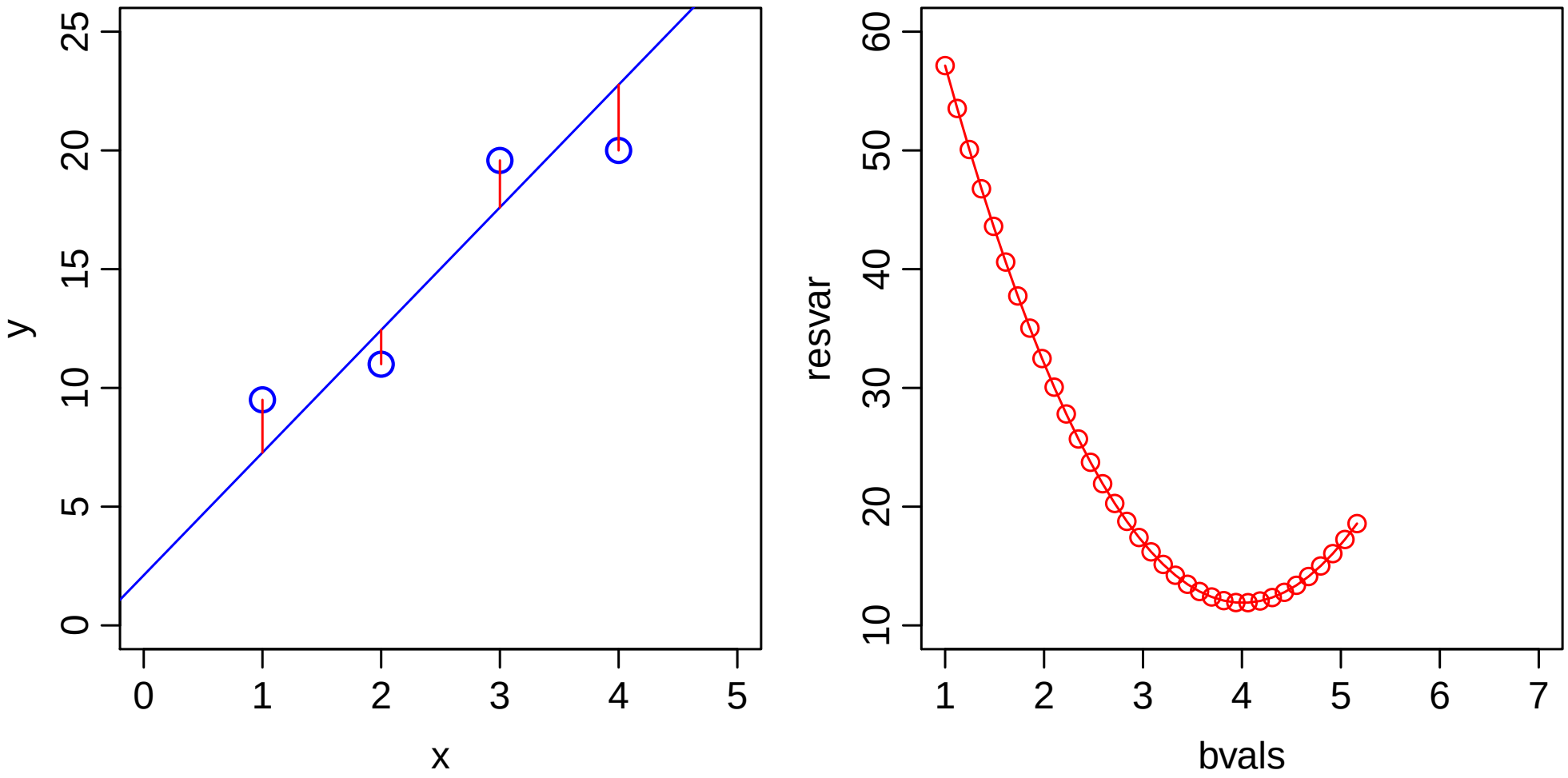
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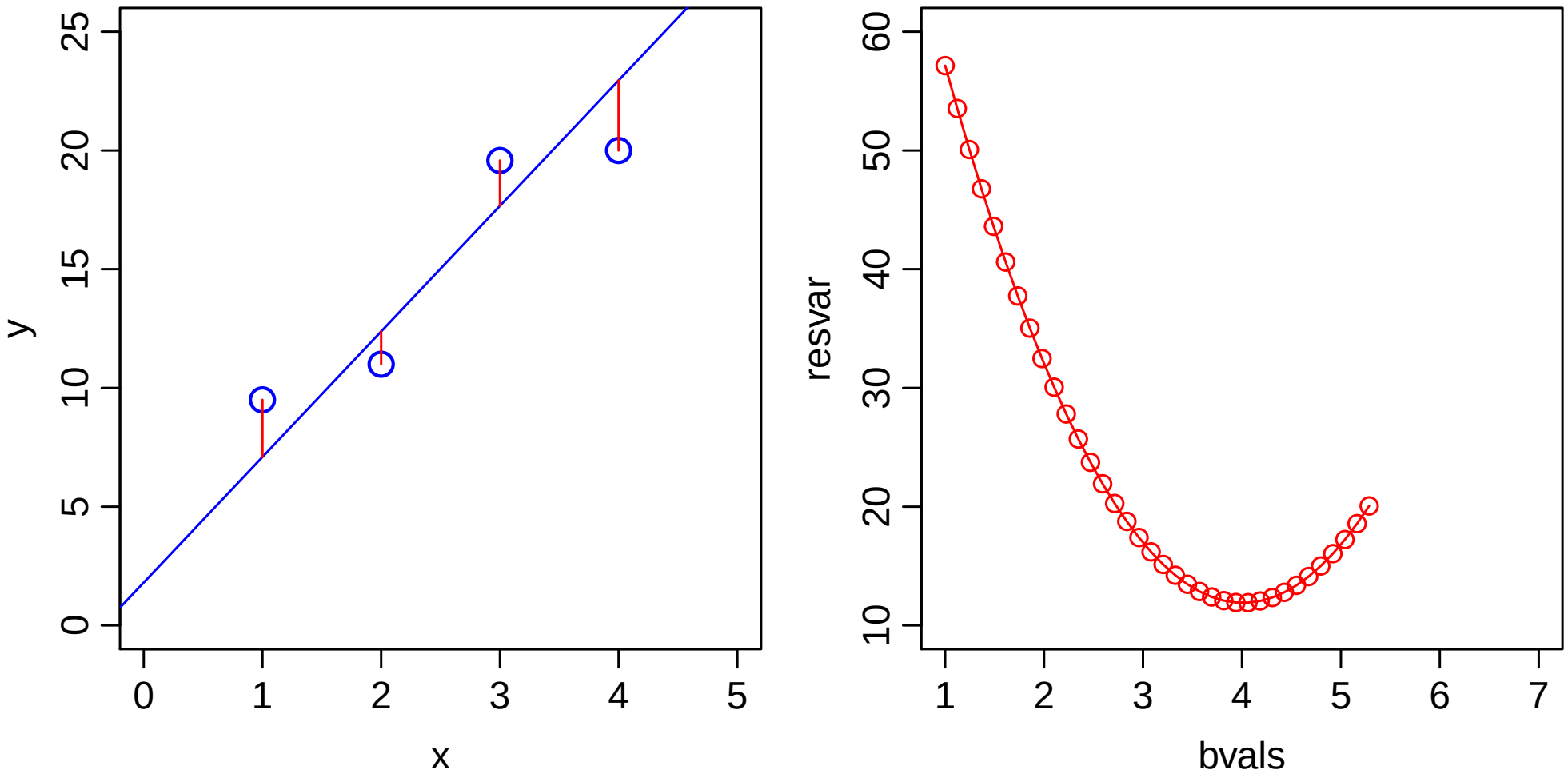
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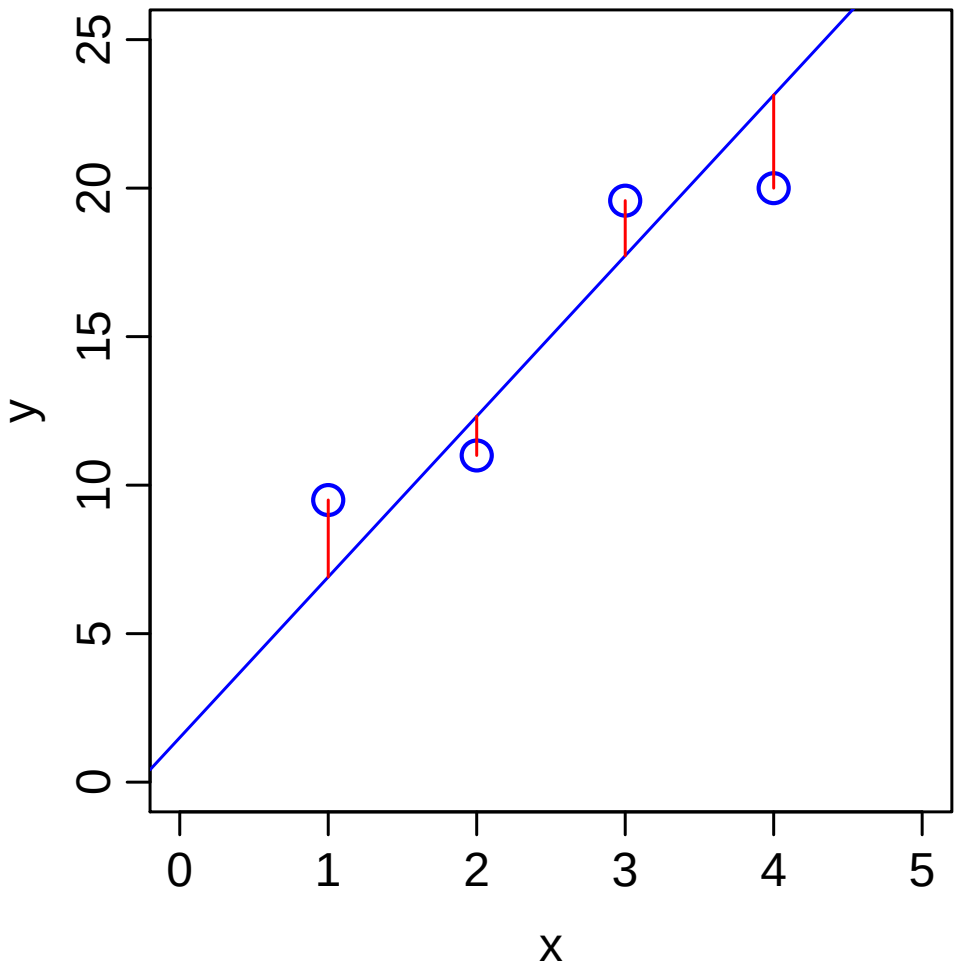
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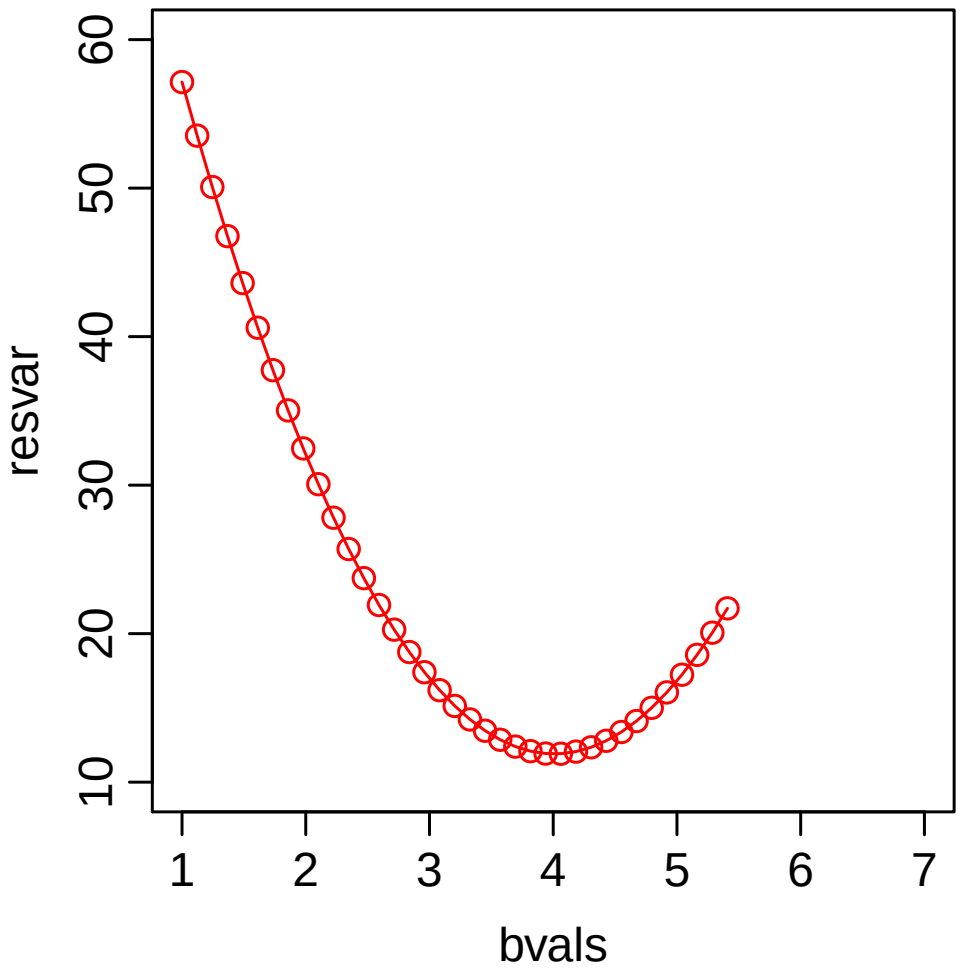
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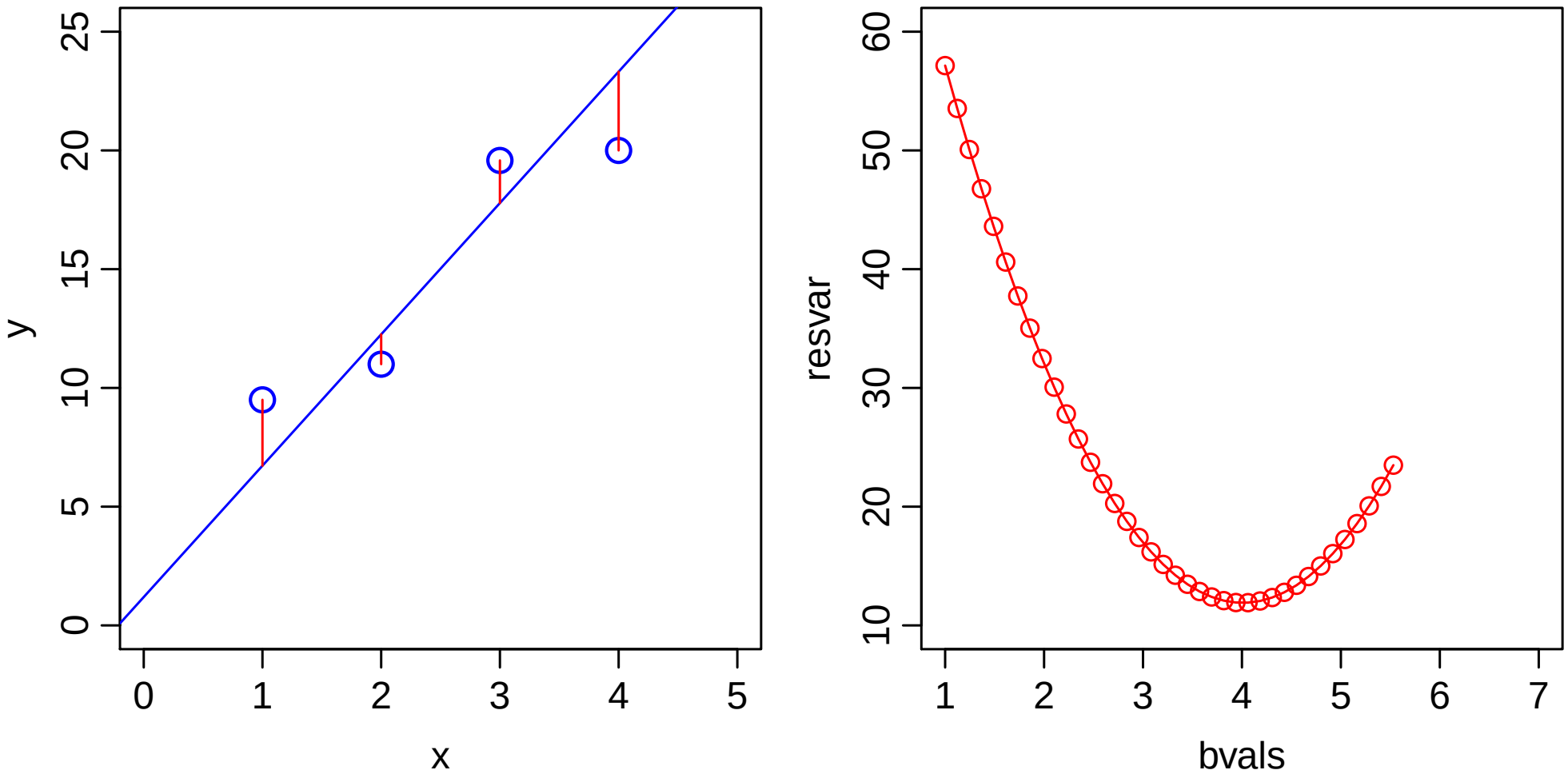
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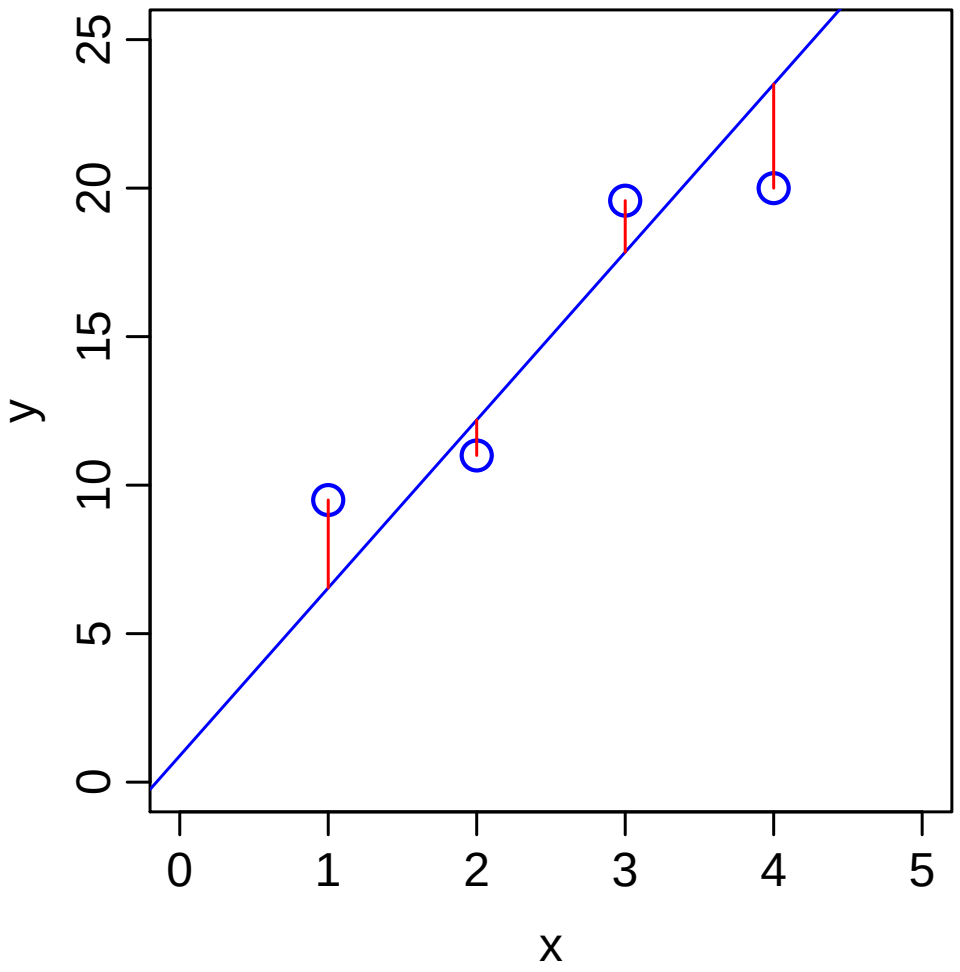
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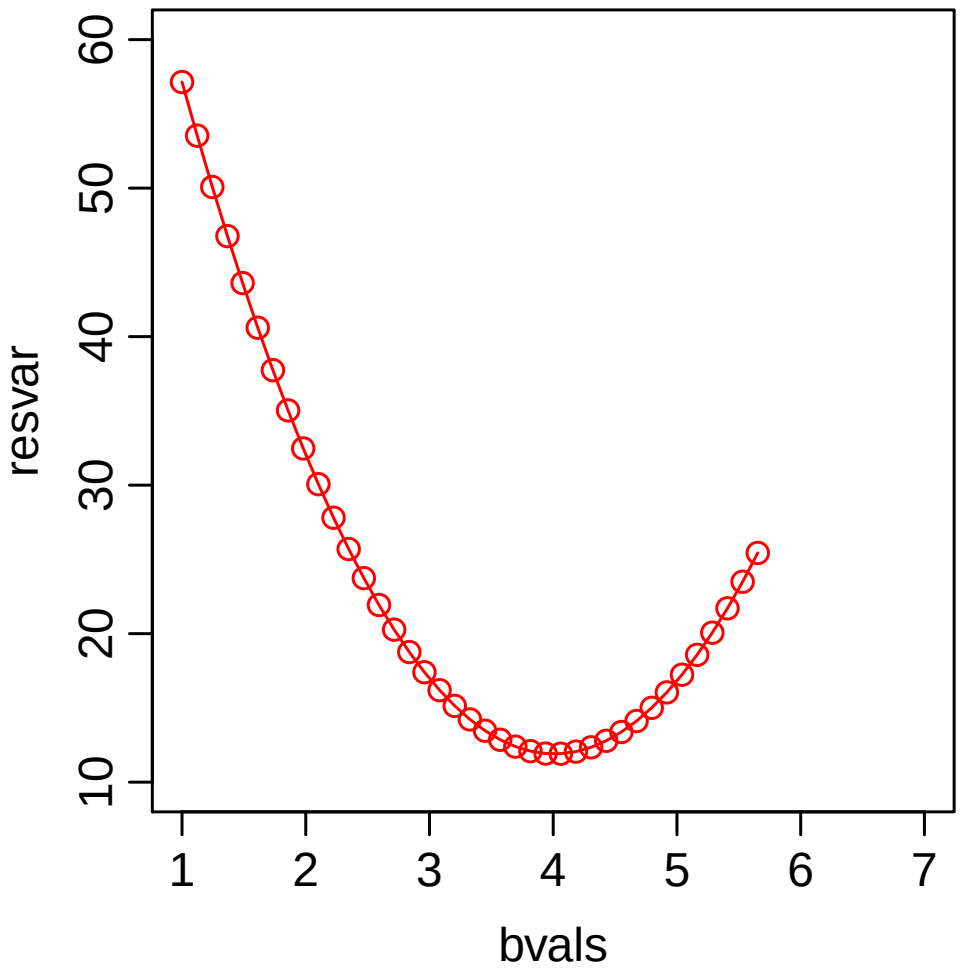
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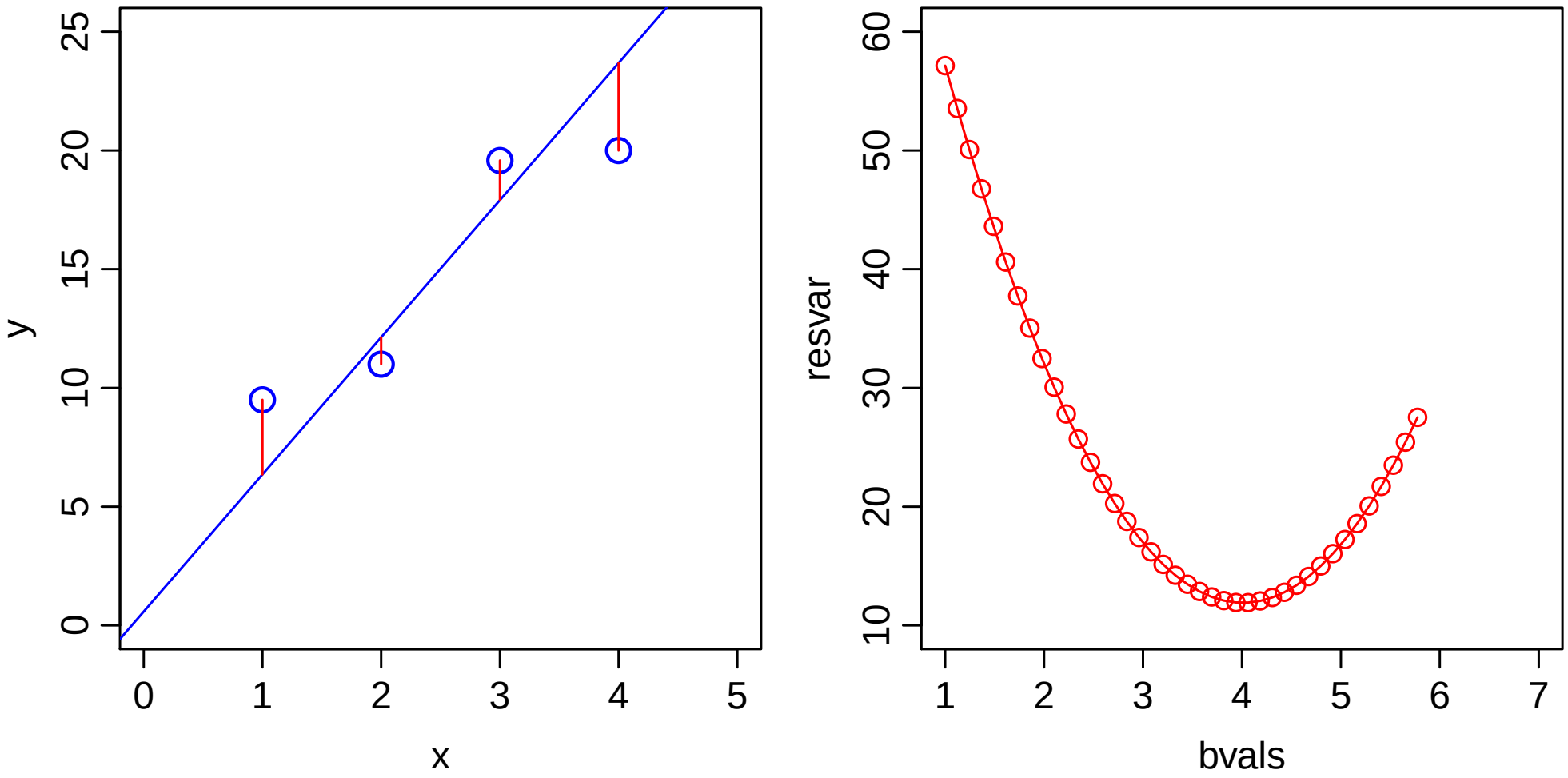
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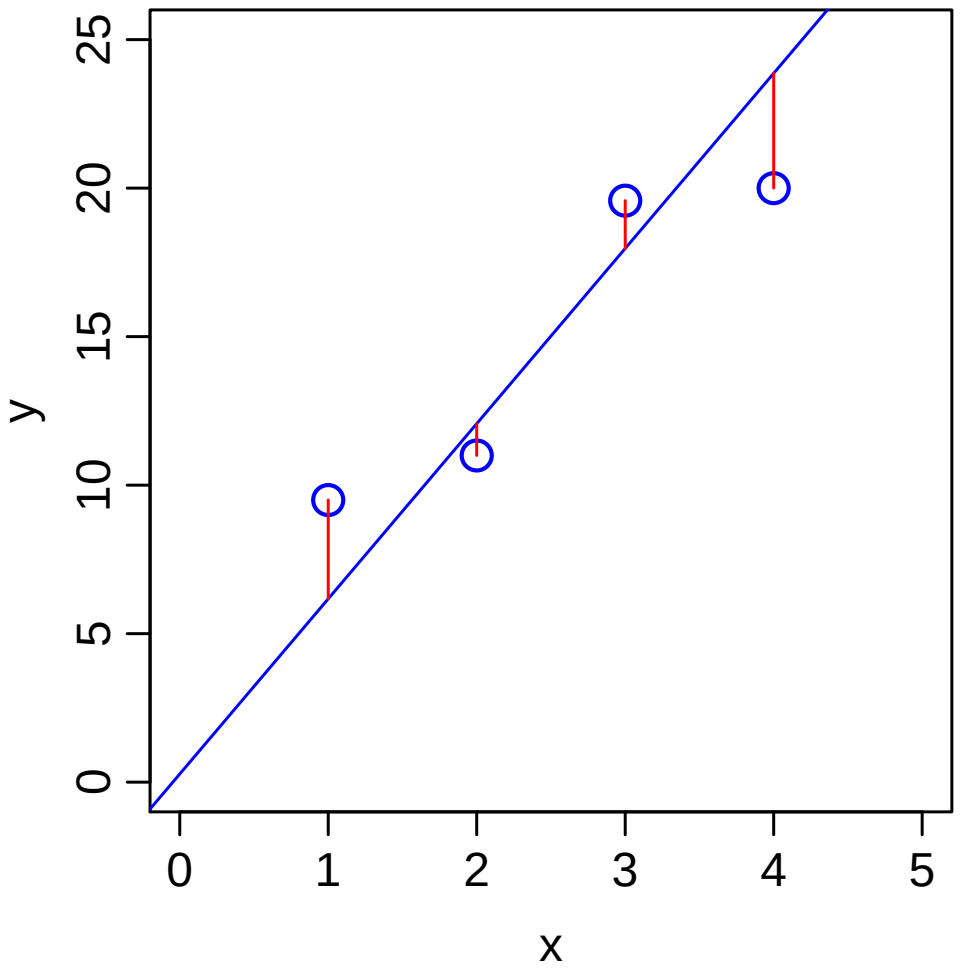
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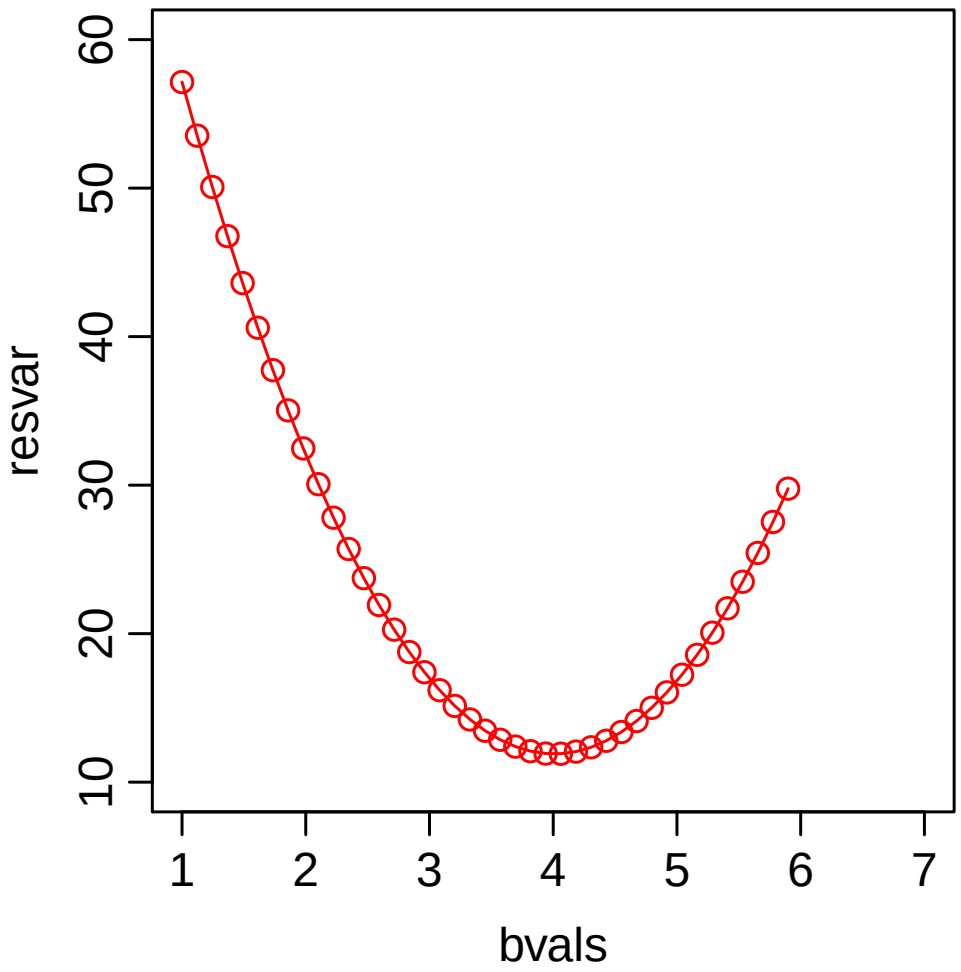
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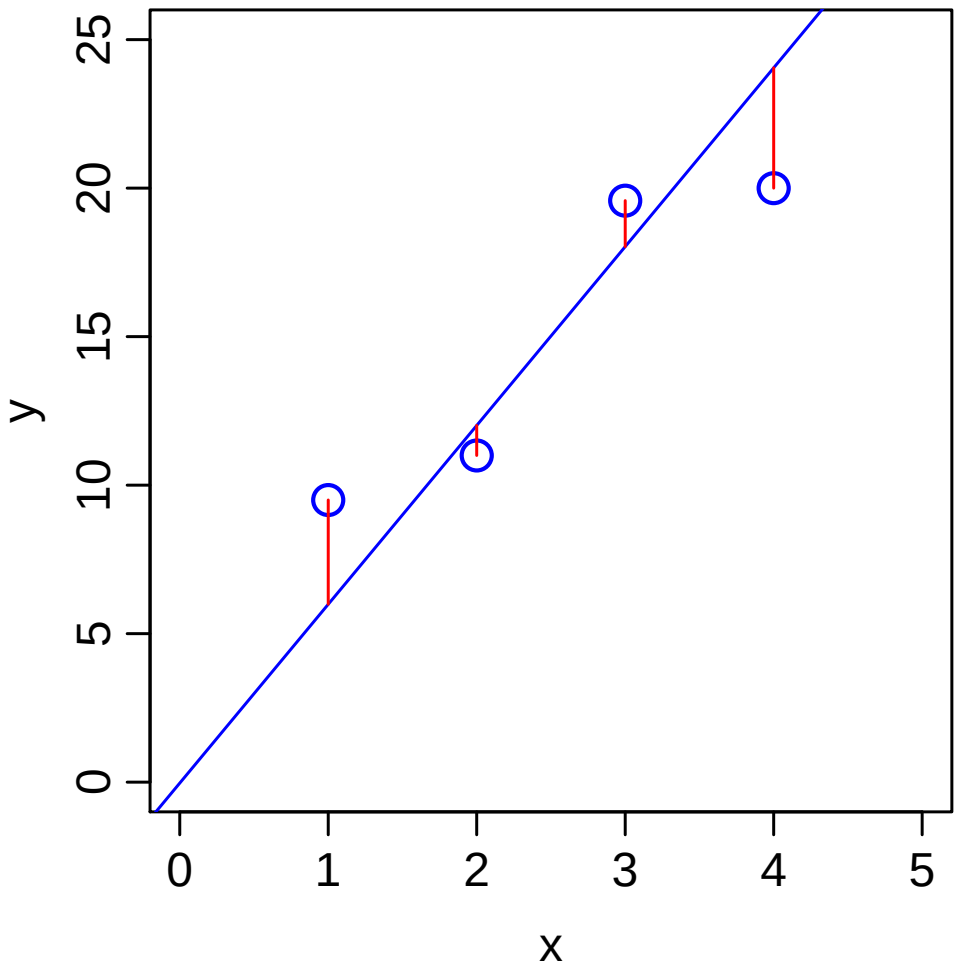
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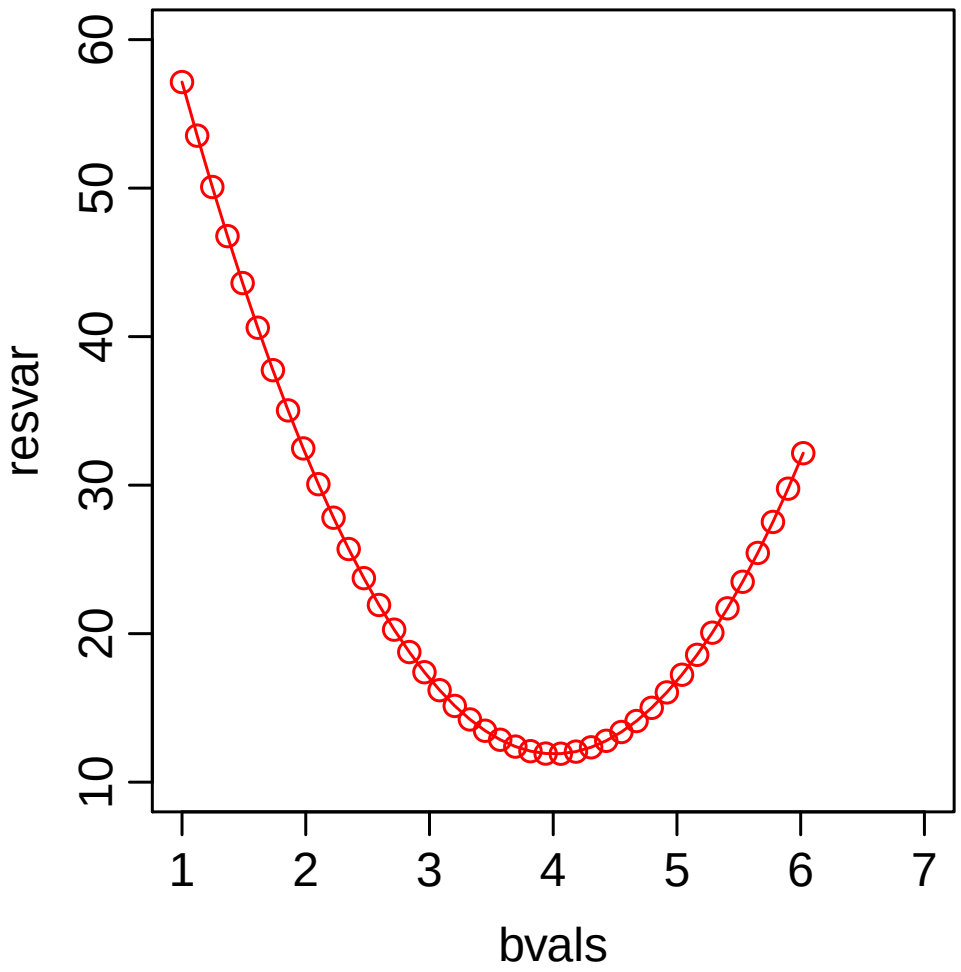
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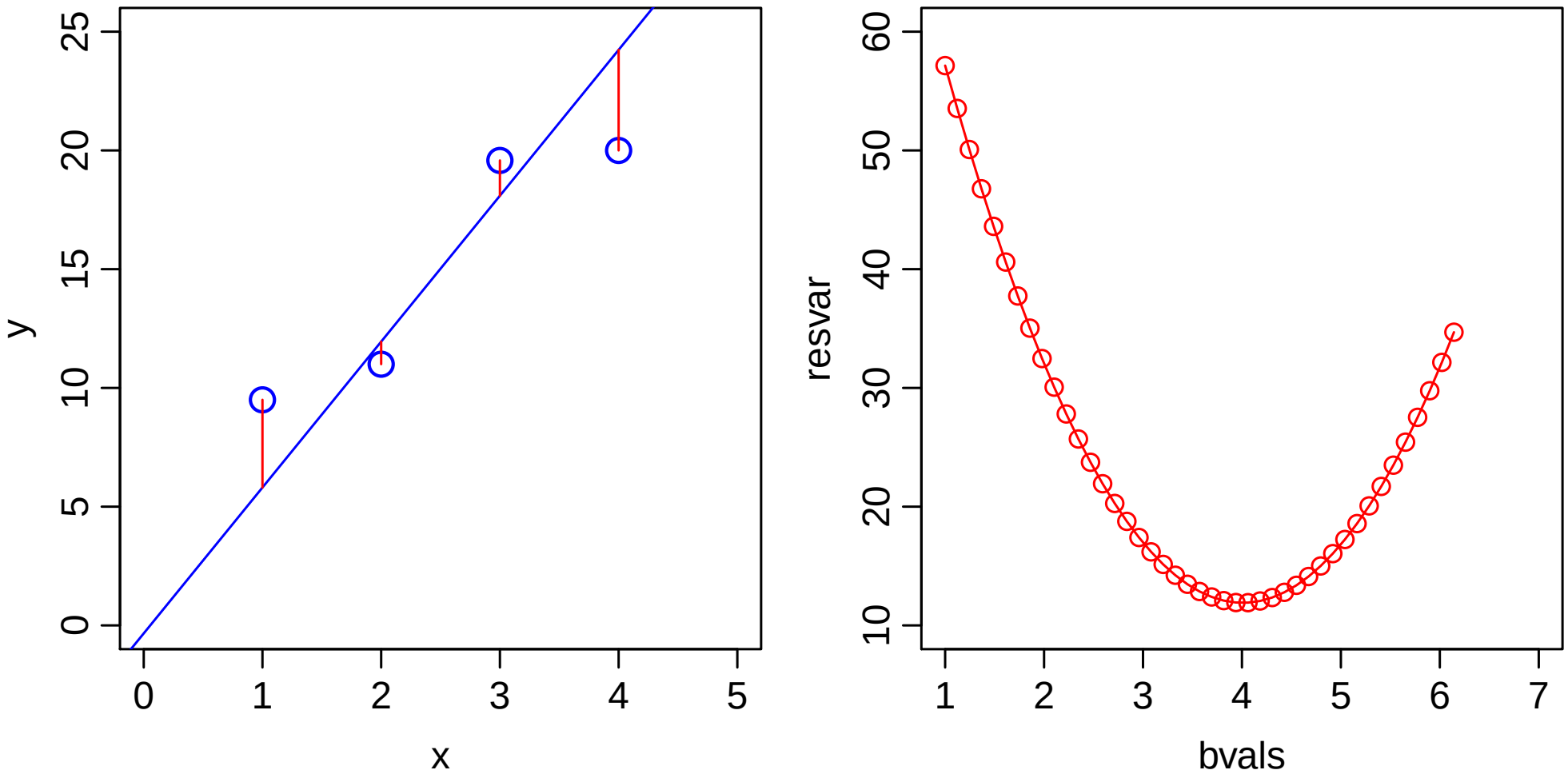
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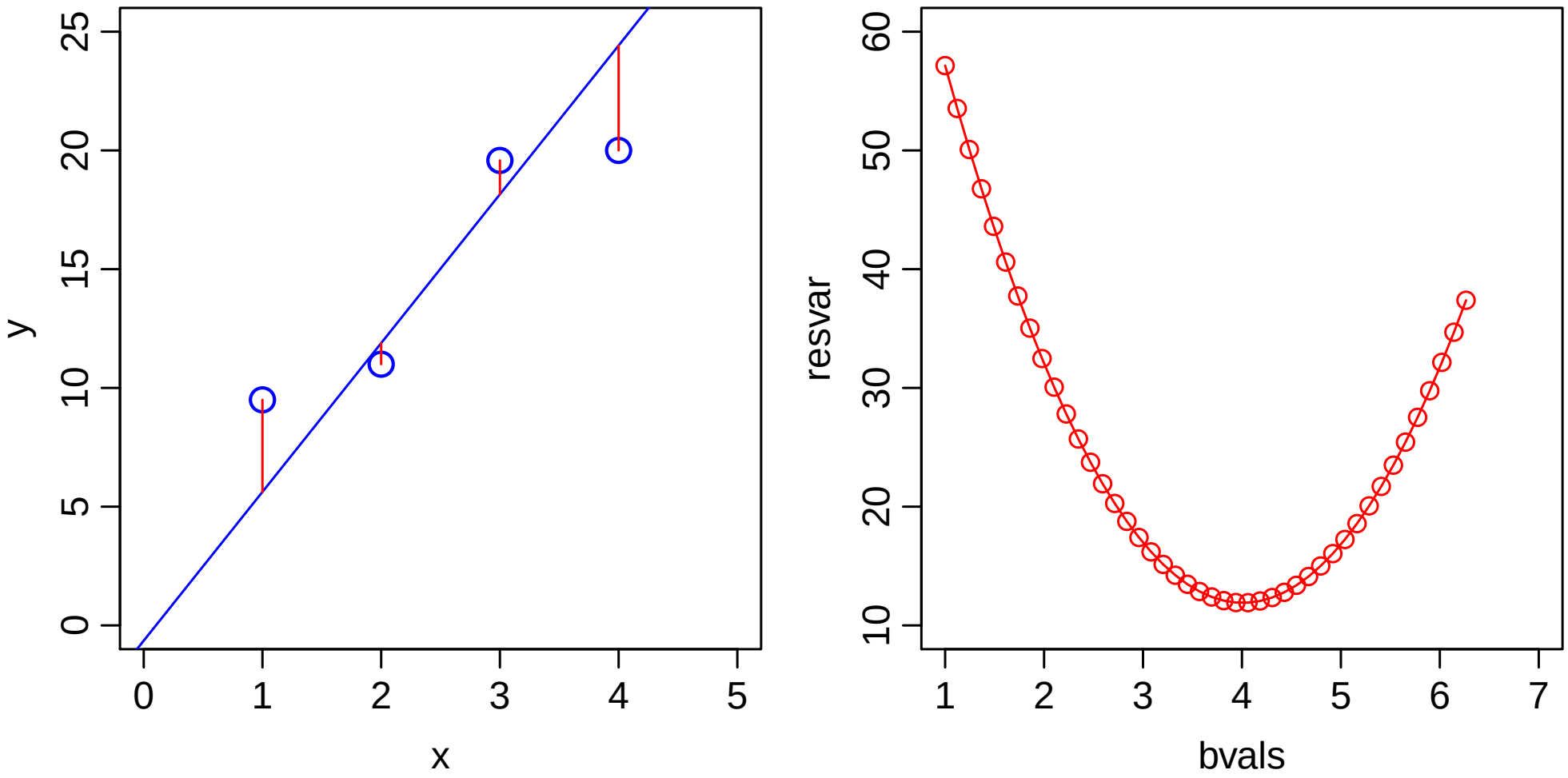
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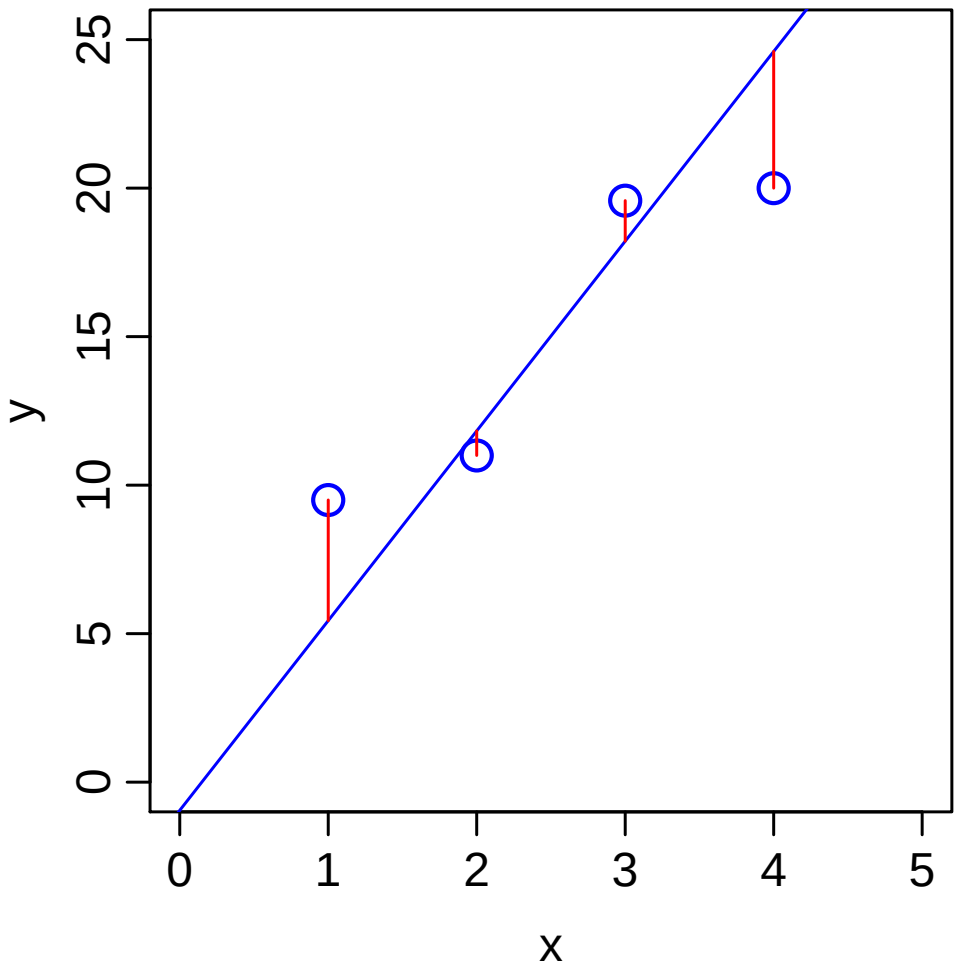
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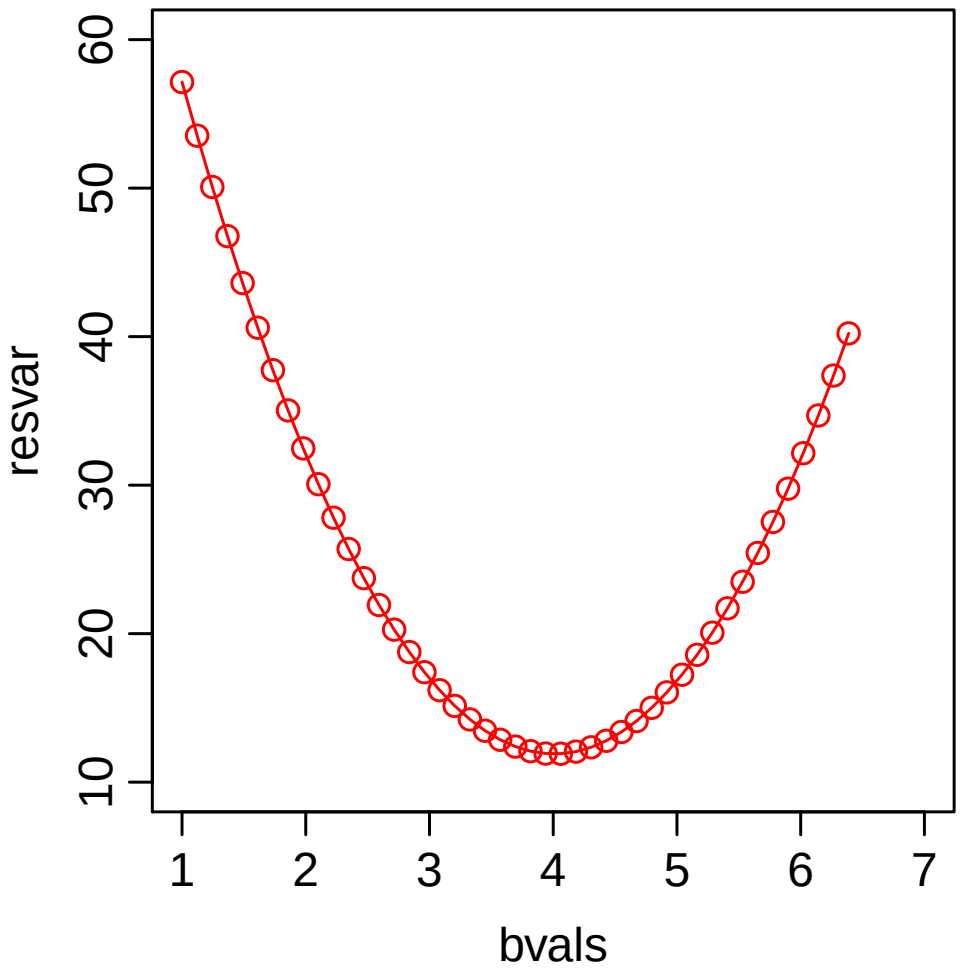
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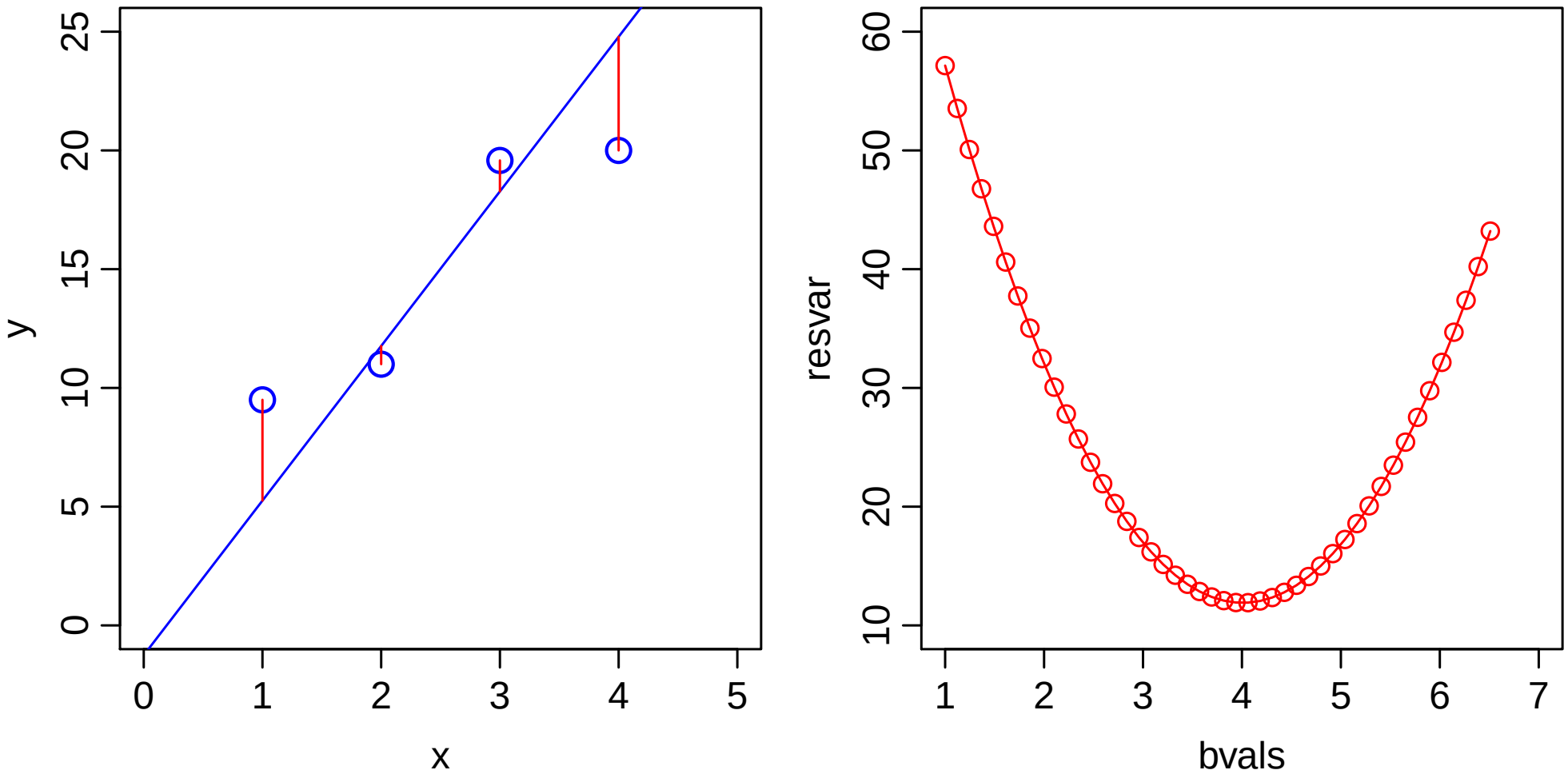
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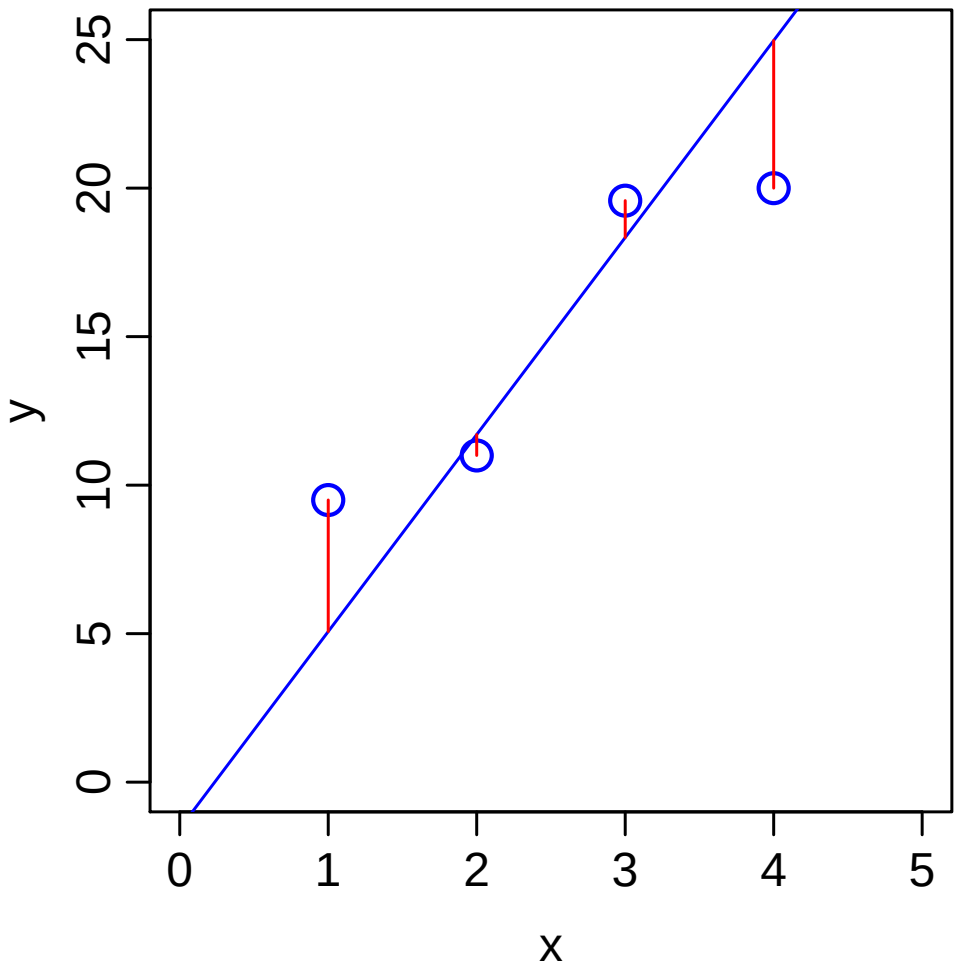
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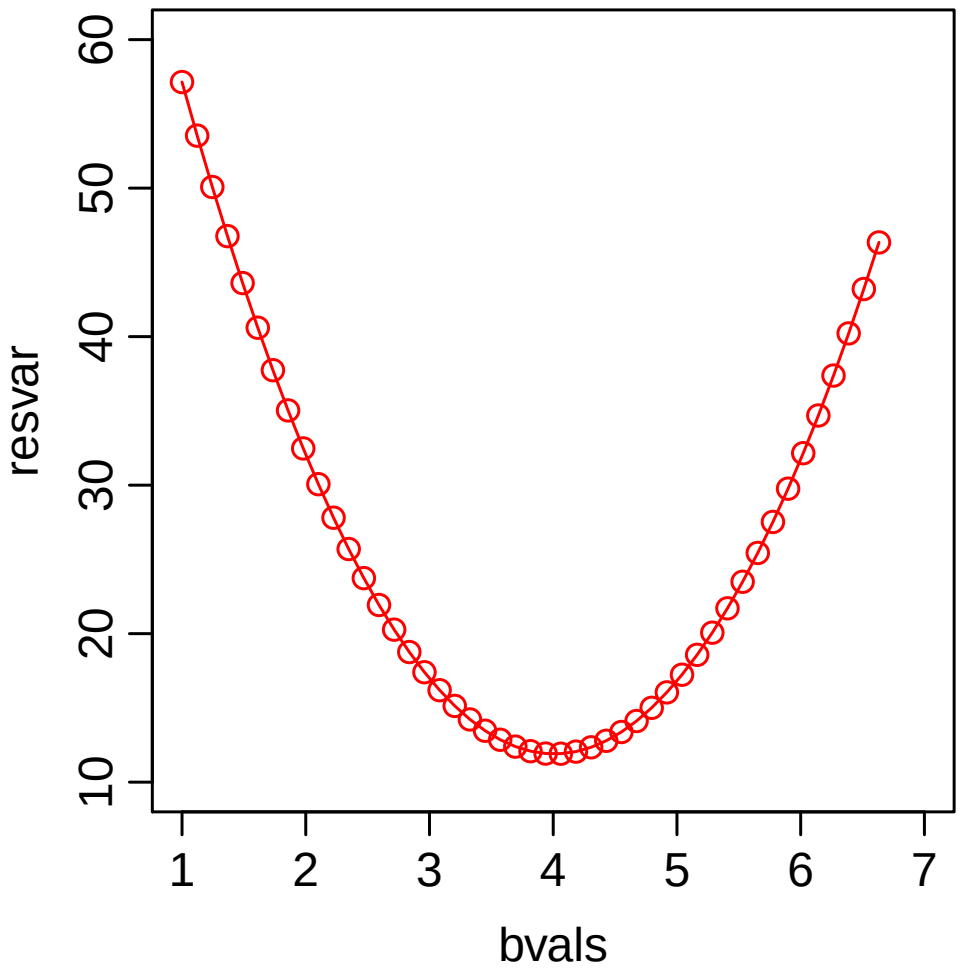
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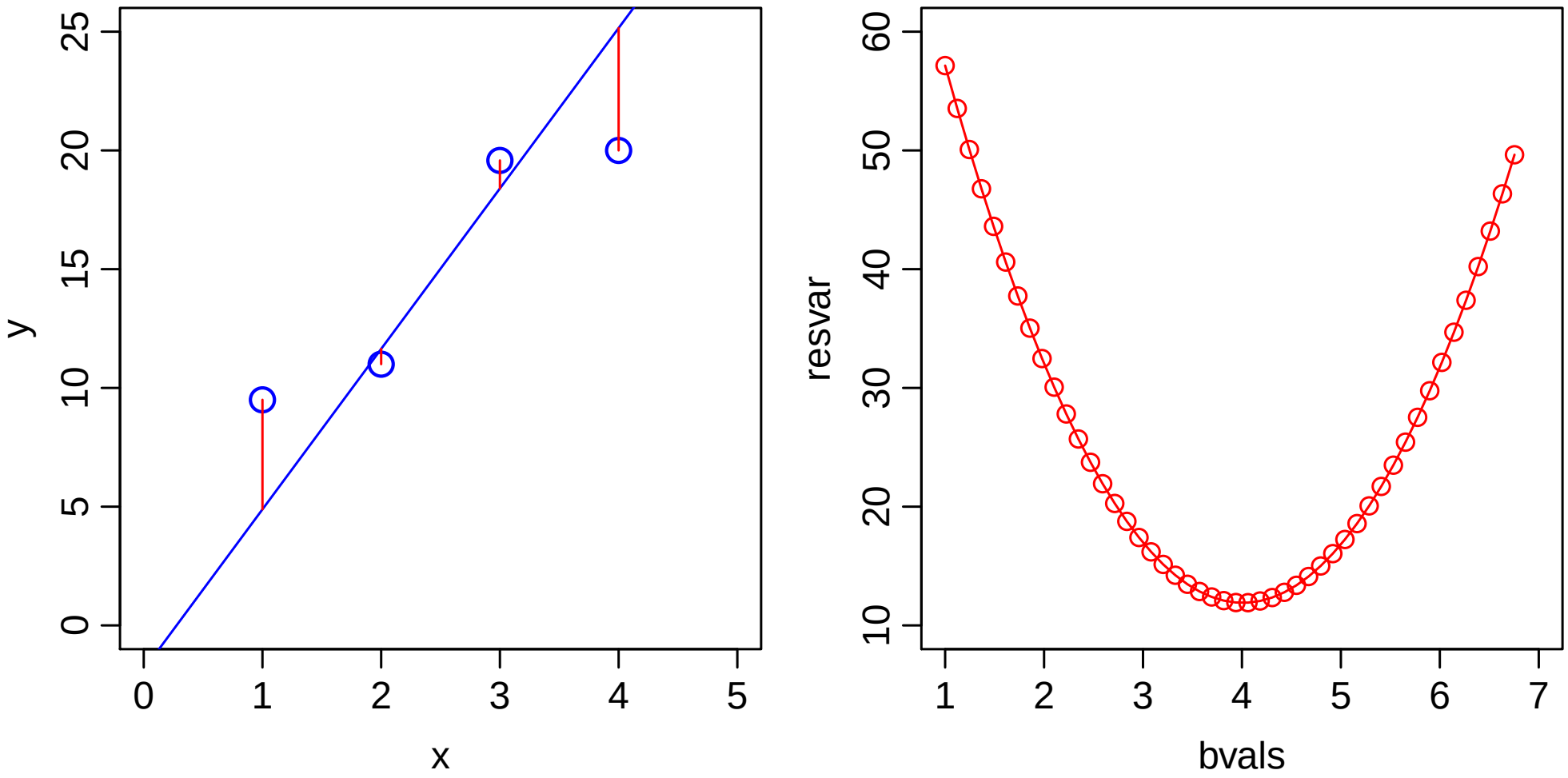
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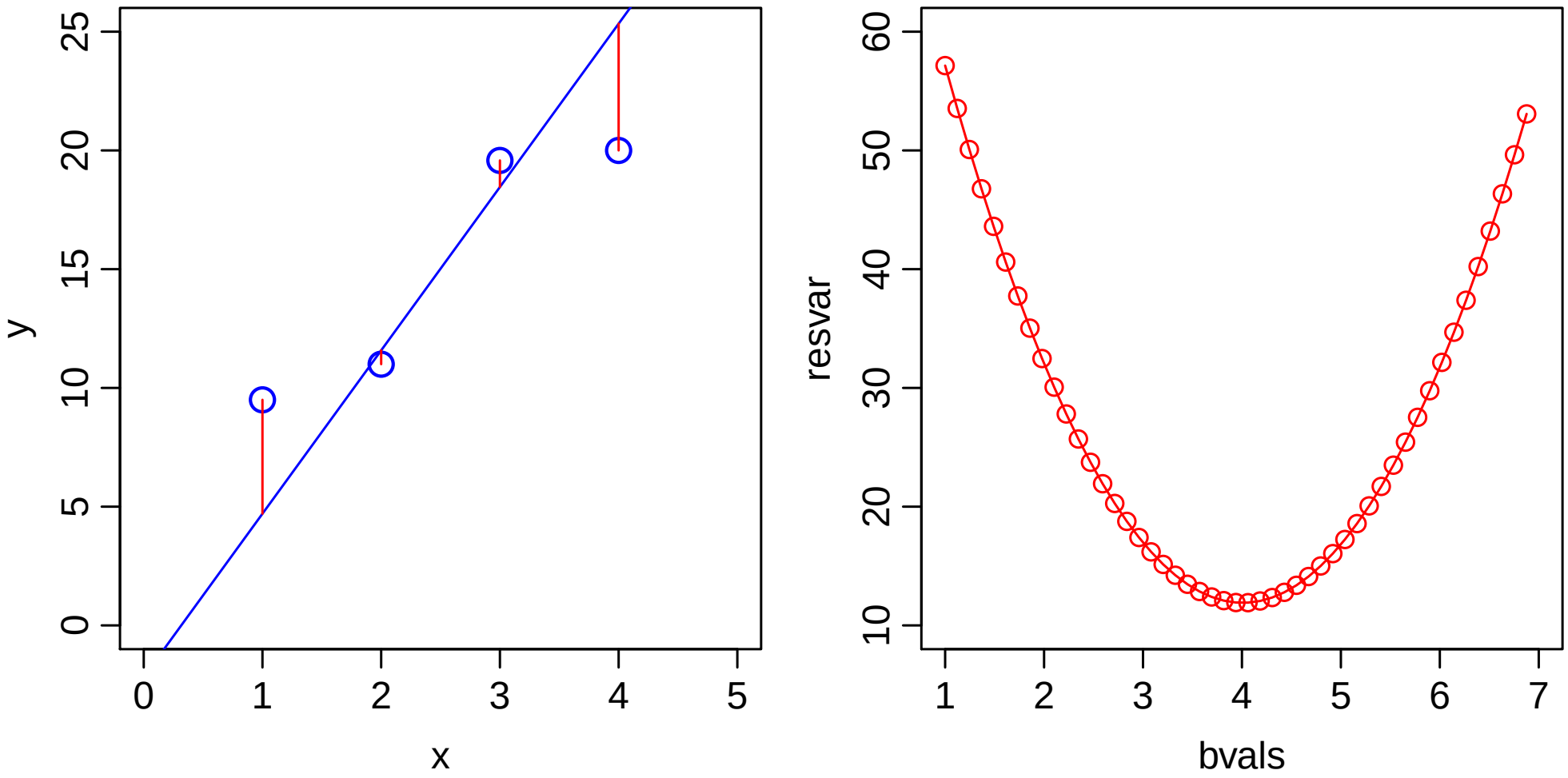
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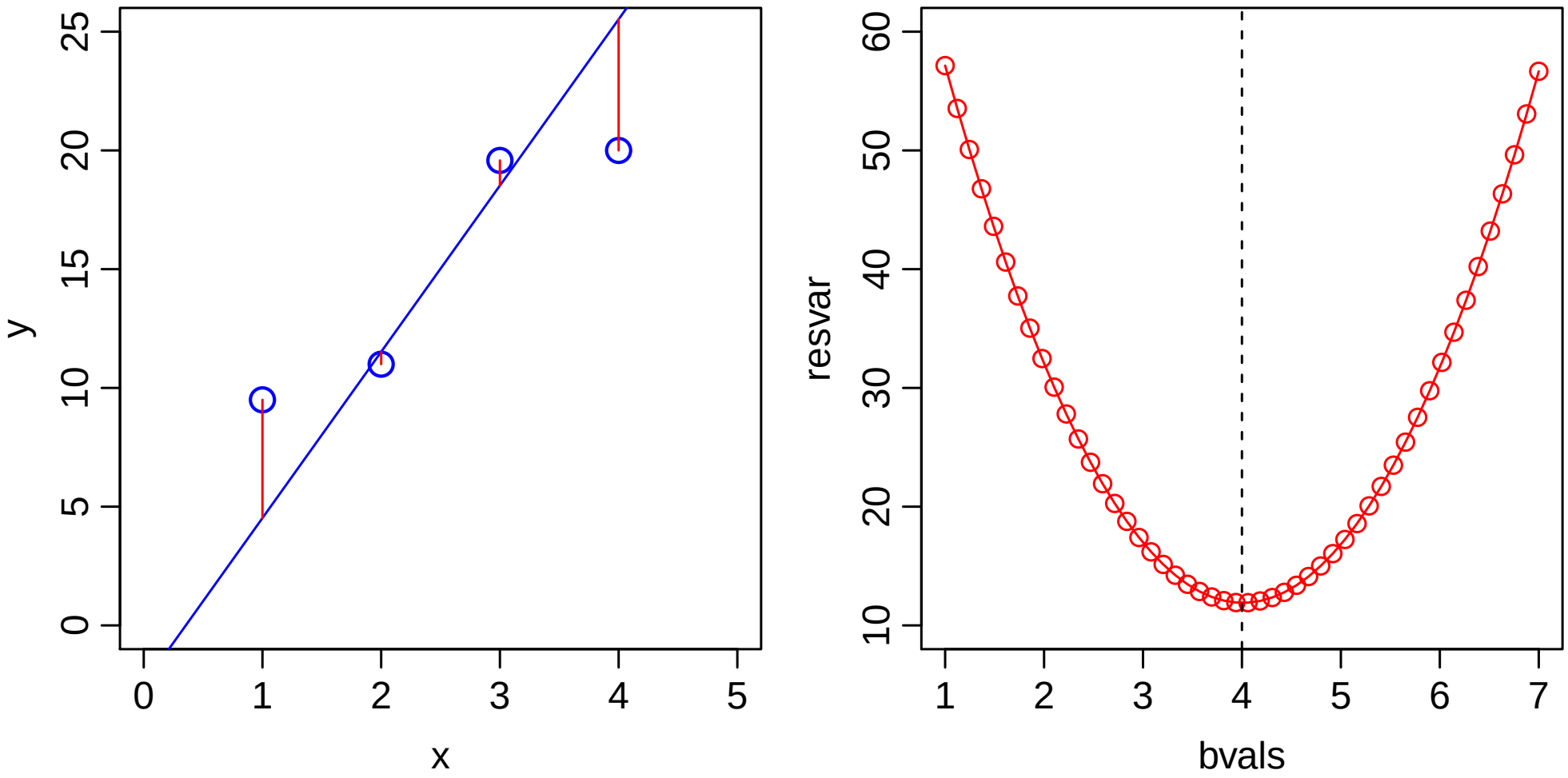
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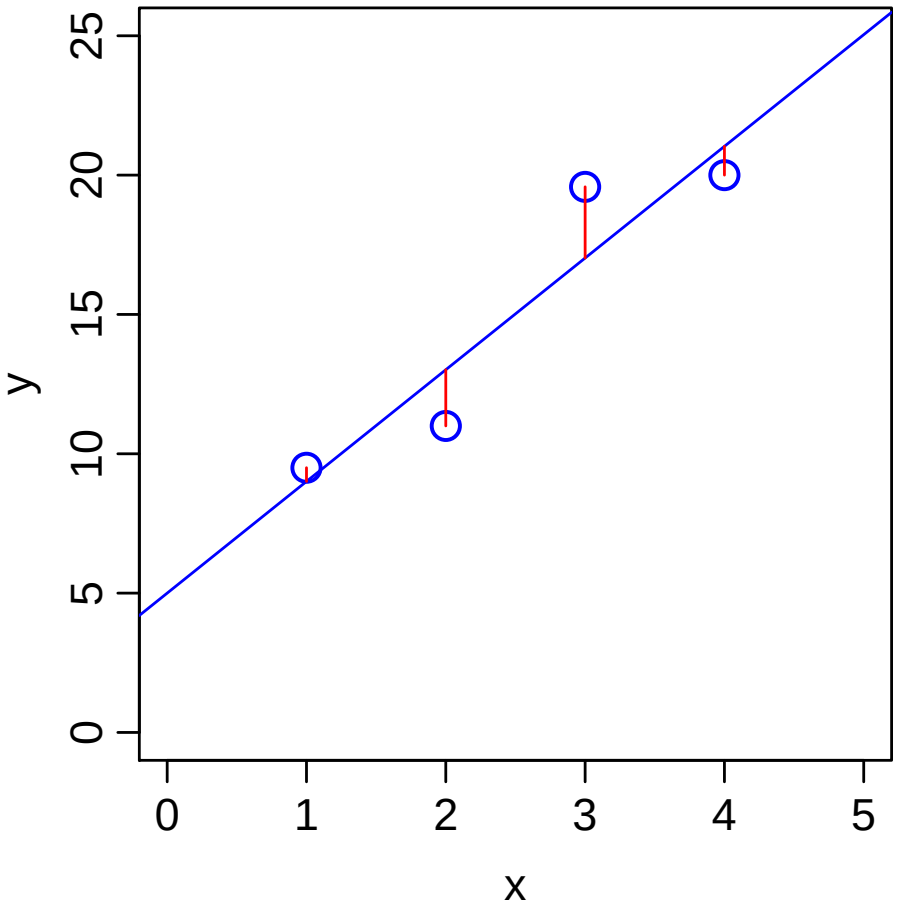

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$9.50 = 5 + 4 \times 1 + 0.50$$
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The least squares solution here is:
$\beta_0 = 5; \beta_1 = 4$

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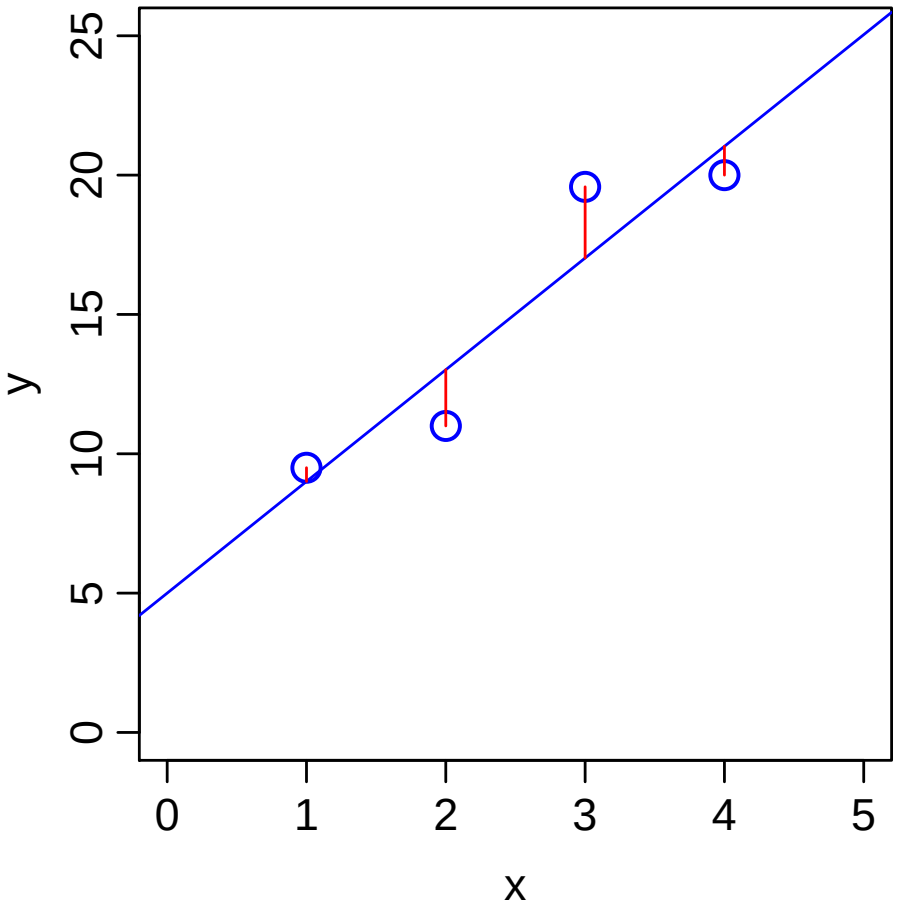

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- *This system of (linear) equations can be compactly represented (and solved using matrix algebra) as* $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

Intrinsic non-linearity makes Least-Aquares model fitting difficult

- In an intrinsically non-linear model such as $y_i = \beta_0 e^{\beta_2 x_i} + \varepsilon_i$, the nice trick of solving $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ *exactly* is impossible

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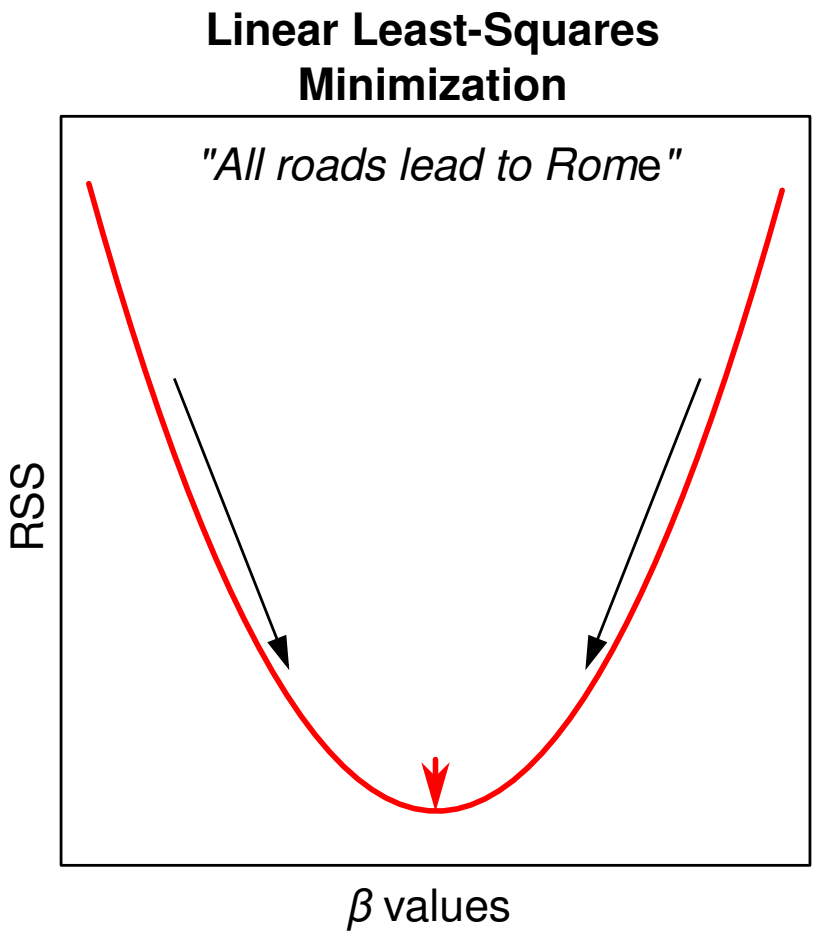

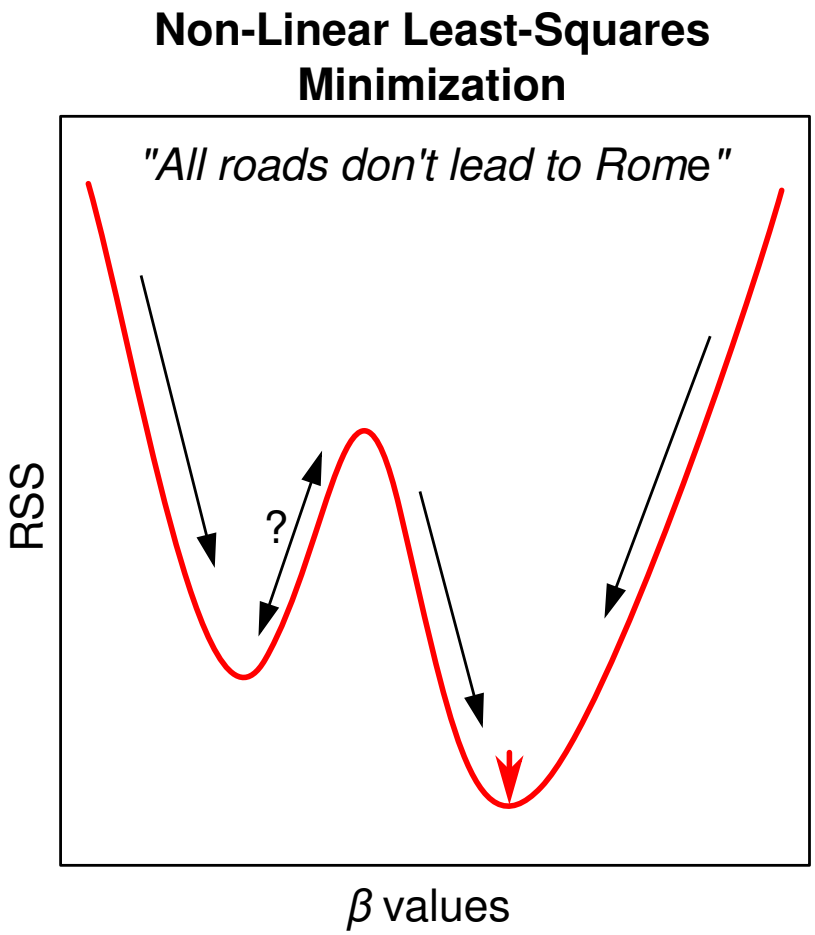

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- *Can you think of some examples?*

Non-Linear Model Example: Temperature and metabolism

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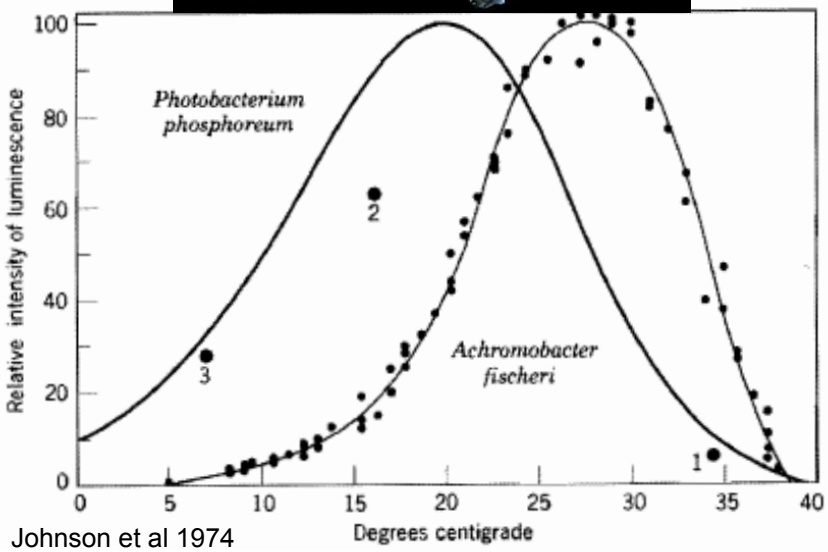

Johnson et al 1974

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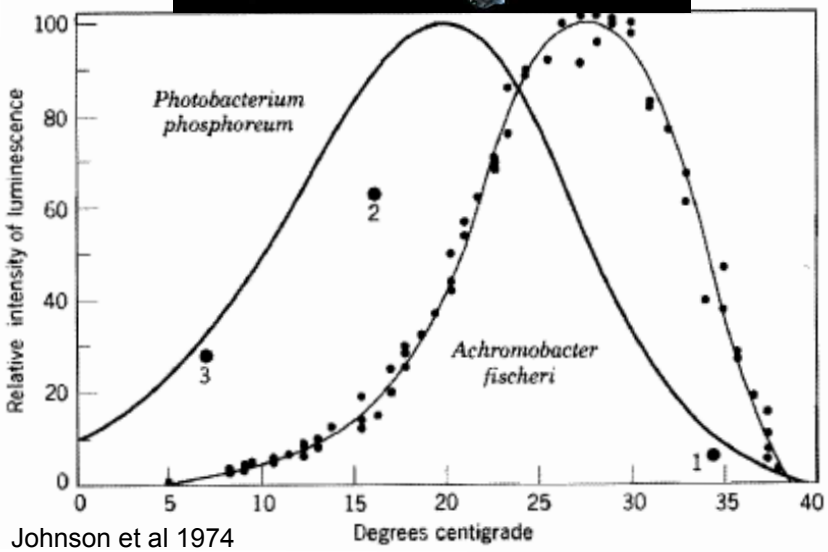

Johnson et al 1974

$$B = B_0 \boxed{e^{-\frac{E}{kT}}} f(T, T_{pk}, E_D)$$

T = temperature (K)
k = Boltzmann constant (eV K^{-1})
E = Activation energy (eV)
T_{pk} = Temperature of peak performance
E_D = Deactivation energy (eV)
(J H van't Hoff 1884, S Arrhenius 1889)

The nlls Fitting Method

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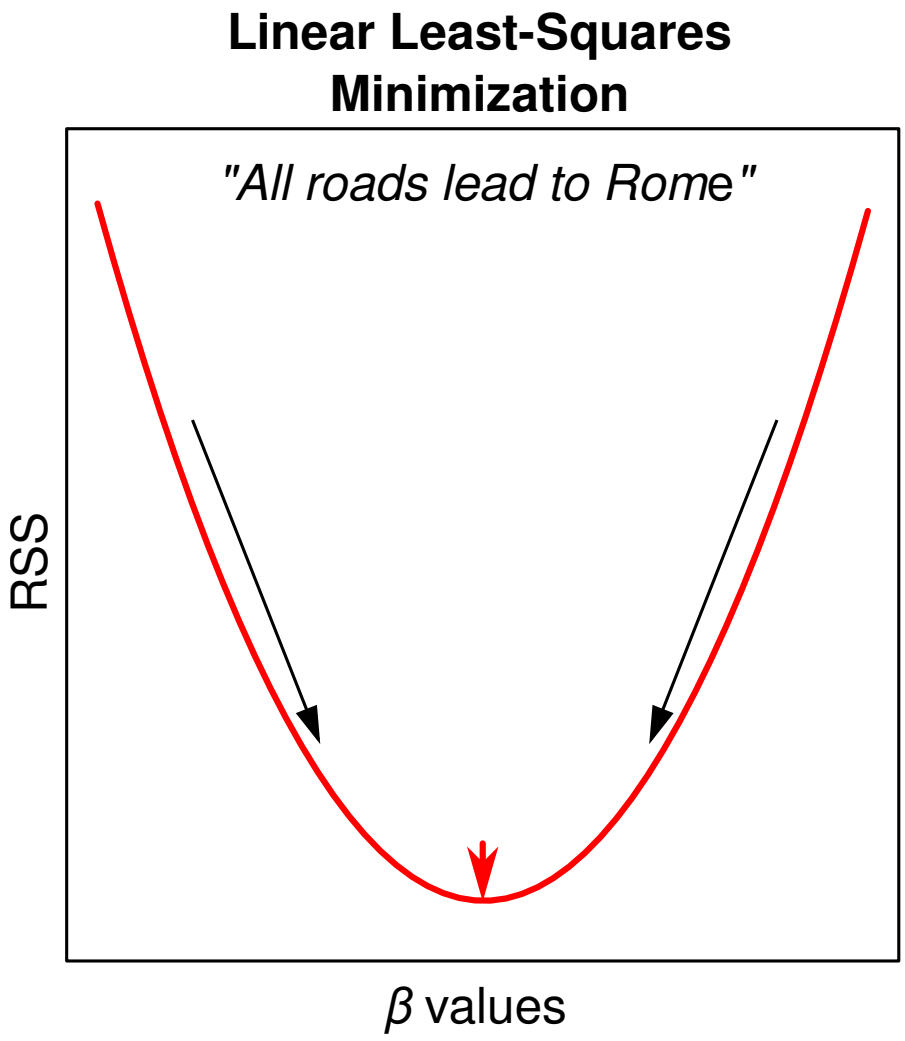
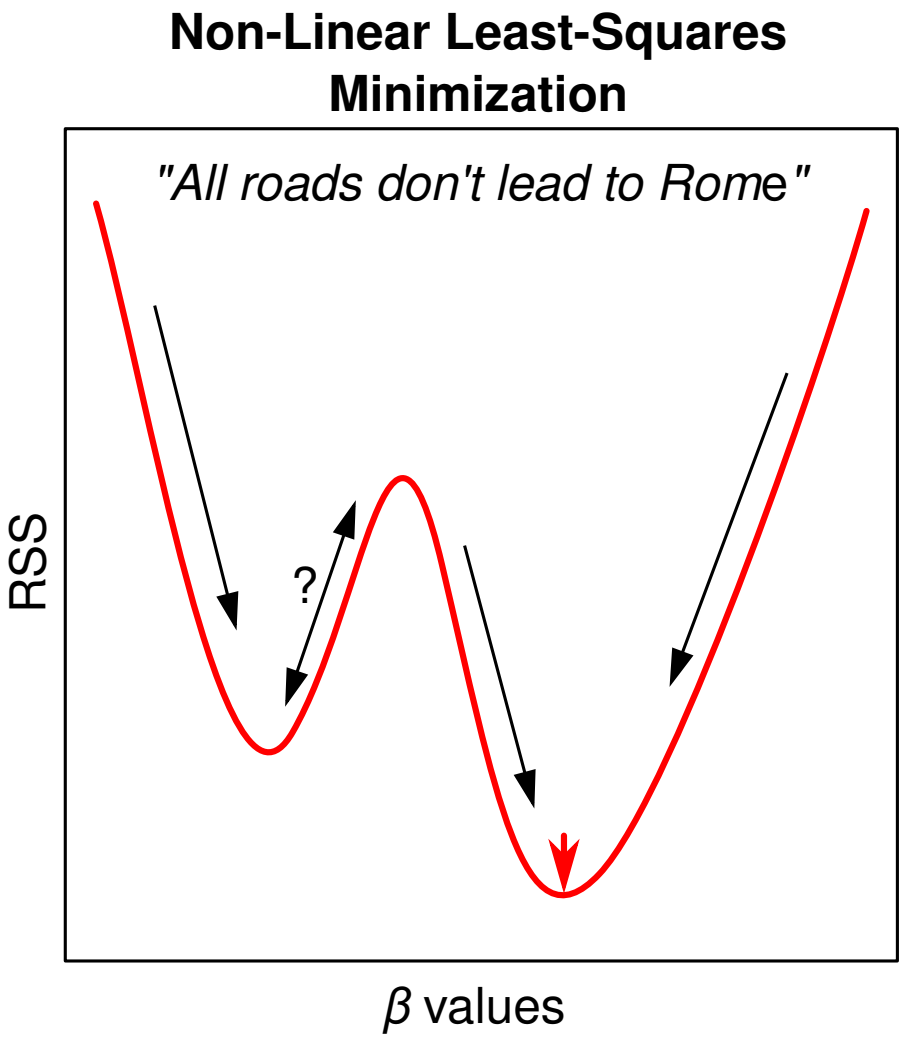
  - Eventually, if it all goes well, a combination of β_j's that is *very close* to the desired solution (where the rSS is *approximately* minimized) can be found

The NLLS fitting / optimization process





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7. Stop simulations when the adjustments make virtually no difference to the rSS

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The *tricky part — adjust parameters to make curve come closer to the data points* (step 4) — has two main algorithms that are generally used:

- The **Gauss-Newton** algorithm is often used, but doesn't work very well if the model to be fitted is mathematically complicated (the parameter search "landscape" is difficult) and the *starting values* for parameters are far-off-optimal

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- The **Levenberg-Marquardt** algorithm switches between Gauss-Newton and "gradient descent" and is more robust against starting values that are far-off-optimal and is more reliable in most scenarios.

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- Unlike in Linear Models, r^2 values *should not* be used to interpret the quality of a nlls fit (more on this in the practicals).

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- What if the errors are not normal? — Interpret results cautiously, and use Maximum Likelihood or bayesian fitting methods instead

Practicals Overview

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  - The command is `nlsLM()`
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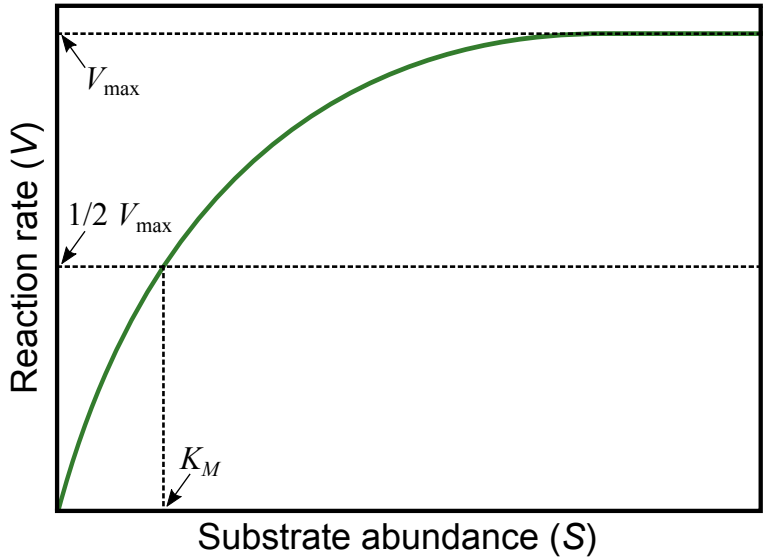
  - It offers additional features like the ability to "bound" parameters to realistic values

NLLS Fitting Practicals

- We will start with nlls fitting of the Michaelis-Menten model of biochemical reaction kinetics:

$$V = \frac{V_{\max}[S]}{K_m + [S]}$$



- S = Substrate density
- $V_{\max}$ = Maximum reaction rate (at saturating substrate concentration)
- K_M = Half-saturation constant; the S at which reaction rate reaches half of possible maximum ($= \frac{1}{2} V_{\max}$)

- You will use nlls fitting to obtain estimates of $V_{\max}$ and K_M

- Note that $V_{\max} \leq 0$ and $K_M \leq 0$ are physically impossible (useful fir picking starting values)

READINGS

- Motulsky, Harvey, and Arthur Christopoulos. Fitting models to biological data using linear and nonlinear regression: a practical guide to curve fitting. OUP USA, 2004.